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Encoder, decoder, encoding method, and decoding method

Abstract

An encoder includes: circuitry; and memory coupled to the circuitry. In operation, the circuitry: controls whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in encoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and encodes an image in the inter-predicted picture using the reference image resampled.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a U.S. continuation application of PCT International Patent Application Number PCT/JP2022/017641 filed on Apr. 12, 2022, claiming the benefit of priority of European Patent Application Number 21172831.6 filed on May 7, 2021, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

(1) The present disclosure relates to an encoder, a decoder, an encoding method, and a decoding method.

2. Description of the Related Art

(2) With advancement in video coding technology, from H.261 and MPEG-1 to H.264/AVC (Advanced Video Coding), MPEG-LA, H.265/HEVC (High Efficiency Video Coding) and H.266/VVC (Versatile Video Codec), there remains a constant need to provide improvements and optimizations to the video coding technology to process an ever-increasing amount of digital video data in various applications. The present disclosure relates to further advancements, improvements and optimizations in video coding.

(3) Note that H.265 (ISO/IEC 23008-2 HEVC)/HEVC (High Efficiency Video Coding) relates to one example of a conventional standard regarding the above-described video coding technology.

SUMMARY

(4) For example, an encoder according to one aspect of the present disclosure includes: circuitry; and memory coupled to the circuitry. In operation, the circuitry: controls whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in encoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and encodes an image in the inter-predicted picture using the reference image resampled.

(5) Each of embodiments, or each of part of constituent elements and methods in the present disclosure enables, for example, at least one of the following: improvement in coding efficiency, enhancement in image quality, reduction in processing amount of encoding/decoding, reduction in circuit scale, improvement in processing speed of encoding/decoding, etc. Alternatively, each of embodiments, or each of part of constituent elements and methods in the present disclosure enables, in encoding and decoding, appropriate selection of an element or an operation. The element is, for example, a filter, a block, a size, a motion vector, a reference picture, or a reference block. It is to be noted that the present disclosure includes disclosure regarding configurations and methods which may provide advantages other than the above-described ones. Examples of such configurations and methods include a configuration or method for improving coding efficiency while reducing increase in processing amount.

(6) Additional benefits and advantages according to an aspect of the present disclosure will become apparent from the specification and drawings. The benefits and/or advantages may be individually obtained by the various embodiments and features of the specification and drawings, and not all of which need to be provided in order to obtain one or more of such benefits and/or advantages.

(7) It is to be noted that these general or specific aspects may be implemented using a system, an integrated circuit, a computer program, or a computer readable medium (recording medium) such as a CD-ROM, or any combination of systems, methods, integrated circuits, computer programs, and media.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) These and other objects, advantages and features of the disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.
- (2) FIG. 1 is a schematic diagram illustrating one example of a configuration of a transmission system according to an embodiment;
- (3) FIG. 2 is a diagram illustrating one example of a hierarchical structure of data in a stream;
- (4) FIG. 3 is a diagram illustrating one example of a slice configuration;
- (5) FIG. 4 is a diagram illustrating one example of a tile configuration;
- (6) FIG. 5 is a diagram illustrating one example of an encoding structure in scalable encoding;
- (7) FIG. 6 is a diagram illustrating one example of an encoding structure in scalable encoding;
- (8) FIG. 7 is a block diagram illustrating one example of a configuration of an encoder according to an embodiment;
- (9) FIG. 8 is a block diagram illustrating a mounting example of the encoder;
- (10) FIG. 9 is a flow chart illustrating one example of an overall encoding process performed by the encoder;
- (11) FIG. 10 is a diagram illustrating one example of block splitting;
- (12) FIG. 11 is a diagram illustrating one example of a configuration of a splitter;
- (13) FIG. 12 is a diagram illustrating examples of splitting patterns;
- (14) FIG. 13A is a diagram illustrating one example of a syntax tree of a splitting pattern;
- (15) FIG. 13B is a diagram illustrating another example of a syntax tree of a splitting pattern;
- (16) FIG. 14 is a chart illustrating transform basis functions for each transform type;
- (17) FIG. 15 is a diagram illustrating examples of SVT;
- (18) FIG. 16 is a flow chart illustrating one example of a process performed by a transformer;
- (19) FIG. 17 is a flow chart illustrating another example of a process performed by the transformer;
- (20) FIG. 18 is a block diagram illustrating one example of a configuration of a quantizer;
- (21) FIG. 19 is a flow chart illustrating one example of quantization performed by the quantizer;
- (22) FIG. 20 is a block diagram illustrating one example of a configuration of an entropy encoder;
- (23) FIG. 21 is a diagram illustrating a flow of CABAC in the entropy encoder;
- (24) FIG. 22 is a block diagram illustrating one example of a configuration of a loop filter;
- (25) FIG. 23A is a diagram illustrating one example of a filter shape used in an adaptive loop filter (ALF);
- (26) FIG. 23B is a diagram illustrating another example of a filter shape used in an ALF;
- (27) FIG. 23C is a diagram illustrating another example of a filter shape used in an ALF;
- (28) FIG. 23D is a diagram illustrating an example where Y samples (first component) are used for a cross component ALF (CCALF) for Cb and a CCALF for Cr (components different from the first component);
- (29) FIG. 23E is a diagram illustrating a diamond shaped filter;

- (30) FIG. 23F is a diagram illustrating an example for a joint chroma CCALF (JC-CCALF);
- (31) FIG. 23G is a diagram illustrating an example for JC-CCALF weight index candidates;
- (32) FIG. 24 is a block diagram illustrating one example of a specific configuration of a loop filter which functions as a DBF;
- (33) FIG. 25 is a diagram illustrating an example of a deblocking filter having a symmetrical filtering characteristic with respect to a block boundary;
- (34) FIG. 26 is a diagram for illustrating a block boundary on which a deblocking filter process is performed;
- (35) FIG. 27 is a diagram illustrating examples of Bs values;
- (36) FIG. 28 is a flow chart illustrating one example of a process performed by a predictor of the encoder;
- (37) FIG. 29 is a flow chart illustrating another example of a process performed by the predictor of the encoder;
- (38) FIG. 30 is a flow chart illustrating another example of a process performed by the predictor of the encoder;
- (39) FIG. 31 is a diagram illustrating one example of sixty-seven intra prediction modes used in intra prediction;
- (40) FIG. 32 is a flow chart illustrating one example of a process performed by an intra predictor;
- (41) FIG. 33 is a diagram illustrating examples of reference pictures;
- (42) FIG. 34 is a diagram illustrating examples of reference picture lists;
- (43) FIG. 35 is a flow chart illustrating a basic processing flow of inter prediction;
- (44) FIG. 36 is a flow chart illustrating one example of MV derivation;
- (45) FIG. 37 is a flow chart illustrating another example of MV derivation;
- (46) FIG. 38A is a diagram illustrating one example of categorization of modes for MV derivation;
- (47) FIG. 38B is a diagram illustrating one example of categorization of modes for MV derivation;
- (48) FIG. 39 is a flow chart illustrating an example of inter prediction by normal inter mode;
- (49) FIG. 40 is a flow chart illustrating an example of inter prediction by normal merge mode;
- (50) FIG. 41 is a diagram for illustrating one example of an MV derivation process by normal merge mode;
- (51) FIG. 42 is a diagram for illustrating one example of an MV derivation process by a history-based motion vector prediction/predictor (HMVP) mode;
- (52) FIG. 43 is a flow chart illustrating one example of frame rate up conversion (FRUC);
- (53) FIG. 44 is a diagram for illustrating one example of pattern matching (bilateral matching) between two blocks located along a motion trajectory;
- (54) FIG. 45 is a diagram for illustrating one example of pattern matching (template matching) between a template in a current picture and a block in a reference picture;
- (55) FIG. 46A is a diagram for illustrating one example of MV derivation in units of a sub-block in affine mode in which two control points are used;
- (56) FIG. 46B is a diagram for illustrating one example of MV derivation in units of a sub-block in affine mode in which three control points are used;
- (57) FIG. 47A is a conceptual diagram for illustrating one example of MV derivation at control points in an affine mode;
- (58) FIG. 47B is a conceptual diagram for illustrating one example of MV derivation at control points in an affine mode;
- (59) FIG. 47C is a conceptual diagram for illustrating one example of MV derivation at control points in an affine mode;
- (60) FIG. 48A is a diagram for illustrating an affine mode in which two control points are used;
- (61) FIG. 48B is a diagram for illustrating an affine mode in which three control points are used;
- (62) FIG. 49A is a conceptual diagram for illustrating one example of a method for MV derivation at control points when the number of control points for an encoded block and the number of control

points for a current block are different from each other;

(63) FIG. **49B** is a conceptual diagram for illustrating another example of a method for MV derivation at control points when the number of control points for an encoded block and the number of control points for a current block are different from each other;

(64) FIG. **50** is a flow chart illustrating one example of a process in affine merge mode;

(65) FIG. **51** is a flow chart illustrating one example of a process in affine inter mode;

(66) FIG. **52A** is a diagram for illustrating generation of two triangular prediction images;

(67) FIG. **52B** is a conceptual diagram illustrating examples of a first portion of a first partition and first and second sets of samples;

(68) FIG. **52C** is a conceptual diagram illustrating a first portion of a first partition;

(69) FIG. **53** is a flow chart illustrating one example of a triangle mode;

(70) FIG. **54** is a diagram illustrating one example of an advanced temporal motion vector prediction/predictor (ATMVP) mode in which an MV is derived in units of a sub-block;

(71) FIG. **55** is a diagram illustrating a relationship between a merge mode and dynamic motion vector refreshing (DMVR);

(72) FIG. **56** is a conceptual diagram for illustrating one example of DMVR;

(73) FIG. **57** is a conceptual diagram for illustrating another example of DMVR for determining an MV;

(74) FIG. **58A** is a diagram illustrating one example of motion estimation in DMVR;

(75) FIG. **58B** is a flow chart illustrating one example of motion estimation in DMVR;

(76) FIG. **59** is a flow chart illustrating one example of generation of a prediction image;

(77) FIG. **60** is a flow chart illustrating another example of generation of a prediction image;

(78) FIG. **61** is a flow chart illustrating one example of a correction process of a prediction image by overlapped block motion compensation (OBMC);

(79) FIG. **62** is a conceptual diagram for illustrating one example of a prediction image correction process by OBMC;

(80) FIG. **63** is a diagram for illustrating a model assuming uniform linear motion;

(81) FIG. **64** is a flow chart illustrating one example of inter prediction according to BIO;

(82) FIG. **65** is a diagram illustrating one example of a configuration of an inter predictor which performs inter prediction according to BIO;

(83) FIG. **66A** is a diagram for illustrating one example of a prediction image generation method using a luminance correction process by local illumination compensation (LIC);

(84) FIG. **66B** is a flow chart illustrating one example of a prediction image generation method using a luminance correction process by LIC;

(85) FIG. **67** is a block diagram illustrating a configuration of a decoder according to an embodiment;

(86) FIG. **68** is a block diagram illustrating a mounting example of a decoder;

(87) FIG. **69** is a flow chart illustrating one example of an overall decoding process performed by the decoder;

(88) FIG. **70** is a diagram illustrating a relationship between a splitting determiner and other constituent elements;

(89) FIG. **71** is a block diagram illustrating one example of a configuration of an entropy decoder;

(90) FIG. **72** is a diagram illustrating a flow of CABAC in the entropy decoder;

(91) FIG. **73** is a block diagram illustrating one example of a configuration of an inverse quantizer;

(92) FIG. **74** is a flow chart illustrating one example of inverse quantization performed by the inverse quantizer;

(93) FIG. **75** is a flow chart illustrating one example of a process performed by an inverse transformer;

(94) FIG. **76** is a flow chart illustrating another example of a process performed by the inverse transformer;

(95) FIG. **77** is a block diagram illustrating one example of a configuration of a loop filter;

(96) FIG. **78** is a flow chart illustrating one example of a process performed by a predictor of the decoder;

(97) FIG. **79** is a flow chart illustrating another example of a process performed by the predictor of the decoder;

(98) FIG. **80A** is a flow chart illustrating a portion of other example of a process performed by the predictor of the decoder;

(99) FIG. **80B** is a flow chart illustrating the remaining portion of the other example of the process performed by the predictor of the decoder;

(100) FIG. **81** is a diagram illustrating one example of a process performed by an intra predictor of the decoder;

(101) FIG. **82** is a flow chart illustrating one example of MV derivation in the decoder;

(102) FIG. **83** is a flow chart illustrating another example of MV derivation in the decoder;

(103) FIG. **84** is a flow chart illustrating an example of inter prediction by normal inter mode in the decoder;

(104) FIG. **85** is a flow chart illustrating an example of inter prediction by normal merge mode in the decoder;

(105) FIG. **86** is a flow chart illustrating an example of inter prediction by FRUC mode in the decoder;

(106) FIG. **87** is a flow chart illustrating an example of inter prediction by affine merge mode in the decoder;

(107) FIG. **88** is a flow chart illustrating an example of inter prediction by affine inter mode in the decoder;

(108) FIG. **89** is a flow chart illustrating an example of inter prediction by triangle mode in the decoder;

(109) FIG. **90** is a flow chart illustrating an example of motion estimation by DMVR in the decoder;

(110) FIG. **91** is a flow chart illustrating one specific example of motion estimation by DMVR in the decoder;

(111) FIG. **92** is a flow chart illustrating one example of generation of a prediction image in the decoder;

(112) FIG. **93** is a flow chart illustrating another example of generation of a prediction image in the decoder;

(113) FIG. **94** is a flow chart illustrating another example of correction of a prediction image by OBMC in the decoder;

(114) FIG. **95** is a flow chart illustrating another example of correction of a prediction image by BIO in the decoder;

(115) FIG. **96** is a flow chart illustrating another example of correction of a prediction image by LIC in the decoder;

(116) FIG. **97** is a schematic drawing illustrating change of resolution from a higher resolution to a lower resolution;

(117) FIG. **98A** is a schematic drawing illustrating an example of resolution change handling;

(118) FIG. **98B** is a schematic drawing illustrating another example of the resolution change handling;

(119) FIG. **98C** is a schematic drawing illustrating another example of the resolution change handling;

(120) FIG. **99** is a schematic drawing illustrating a relationship between parameters;

(121) FIG. **100A** is a schematic drawing illustrating an exemplary video stream complying with the constraints of $k > 1$;

(122) FIG. **100B** is a schematic drawing illustrating an exemplary video stream not complying with

the constraints of $k > 1$;

(123) FIG. **101** is a flow chart illustrating exemplary methods for encoding and decoding pictures;

(124) FIG. **102** is a flow chart illustrating an exemplary encoding method for encoding a sequence of pictures into a bitstream;

(125) FIG. **103** is a schematic drawing illustrating a server and a receiver compliant with DASH protocol;

(126) FIG. **104A** is a flow chart illustrating a part of an exemplary encoding method employing rate control;

(127) FIG. **104B** is a flow chart illustrating the remaining part of the exemplary encoding method employing rate control;

(128) FIG. **105** is a flow chart illustrating an exemplary decoding method for a broadcast receiver;

(129) FIG. **106A** is a flow chart illustrating a part of an exemplary encoding method employing a set of predefined resolutions;

(130) FIG. **106B** is a flow chart illustrating the remaining part of the exemplary encoding method employing a set of predefined resolutions;

(131) FIG. **107A** is a flow chart illustrating a part of an exemplary encoding method employing rate control and a set of predefined resolutions;

(132) FIG. **107B** is a flow chart illustrating the remaining part of the exemplary encoding method employing rate control and a set of predefined resolutions;

(133) FIG. **108** is a flow chart indicating an operation performed by an encoder according to the embodiment;

(134) FIG. **109** is a flow chart indicating an operation performed by a decoder according to the embodiment;

(135) FIG. **110** is a diagram illustrating an overall configuration of a content providing system for implementing a content distribution service;

(136) FIG. **111** is a diagram illustrating an example of a display screen of a web page;

(137) FIG. **112** is a diagram illustrating an example of a display screen of a web page;

(138) FIG. **113** is a diagram illustrating one example of a smartphone; and

(139) FIG. **114** is a block diagram illustrating an example of a configuration of a smartphone.

DETAILED DESCRIPTION OF THE EMBODIMENT(S)

(140) [Introduction]

(141) In sequentially encoding multiple pictures included in a video, the resolution of a picture may be changed from the resolution of a previous picture preceding the picture in encoding order or display order. When the resolution of a reference picture to be used in encoding of an inter-predicted picture is different from the resolution of the inter-predicted picture, a reference image in the reference picture is resampled, and an image in the inter-predicted picture can be encoded using the resampled reference image.

(142) In this manner, even when the resolution of a reference picture to be used in encoding of an inter-predicted picture is different from the resolution of the inter-predicted picture, it is possible to use the inter prediction.

(143) However, frequent resolution change increases the consumption of a computational resource and a memory resource, and also increases the load of the encoder and the decoder. Moreover, it is difficult to constantly secure the computational resource and the memory resource supporting the resolution change. Furthermore, when the resolution change occur at random, it is also difficult to conduct a test on the encoder and the decoder.

(144) In view of this, for example, an encoder according to one aspect of the present disclosure includes: circuitry; and memory coupled to the circuitry. In operation, the circuitry: controls whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access

pictures; and when a resolution of a reference picture to be used in encoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and encodes an image in the inter-predicted picture using the reference image resampled.

(145) With this, in the encoder, it may be possible to prevent the frequent resolution change. Accordingly, in the encoder, it may be possible to reduce the consumption of the computational resource and memory resource, and also reduce the load of the encoder. Moreover, in the encoder, it may be possible to prevent the computational resource and the memory resource supporting the resolution change from being constantly secured. Moreover, in the encoder, it may be possible to reduce the difficulty of the test.

(146) Moreover, for example, the constraint allows the changing of the resolution of the picture only when the picture is a random access picture among the one or more random access pictures at every k-th random access picture, where k is an integer greater than 1.

(147) With this, in the encoder, it may be possible to further prevent the frequent resolution change. Accordingly, in the encoder, it may be possible to further reduce the consumption of the computational resource and memory resource, and also further reduce the load of the encoder.

(148) Moreover, for example, the circuitry: sequentially encodes a plurality of pictures into a bitstream; monitors an amount of bits in the bitstream; and controls the changing of the resolution of the picture according to the amount of bits monitored.

(149) With this, in the encoder, it may be possible to change the resolution according to the amount of bits in the bitstream and adjust the amount of bits in the bitstream according to the resolution change.

(150) Moreover, for example, the circuitry: encodes at least one first trailing picture belonging to a first group of pictures; subsequently encodes a random access picture belonging to a second group of pictures following the first group of pictures in the display order, the random access picture being included in the one or more random access pictures; subsequently encodes at least one leading picture belonging to the first group of pictures; and subsequently encodes at least one second trailing picture belonging to the second group of pictures, and the constraint: does not allow a resolution of the at least one leading picture to be different from a resolution of the at least one first trailing picture; allows a resolution of the random access picture to be different from the resolution of the at least one leading picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the random access picture.

(151) With this, in the encoder, it may be possible to change the resolution only at a random access picture in the display order. Accordingly, in the encoder, it may be possible to smoothly perform the resolution change along the display order.

(152) Moreover, for example, the circuitry: encodes at least one first trailing picture belonging to a first group of pictures; subsequently encodes a random access picture belonging to a second group of pictures following the first group of pictures in the display order, the random access picture being included in the one or more random access pictures; subsequently encodes at least one leading picture belonging to the first group of pictures; and subsequently encodes at least one second trailing picture belonging to the second group of pictures, and the constraint: allows a resolution of the random access picture to be different from a resolution of the at least one first trailing picture; does not allow a resolution of the at least one leading picture to be different from the resolution of the random access picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the at least one leading picture.

(153) With this, in the encoder, it may be possible to change the resolution only at a random access picture in the encoding order. Accordingly, in the encoder, it may be possible to smoothly perform the resolution change along the encoding order.

(154) Moreover, for example, for each of segments determined using Dynamic Adaptive Streaming

over HTTP (DASH) protocol, the constraint does not allow the changing of the resolution of the picture in the segment.

(155) With this, in the encoder, it may be possible to constrain the resolution change in units of segments. Accordingly, with this, in the encoder, it may be possible to prevent the frequent resolution change.

(156) Moreover, for example, for each of representations determined using DASH protocol, the constraint does not allow the changing of the resolution of the picture to be performed at intervals shorter than a threshold.

(157) With this, in the encoder, for the same representation, it may be possible to prevent the resolution change from being performed in short intervals. Accordingly, with this, in the encoder, it may be possible to further prevent the frequent resolution change.

(158) Moreover, for example, the constraint limits the resolution of the picture to a resolution candidate among a plurality of resolution candidates.

(159) With this, in the encoder, it may be possible to limit the resolution to any of resolution candidates. Accordingly, in the encoder, it may be possible to reduce the complexity of the processing. Moreover, in the encoder, it may be possible to reduce the difficulty of the test.

(160) Moreover, for example, the plurality of resolution candidates include resolution candidates determined by scaling the resolution of the previous picture by a plurality of fixed ratios.

(161) With this, in the encoder, it may be possible to smoothly change the resolution from the resolution before the change using the fixed ratio. Moreover, in the encoder, it may be possible to smoothly resample the reference image using the fixed ratio. Accordingly, in the encoder, it may be possible to reduce the complexity of the processing.

(162) Moreover, for example, the plurality of fixed ratios include at least one of 2 times, $3/2$ times, $4/3$ times, $3/4$ times, $2/3$ times, or $1/2$ times.

(163) With this, in the encoder, it may be possible to change the resolution using a ratio easy to resample the reference image. Accordingly, in the encoder, it may be possible to reduce the complexity of the processing.

(164) Moreover, for example, the plurality of resolution candidates include resolution candidates determined as 7680×4320 pixels, 5120×2880 pixels, 3840×2160 pixels, 2560×1440 pixels, 1920×1080 pixels, 1280×720 pixels, and 960×540 pixels.

(165) With this, in the encoder, it may be possible to use a general resolution, and maintain the versatility of the encoder.

(166) Moreover, for example, the plurality of resolution candidates include resolution candidates determined by applying addition or subtraction of fixed difference values to the resolution of the previous picture.

(167) With this, in the encoder, it may be possible to appropriately limit the resolution after the change according to the resolution before the change and the fixed difference value, and reduce the complexity of the processing.

(168) Moreover, for example, the constraint limits a maximum resolution specified as an upper limit in a sequence parameter set to be equal to a resolution specified as a resolution of at least one picture in a picture parameter set applying to the at least one picture among pictures to which the sequence parameter set is applied, and the circuitry: encodes, into the sequence parameter set, the maximum resolution determined according to the constraint; and encodes, into the picture parameter set, the resolution specified as the resolution of the at least one picture.

(169) With this, in the encoder, it may be possible to prevent a too high maximum resolution from being specified in the sequence parameter set. Accordingly, in the encoder, it may be possible to prevent too much computational resource and memory resource from being secured.

(170) Moreover, for example, a decoder according to one aspect of the present disclosure includes: circuitry; and memory coupled to the circuitry. In operation, the circuitry: controls whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one

of display order and decoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in decoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and decodes an image in the inter-predicted picture using the reference image resampled.

(171) With this, in the decoder, it may be possible to prevent the frequent resolution change.

Accordingly, in the decoder, it may be possible to reduce the consumption of the computational resource and memory resource, and also reduce the load of the decoder. Moreover, in the decoder, it may be possible to prevent the computational resource and the memory resource supporting the resolution change from being constantly secured. Moreover, in the decoder, it may be possible to reduce the difficulty of the test.

(172) Moreover, for example, the circuitry allocates, according to the constraint, at least one of a computational resource or a memory resource to post-processing for displaying a decoded image.

(173) With this, in the decoder, it may be possible to prepare, according to the constraint, computational resource and memory resource necessary and sufficient for the post-processing for displaying the decoded image.

(174) Moreover, for example, the constraint allows the changing of the resolution of the picture only when the picture is a random access picture among the one or more random access pictures at every k -th random access picture, where k is an integer greater than 1.

(175) With this, in the decoder, it may be possible to further prevent the frequent resolution change. Accordingly, in the decoder, it may be possible to further reduce the consumption of the computational resource and memory resource, and also further reduce the load of the decoder.

(176) Moreover, for example, the circuitry: sequentially decodes a plurality of pictures from a bitstream; monitors an amount of bits in the bitstream; and controls the changing of the resolution of the picture according to the amount of bits monitored.

(177) With this, in the decoder, it may be possible to change the resolution according to the amount of bits in the bitstream and adjust the amount of bits in the bitstream according to the resolution change.

(178) Moreover, for example, the circuitry: decodes at least one first trailing picture belonging to a first group of pictures; subsequently decodes a random access picture belonging to a second group of pictures following the first group of pictures in the display order, the random access picture being included in the one or more random access pictures; subsequently decodes at least one leading picture belonging to the first group of pictures; and subsequently decodes at least one second trailing picture belonging to the second group of pictures, and the constraint: does not allow a resolution of the at least one leading picture to be different from a resolution of the at least one first trailing picture; allows a resolution of the random access picture to be different from the resolution of the at least one leading picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the random access picture.

(179) With this, in the decoder, it may be possible to change the resolution only at a random access picture in the display order. Accordingly, in the decoder, it may be possible to smoothly perform the resolution change along the display order.

(180) Moreover, for example, the circuitry: decodes at least one first trailing picture belonging to a first group of pictures; subsequently decodes a random access picture belonging to a second group of pictures following the first group of pictures in the display order, the random access picture being included in the one or more random access pictures; subsequently decodes at least one leading picture belonging to the first group of pictures; and subsequently decodes at least one second trailing picture belonging to the second group of pictures, and the constraint: allows a resolution of the random access picture to be different from a resolution of the at least one first

trailing picture; does not allow a resolution of the at least one leading picture to be different from the resolution of the random access picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the at least one leading picture.

(181) With this, in the decoder, it may be possible to change the resolution only at a random access picture in the decoding order. Accordingly, in the decoder, it may be possible to smoothly perform the resolution change along the decoding order.

(182) Moreover, for example, for each of segments determined using Dynamic Adaptive Streaming over HTTP (DASH) protocol, the constraint does not allow the changing of the resolution of the picture in the segment.

(183) With this, in the decoder, it may be possible to constrain the resolution change in units of segments. Accordingly, with this, in the decoder, it may be possible to prevent the frequent resolution change.

(184) Moreover, for example, for each of representations determined using DASH protocol, the constraint does not allow the changing of the resolution of the picture to be performed at intervals shorter than a threshold.

(185) With this, in the decoder, for the same representation, it may be possible to prevent the resolution change from being performed in short intervals. Accordingly, with this, in the decoder, it may be possible to further prevent the frequent resolution change.

(186) Moreover, for example, the constraint limits the resolution of the picture to a resolution candidate among a plurality of resolution candidates.

(187) With this, in the decoder, it may be possible to limit the resolution to any of resolution candidates. Accordingly, in the decoder, it may be possible to reduce the complexity of the processing. Moreover, in the decoder, it may be possible to reduce the difficulty of the test.

(188) Moreover, for example, the plurality of resolution candidates include resolution candidates determined by scaling the resolution of the previous picture by a plurality of fixed ratios.

(189) With this, in the decoder, it may be possible to smoothly change the resolution from the resolution before the change using the fixed ratio. Moreover, in the decoder, it may be possible to smoothly resample the reference image using the fixed ratio. Accordingly, in the decoder, it may be possible to reduce the complexity of the processing.

(190) Moreover, for example, the plurality of fixed ratios include at least one of 2 times, $3/2$ times, $4/3$ times, $3/4$ times, $2/3$ times, or $1/2$ times.

(191) With this, in the decoder, it may be possible to change the resolution using a ratio easy to resample the reference image. Accordingly, in the decoder, it may be possible to reduce the complexity of the processing.

(192) Moreover, for example, the plurality of resolution candidates include resolution candidates determined as 7680×4320 pixels, 5120×2880 pixels, 3840×2160 pixels, 2560×1440 pixels, 1920×1080 pixels, 1280×720 pixels, and 960×540 pixels.

(193) With this, in the decoder, it may be possible to use a general resolution, and maintain the versatility of the decoder.

(194) Moreover, for example, the plurality of resolution candidates include resolution candidates determined by applying addition or subtraction of fixed difference values to the resolution of the previous picture.

(195) With this, in the decoder, it may be possible to appropriately limit the resolution after the change according to the resolution before the change and the fixed difference value, and reduce the complexity of the processing.

(196) Moreover, for example, the constraint limits a maximum resolution specified as an upper limit in a sequence parameter set to be equal to a resolution specified as a resolution of at least one picture in a picture parameter set applying to the at least one picture among pictures to which the sequence parameter set is applied, and the circuitry: decodes, from the sequence parameter set, the maximum resolution determined according to the constraint; and decodes, from the picture

parameter set, the resolution specified as the resolution of the at least one picture.

(197) With this, in the decoder, it may be possible to prevent a too high maximum resolution from being specified in the sequence parameter set. Accordingly, in the decoder, it may be possible to prevent too much computational resource and memory resource from being secured.

(198) Moreover, for example, an encoding method according to one aspect of the present disclosure includes: controlling whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in encoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resampling a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and encoding an image in the inter-predicted picture using the reference image resampled.

(199) With this, it may be possible to prevent the frequent resolution change. Accordingly, it may be possible to reduce the consumption of the computational resource and memory resource, and also reduce the load. Moreover, it may be possible to prevent the computational resource and the memory resource supporting the resolution change from being constantly secured. Moreover, it may be possible to reduce the difficulty of the test.

(200) Moreover, for example, a decoding method according to one aspect of the present disclosure includes: controlling whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and decoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in decoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resampling a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and decoding an image in the inter-predicted picture using the reference image resampled.

(201) With this, it may be possible to prevent the frequent resolution change. Accordingly, it may be possible to reduce the consumption of the computational resource and memory resource, and also reduce the load. Moreover, it may be possible to prevent the computational resource and the memory resource supporting the resolution change from being constantly secured. Moreover, it may be possible to reduce the difficulty of the test.

(202) Moreover, for example, a non-transitory computer readable medium according to one aspect of the present disclosure is a non-transitory computer readable medium storing a bitstream. The bitstream includes: a plurality of pictures; and a plurality of parameters to perform a decoding process for decoding the plurality of pictures from the bitstream. In the decoding process, in operation according to the plurality of parameters, whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and decoding order is controlled according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in decoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, a reference image in the reference picture is resampled according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and an image in the inter-predicted picture is decoded using the reference image resampled.

(203) With this, it may be possible to prevent the frequent resolution change. Accordingly, it may be possible to reduce the consumption of the computational resource and memory resource, and also reduce the load. Moreover, it may be possible to prevent the computational resource and the memory resource supporting the resolution change from being constantly secured. Moreover, it may be possible to reduce the difficulty of the test.

(204) Moreover, for example, the plurality of parameters include: a first parameter indicating

whether changing of a resolution of one or more pictures among the plurality of pictures is allowed; and a second parameter indicating the resolution of the picture, and when (i) the changing of the resolution of the one or more pictures is allowed according to the first parameter, (ii) the picture is included in the one or more pictures, and (iii) the changing of the resolution of the picture is allowed according to the constraint, the resolution of the picture is changed according to the second parameter.

(205) With this, it may be possible to change the resolution according to the parameters and the constraint related to the resolution change while preventing the frequent resolution change.

(206) Moreover, for example, an encoder according to one aspect of the present disclosure may include an inputter, a splitter, an intra predictor, an inter predictor, a loop filter, a transformer, a quantizer, an entropy encoder, and an outputter.

(207) The inputter receives a current picture. The splitter splits the current picture into multiple blocks.

(208) The intra predictor generates prediction signals of a current block included in the current picture, using a reference image included in the current picture. The inter predictor generates prediction signals of a current block included in the current picture, using a reference image included in a reference picture different from the current picture. The loop filter unit applies a filter to a reconstructed block in a current block included in the current picture.

(209) The transformer generates transformed coefficients by transforming prediction errors between original signals of the current block included in the current picture and prediction signals generated by either the intra predictor or the inter predictor. The quantizer quantizes the transform coefficients to generate quantized coefficients. The entropy encoder applies variable length encoding on the quantized coefficients to generate an encoded bitstream. The quantized coefficients to which the variable length encoding has been applied and the encoded bitstream including control information are then output from the outputter.

(210) Moreover, for example, in operation, the inter predictor: controls whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in encoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and encodes an image in the inter-predicted picture using the reference image resampled.

(211) Moreover, for example, a decoder according to one aspect of the present disclosure includes an inputter, an entropy decoder, an inverse quantizer, an inverse transformer, an intra predictor, an inter predictor, a loop filter unit, and outputter.

(212) The inputter receives an encoded bitstream. The entropy decoder applies variable length decoding on the encoded bitstream to derive quantized coefficients. The inverse quantizer inverse quantizes the quantized coefficients to derive transform coefficients. The inverse transformer inverse transforms the transformed coefficients to derive prediction errors.

(213) The intra predictor generates prediction signals of a current block included in the current picture, using a reference image included in the current picture. The inter predictor generates prediction signals of a current block included in the current picture, using a reference image included in a reference picture different from the current picture.

(214) The loop filter unit applies a filter to a reconstructed block in a current block included in the current picture. The current picture is then output from the outputter.

(215) Moreover, for example, in operation, the inter predictor: controls whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and decoding order, according to a constraint that allows the changing only when the

picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in decoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and decodes an image in the inter-predicted picture using the reference image resampled.

(216) Furthermore, these general or specific aspects may be implemented using a system, an apparatus, a method, an integrated circuit, a computer program, or a non-transitory computer readable medium such as a CD-ROM, or any combination of systems, apparatuses, methods, integrated circuits, computer programs, or media.

(217) [Definitions of Terms]

(218) The respective terms may be defined as indicated below as examples.

(219) (1) Image

(220) An image is a data unit configured with a set of pixels, is a picture or includes blocks smaller than a picture. Images include a still image in addition to a video.

(221) (2) Picture

(222) A picture is an image processing unit configured with a set of pixels, and is also referred to as a frame or a field.

(223) (3) Block

(224) A block is a processing unit which is a set of a particular number of pixels. The block is also referred to as indicated in the following examples. The shapes of blocks are not limited. Examples include a rectangle shape of $M \times N$ pixels and a square shape of $M \times M$ pixels for the first place, and also include a triangular shape, a circular shape, and other shapes.

(225) (Examples of Blocks)

(226) slice/tile/brick CTU/super block/basic splitting unit VPDU/processing splitting unit for hardware CU/processing block unit/prediction block unit (PU)/orthogonal transform block unit (TU)/unit sub-block

(4) Pixel/Sample

(227) A pixel or sample is a smallest point of an image. Pixels or samples include not only a pixel at an integer position but also a pixel at a sub-pixel position generated based on a pixel at an integer position.

(228) (5) Pixel Value/Sample Value

(229) A pixel value or sample value is an eigen value of a pixel. Pixel or sample values naturally include a luma value, a chroma value, an RGB gradation level and also covers a depth value, or a binary value of 0 or 1.

(230) (6) Flag

(231) A flag indicates one or more bits, and may be, for example, a parameter or index represented by two or more bits. Alternatively, the flag may indicate not only a binary value represented by a binary number but also a multiple value represented by a number other than the binary number.

(232) (7) Signal

(233) A signal is the one symbolized or encoded to convey information. Signals include a discrete digital signal and an analog signal which takes a continuous value.

(234) (8) Stream/Bitstream

(235) A stream or bitstream is a digital data string or a digital data flow. A stream or bitstream may be one stream or may be configured with a plurality of streams having a plurality of hierarchical layers. A stream or bitstream may be transmitted in serial communication using a single transmission path, or may be transmitted in packet communication using a plurality of transmission paths.

(236) (9) Difference

(237) In the case of scalar quantity, it is only necessary that a simple difference ($x-y$) and a

difference calculation be included. Differences include an absolute value of a difference ($|x-y|$), a squared difference ($(x-y)^2$), a square root of a difference ($\sqrt{x-y}$), a weighted difference ($ax-by$: a and b are constants), an offset difference ($x-y+a$: a is an offset).

(238) (10) Sum

(239) In the case of scalar quantity, it is only necessary that a simple sum ($x+y$) and a sum calculation be included. Sums include an absolute value of a sum ($|x+y|$), a squared sum ($(x+y)^2$), a square root of a sum ($\sqrt{x+y}$), a weighted difference ($ax+by$: a and b are constants), an offset sum ($x+y+a$: a is an offset).

(240) (11) Based on

(241) A phrase “based on something” means that a thing other than the something may be considered. In addition, “based on” may be used in a case in which a direct result is obtained or a case in which a result is obtained through an intermediate result.

(242) (12) Used, Using

(243) A phrase “something used” or “using something” means that a thing other than the something may be considered. In addition, “used” or “using” may be used in a case in which a direct result is obtained or a case in which a result is obtained through an intermediate result.

(244) (13) Prohibit, Forbid

(245) The term “prohibit” or “forbid” can be rephrased as “does not permit” or “does not allow”. In addition, “being not prohibited/forbidden” or “being permitted/allowed” does not always mean “obligation”.

(246) (14) Limit, Restriction/Restrict/Restricted

(247) The term “limit” or “restriction/restrict/restricted” can be rephrased as “does not permit/allow” or “being not permitted/allowed”. In addition, “being not prohibited/forbidden” or “being permitted/allowed” does not always mean “obligation”. Furthermore, it is only necessary that part of something be prohibited/forbidden quantitatively or qualitatively, and something may be fully prohibited/forbidden.

(248) (15) Chroma

(249) An adjective, represented by the symbols C_b and C_r , specifying that a sample array or single sample is representing one of the two color difference signals related to the primary colors. The term chroma may be used instead of the term chrominance.

(250) (16) Luma

(251) An adjective, represented by the symbol or subscript Y or L , specifying that a sample array or single sample is representing the monochrome signal related to the primary colors. The term luma may be used instead of the term luminance.

(252) [Notes Related to the Descriptions]

(253) In the drawings, same reference numbers indicate same or similar components. The sizes and relative locations of components are not necessarily drawn by the same scale.

(254) Hereinafter, embodiments will be described with reference to the drawings. Note that the embodiments described below each show a general or specific example. The numerical values, shapes, materials, components, the arrangement and connection of the components, steps, the relation and order of the steps, etc., indicated in the following embodiments are mere examples, and are not intended to limit the scope of the claims.

(255) Embodiments of an encoder and a decoder will be described below. The embodiments are examples of an encoder and a decoder to which the processes and/or configurations presented in the description of aspects of the present disclosure are applicable. The processes and/or configurations can also be implemented in an encoder and a decoder different from those according to the embodiments. For example, regarding the processes and/or configurations as applied to the embodiments, any of the following may be implemented: (1) Any of the components of the encoder or the decoder according to the embodiments presented in the description of aspects of the present

disclosure may be substituted or combined with another component presented anywhere in the description of aspects of the present disclosure. (2) In the encoder or the decoder according to the embodiments, discretionary changes may be made to functions or processes performed by one or more components of the encoder or the decoder, such as addition, substitution, removal, etc., of the functions or processes. For example, any function or process may be substituted or combined with another function or process presented anywhere in the description of aspects of the present disclosure. (3) In methods implemented by the encoder or the decoder according to the embodiments, discretionary changes may be made such as addition, substitution, and removal of one or more of the processes included in the method. For example, any process in the method may be substituted or combined with another process presented anywhere in the description of aspects of the present disclosure. (4) One or more components included in the encoder or the decoder according to embodiments may be combined with a component presented anywhere in the description of aspects of the present disclosure, may be combined with a component including one or more functions presented anywhere in the description of aspects of the present disclosure, and may be combined with a component that implements one or more processes implemented by a component presented in the description of aspects of the present disclosure. (5) A component including one or more functions of the encoder or the decoder according to the embodiments, or a component that implements one or more processes of the encoder or the decoder according to the embodiments, may be combined or substituted with a component presented anywhere in the description of aspects of the present disclosure, with a component including one or more functions presented anywhere in the description of aspects of the present disclosure, or with a component that implements one or more processes presented anywhere in the description of aspects of the present disclosure. (6) In methods implemented by the encoder or the decoder according to the embodiments, any of the processes included in the method may be substituted or combined with a process presented anywhere in the description of aspects of the present disclosure or with any corresponding or equivalent process. (7) One or more processes included in methods implemented by the encoder or the decoder according to the embodiments may be combined with a process presented anywhere in the description of aspects of the present disclosure. (8) The implementation of the processes and/or configurations presented in the description of aspects of the present disclosure is not limited to the encoder or the decoder according to the embodiments. For example, the processes and/or configurations may be implemented in a device used for a purpose different from the moving picture encoder or the moving picture decoder disclosed in the embodiments.

[System Configuration]

(256) FIG. 1 is a schematic diagram illustrating one example of a configuration of a transmission system according to an embodiment.

(257) Transmission system Trs is a system which transmits a stream generated by encoding an image and decodes the transmitted stream. Transmission system Trs like this includes, for example, encoder **100**, network Nw, and decoder **200** as illustrated in FIG. 1.

(258) An image is input to encoder **100**. Encoder **100** generates a stream by encoding the input image, and outputs the stream to network Nw. The stream includes, for example, the encoded image and control information for decoding the encoded image. The image is compressed by the encoding.

(259) It is to be noted that a previous image before being encoded and being input to encoder **100** is also referred to as the original image, the original signal, or the original sample. The image may be a video or a still image. The image is a generic concept of a sequence, a picture, and a block, and thus is not limited to a spatial region having a particular size and to a temporal region having a particular size unless otherwise specified. The image is an array of pixels or pixel values, and the signal representing the image or pixel values are also referred to as samples. The stream may be referred to as a bitstream, an encoded bitstream, a compressed bitstream, or an encoded signal. Furthermore, the encoder may be referred to as an image encoder or a video encoder. The encoding

method performed by encoder **100** may be referred to as an encoding method, an image encoding method, or a video encoding method.

(260) Network Nw transmits the stream generated by encoder **100** to decoder **200**. Network Nw may be the Internet, the Wide Area Network (WAN), the Local Area Network (LAN), or any combination of these networks. Network Nw is not always limited to a bi-directional communication network, and may be a unidirectional communication network which transmits broadcast waves of digital terrestrial broadcasting, satellite broadcasting, or the like. Alternatively, network Nw may be replaced by a medium such as a Digital Versatile Disc (DVD) and a Blu-Ray Disc (BD)®, etc. on which a stream is recorded.

(261) Decoder **200** generates, for example, a decoded image which is an uncompressed image by decoding a stream transmitted by network Nw. For example, the decoder decodes a stream according to a decoding method corresponding to an encoding method by encoder **100**.

(262) It is to be noted that the decoder may also be referred to as an image decoder or a video decoder, and that the decoding method performed by decoder **200** may also be referred to as a decoding method, an image decoding method, or a video decoding method.

(263) [Data Structure]

(264) FIG. 2 is a diagram illustrating one example of a hierarchical structure of data in a stream. A stream includes, for example, a video sequence. As illustrated in (a) of FIG. 2, the video sequence includes a video parameter set (VPS), a sequence parameter set (SPS), a picture parameter set (PPS), supplemental enhancement information (SET), and a plurality of pictures.

(265) In a video having a plurality of layers, a VPS includes: a coding parameter which is common between some of the plurality of layers; and a coding parameter related to some of the plurality of layers included in the video or an individual layer.

(266) An SPS includes a parameter which is used for a sequence, that is, a coding parameter which decoder **200** refers to in order to decode the sequence. For example, the coding parameter may indicate the width or height of a picture. It is to be noted that a plurality of SPSs may be present.

(267) A PPS includes a parameter which is used for a picture, that is, a coding parameter which decoder **200** refers to in order to decode each of the pictures in the sequence. For example, the coding parameter may include a reference value for the quantization width which is used to decode a picture and a flag indicating application of weighted prediction. It is to be noted that a plurality of PPSs may be present. Each of the SPS and the PPS may be simply referred to as a parameter set.

(268) As illustrated in (b) of FIG. 2, a picture may include a picture header and at least one slice. A picture header includes a coding parameter which decoder **200** refers to in order to decode the at least one slice.

(269) As illustrated in (c) of FIG. 2, a slice includes a slice header and at least one brick. A slice header includes a coding parameter which decoder **200** refers to in order to decode the at least one brick.

(270) As illustrated in (d) of FIG. 2, a brick includes at least one coding tree unit (CTU).

(271) It is to be noted that a picture may not include any slice and may include a tile group instead of a slice. In this case, the tile group includes at least one tile. In addition, a brick may include a slice.

(272) A CTU is also referred to as a super block or a basis splitting unit. As illustrated in (e) of FIG. 2, a CTU like this includes a CTU header and at least one coding unit (CU). A CTU header includes a coding parameter which decoder **200** refers to in order to decode the at least one CU.

(273) A CU may be split into a plurality of smaller CUs. As illustrated in (f) of FIG. 2, a CU includes a CU header, prediction information, and residual coefficient information. Prediction information is information for predicting the CU, and the residual coefficient information is information indicating a prediction residual to be described later. Although a CU is basically the same as a prediction unit (PU) and a transform unit (TU), it is to be noted that, for example, an SBT to be described later may include a plurality of TUs smaller than the CU. In addition, the CU

may be processed for each virtual pipeline decoding unit (VPDU) included in the CU. The VPDU is, for example, a fixed unit which can be processed at one stage when pipeline processing is performed in hardware.

(274) It is to be noted that a stream may not include part of the hierarchical layers illustrated in FIG. 2. The order of the hierarchical layers may be exchanged, or any of the hierarchical layers may be replaced by another hierarchical layer. Here, a picture which is a target for a process which is about to be performed by a device such as encoder **100** or decoder **200** is referred to as a current picture. A current picture means a current picture to be encoded when the process is an encoding process, and a current picture means a current picture to be decoded when the process is a decoding process. Likewise, for example, a CU or a block of CUs which is a target for a process which is about to be performed by a device such as encoder **100** or decoder **200** is referred to as a current block. A current block means a current block to be encoded when the process is an encoding process, and a current block means a current block to be decoded when the process is a decoding process.

(275) [Picture Structure: Slice/Tile]

(276) A picture may be configured with one or more slice units or tile units in order to decode the picture in parallel.

(277) Slices are basic encoding units included in a picture. A picture may include, for example, one or more slices. In addition, a slice includes one or more successive coding tree units (CTUs).

(278) FIG. 3 is a diagram illustrating one example of a slice configuration. For example, a picture includes 11×8 CTUs, and is split into four slices (slices 1 to 4). Slice 1 includes sixteen CTUs, slice 2 includes twenty-one CTUs, slice 3 includes twenty-nine CTUs, and slice 4 includes twenty-two CTUs. Here, each CTU in the picture belongs to one of the slices. The shape of each slice is a shape obtained by splitting the picture horizontally. A boundary of each slice does not need to coincide with an image end, and may coincide with any of the boundaries between CTUs in the image. The processing order of the CTUs in a slice (an encoding order or a decoding order) is, for example, a raster-scan order. A slice includes a slice header and encoded data. Features of the slice may be written in the slice header. The features include a CTU address of a top CTU in the slice, a slice type, etc.

(279) A tile is a unit of a rectangular region included in a picture. Each of tiles may be assigned with a number referred to as TileId in raster-scan order.

(280) FIG. 4 is a diagram illustrating one example of a tile configuration. For example, a picture includes 11×8 CTUs, and is split into four tiles of rectangular regions (tiles 1 to 4). When tiles are used, the processing order of CTUs is changed from the processing order in the case where no tile is used. When no tile is used, a plurality of CTUs in a picture are processed in raster-scan order. When a plurality of tiles are used, at least one CTU in each of the plurality of tiles is processed in raster-scan order. For example, as illustrated in FIG. 4, the processing order of the CTUs included in tile 1 is the order which starts from the left-end of the first column of tile 1 toward the right-end of the first column of tile 1 and then starts from the left-end of the second column of tile 1 toward the right-end of the second column of tile 1.

(281) It is to be noted that one tile may include one or more slices, and one slice may include one or more tiles.

(282) It is to be noted that a picture may be configured with one or more tile sets. A tile set may include one or more tile groups, or one or more tiles. A picture may be configured with only one of a tile set, a tile group, and a tile. For example, an order for scanning a plurality of tiles for each tile set in raster scan order is assumed to be a basic encoding order of tiles. A set of one or more tiles which are continuous in the basic encoding order in each tile set is assumed to be a tile group. Such a picture may be configured by splitter **102** (see FIG. 7) to be described later.

(283) [Scalable Encoding]

(284) FIGS. 5 and 6 are diagrams illustrating examples of scalable stream structures.

(285) As illustrated in FIG. 5, encoder **100** may generate a temporally/spatially scalable stream by dividing each of a plurality of pictures into any of a plurality of layers and encoding the picture in the layer. For example, encoder **100** encodes the picture for each layer, thereby achieving scalability where an enhancement layer is present above a base layer. Such encoding of each picture is also referred to as scalable encoding. In this way, decoder **200** is capable of switching image quality of an image which is displayed by decoding the stream. In other words, decoder **200** determines up to which layer to decode based on internal factors such as the processing ability of decoder **200** and external factors such as a state of a communication bandwidth. As a result, decoder **200** is capable of decoding a content while freely switching between low resolution and high resolution. For example, the user of the stream watches a video of the stream halfway using a smartphone on the way to home, and continues watching the video at home on a device such as a TV connected to the Internet. It is to be noted that each of the smartphone and the device described above includes decoder **200** having the same or different performances. In this case, when the device decodes layers up to the higher layer in the stream, the user can watch the video at high quality at home. In this way, encoder **100** does not need to generate a plurality of streams having different image qualities of the same content, and thus the processing load can be reduced.

(286) Furthermore, the enhancement layer may include meta information based on statistical information on the image. Decoder **200** may generate a video whose image quality has been enhanced by performing super-resolution imaging on a picture in the base layer based on the metadata. Super-resolution imaging may be any of improvement in the Signal-to-Noise (SN) ratio in the same resolution and increase in resolution. Metadata may include information for identifying a linear or a non-linear filter coefficient, as used in a super-resolution process, or information identifying a parameter value in a filter process, machine learning, or a least squares method used in super-resolution processing.

(287) Alternatively, a configuration may be provided in which a picture is divided into, for example, tiles in accordance with, for example, the meaning of an object in the picture. In this case, decoder **200** may decode only a partial region in a picture by selecting a tile to be decoded. In addition, an attribute of the object (person, car, ball, etc.) and a position of the object in the picture (coordinates in identical images) may be stored as metadata. In this case, decoder **200** is capable of identifying the position of a desired object based on the metadata, and determining the tile including the object. For example, as illustrated in FIG. 6, the metadata may be stored using a data storage structure different from image data, such as SEI in HEVC. This metadata indicates, for example, the position, size, or color of a main object.

(288) Metadata may be stored in units of a plurality of pictures, such as a stream, a sequence, or a random access unit. In this way, decoder **200** is capable of obtaining, for example, the time at which a specific person appears in the video, and by fitting the time information with picture unit information, is capable of identifying a picture in which the object is present and determining the position of the object in the picture.

(289) [Encoder]

(290) Next, encoder **100** according to this embodiment is described. FIG. 7 is a block diagram illustrating one example of a configuration of encoder **100** according to this embodiment. Encoder **100** encodes an image in units of a block.

(291) As illustrated in FIG. 7, encoder **100** is an apparatus which encodes an image in units of a block, and includes splitter **102**, subtractor **104**, transformer **106**, quantizer **108**, entropy encoder **110**, inverse quantizer **112**, inverse transformer **114**, adder **116**, block memory **118**, loop filter **120**, frame memory **122**, intra predictor **124**, inter predictor **126**, prediction controller **128**, and prediction parameter generator **130**. It is to be noted that intra predictor **124** and inter predictor **126** are configured as part of a prediction executor.

(292) [Mounting Example of Encoder]

(293) FIG. 8 is a block diagram illustrating a mounting example of encoder **100**. Encoder **100**

includes processor **a1** and memory **a2**. For example, the plurality of constituent elements of encoder **100** illustrated in FIG. 7 are mounted on processor **a1** and memory **a2** illustrated in FIG. 8. (294) Processor **a1** is circuitry which performs information processing and is accessible to memory **a2**. For example, processor **a1** is dedicated or general electronic circuitry which encodes an image. Processor **a1** may be a processor such as a CPU. In addition, processor **a1** may be an aggregate of a plurality of electronic circuits. In addition, for example, processor **a1** may take the roles of two or more constituent elements other than a constituent element for storing information out of the plurality of constituent elements of encoder **100** illustrated in FIG. 7, etc.

(295) Memory **a2** is dedicated or general memory for storing information that is used by processor **a1** to encode the image. Memory **a2** may be electronic circuitry, and may be connected to processor **a1**. In addition, memory **a2** may be included in processor **a1**. In addition, memory **a2** may be an aggregate of a plurality of electronic circuits. In addition, memory **a2** may be a magnetic disc, an optical disc, or the like, or may be represented as storage, a medium, or the like. In addition, memory **a2** may be non-volatile memory, or volatile memory.

(296) For example, memory **a2** may store an image to be encoded or a stream corresponding to an encoded image. In addition, memory **a2** may store a program for causing processor **a1** to encode an image.

(297) In addition, for example, memory **a2** may take the roles of two or more constituent elements for storing information out of the plurality of constituent elements of encoder **100** illustrated in FIG. 7. More specifically, memory **a2** may take the roles of block memory **118** and frame memory **122** illustrated in FIG. 7. More specifically, memory **a2** may store a reconstructed image (specifically, a reconstructed block, a reconstructed picture, or the like).

(298) It is to be noted that, in encoder **100**, not all of the plurality of constituent elements indicated in FIG. 7, etc. may be implemented, and not all the processes described above may be performed. Part of the constituent elements indicated in FIG. 7 may be included in another device, or part of the processes described above may be performed by another device.

(299) Hereinafter, an overall flow of processes performed by encoder **100** is described, and then each of constituent elements included in encoder **100** is described.

(300) [Overall Flow of Encoding Process]

(301) FIG. 9 is a flow chart illustrating one example of an overall encoding process performed by encoder **100**.

(302) First, splitter **102** of encoder **100** splits each of pictures included in an original image into a plurality of blocks having a fixed size (128×128 pixels) (Step Sa_1). Splitter **102** then selects a splitting pattern for the fixed-size block (Step Sa_2). In other words, splitter **102** further splits the fixed-size block into a plurality of blocks which form the selected splitting pattern. Encoder **100** performs, for each of the plurality of blocks, Steps Sa_3 to Sa_9 for the block.

(303) Prediction controller **128** and a prediction executor which is configured with intra predictor **124** and inter predictor **126** generate a prediction image of a current block (Step Sa_3). It is to be noted that the prediction image is also referred to as a prediction signal, a prediction block, or prediction samples.

(304) Next, subtractor **104** generates the difference between a current block and a prediction image as a prediction residual (Step Sa_4). It is to be noted that the prediction residual is also referred to as a prediction error.

(305) Next, transformer **106** transforms the prediction image and quantizer **108** quantizes the result, to generate a plurality of quantized coefficients (Step Sa_5).

(306) Next, entropy encoder **110** encodes (specifically, entropy encodes) the plurality of quantized coefficients and a prediction parameter related to generation of a prediction image to generate a stream (Step Sa_6).

(307) Next, inverse quantizer **112** performs inverse quantization of the plurality of quantized coefficients and inverse transformer **114** performs inverse transform of the result, to restore a

prediction residual (Step Sa_7).

(308) Next, adder **116** adds the prediction image to the restored prediction residual to reconstruct the current block (Step Sa_8). In this way, the reconstructed image is generated. It is to be noted that the reconstructed image is also referred to as a reconstructed block, and, in particular, that a reconstructed image generated by encoder **100** is also referred to as a local decoded block or a local decoded image.

(309) When the reconstructed image is generated, loop filter **120** performs filtering of the reconstructed image as necessary (Step Sa_9).

(310) Encoder **100** then determines whether encoding of the entire picture has been finished (Step Sa_10). When determining that the encoding has not yet been finished (No in Step Sa_10), processes from Step Sa_2 are executed repeatedly.

(311) Although encoder **100** selects one splitting pattern for a fixed-size block, and encodes each block according to the splitting pattern in the above-described example, it is to be noted that each block may be encoded according to a corresponding one of a plurality of splitting patterns. In this case, encoder **100** may evaluate a cost for each of the plurality of splitting patterns, and, for example, may select the stream obtained by encoding according to the splitting pattern which yields the smallest cost as a stream which is output finally.

(312) Alternatively, the processes in Steps Sa_1 to Sa_10 may be performed sequentially by encoder **100**, or two or more of the processes may be performed in parallel or may be reordered.

(313) The encoding process by encoder **100** is hybrid encoding using prediction encoding and transform encoding. In addition, prediction encoding is performed by an encoding loop configured with subtractor **104**, transformer **106**, quantizer **108**, inverse quantizer **112**, inverse transformer **114**, adder **116**, loop filter **120**, block memory **118**, frame memory **122**, intra predictor **124**, inter predictor **126**, and prediction controller **128**. In other words, the prediction executor configured with intra predictor **124** and inter predictor **126** is part of the encoding loop.

(314) [Splitter]

(315) Splitter **102** splits each of pictures included in the original image into a plurality of blocks, and outputs each block to subtractor **104**. For example, splitter **102** first splits a picture into blocks of a fixed size (for example, 128×128 pixels). The fixed-size block is also referred to as a coding tree unit (CTU). Splitter **102** then splits each fixed-size block into blocks of variable sizes (for example, 64×64 pixels or smaller), based on recursive quadtree and/or binary tree block splitting. In other words, splitter **102** selects a splitting pattern. The variable-size block is also referred to as a coding unit (CU), a prediction unit (PU), or a transform unit (TU). It is to be noted that, in various kinds of mounting examples, there is no need to differentiate between CU, PU, and TU; all or some of the blocks in a picture may be processed in units of a CU, a PU, or a TU.

(316) FIG. **10** is a diagram illustrating one example of block splitting according to this embodiment. In FIG. **10**, the solid lines represent block boundaries of blocks split by quadtree block splitting, and the dashed lines represent block boundaries of blocks split by binary tree block splitting.

(317) Here, block **10** is a square block having 128×128 pixels. This block **10** is first split into four square 64×64 pixel blocks (quadtree block splitting).

(318) The upper-left 64×64 pixel block is further vertically split into two rectangle 32×64 pixel blocks, and the left 32×64 pixel block is further vertically split into two rectangle 16×64 pixel blocks (binary tree block splitting). As a result, the upper-left square 64×64 pixel block is split into two 16×64 pixel blocks **11** and **12** and one 32×64 pixel block **13**.

(319) The upper-right square 64×64 pixel block is horizontally split into two rectangle 64×32 pixel blocks **14** and **15** (binary tree block splitting).

(320) The lower-left square 64×64 pixel block is first split into four square 32×32 pixel blocks (quadtree block splitting). The upper-left block and the lower-right block among the four square 32×32 pixel blocks are further split. The upper-left square 32×32 pixel block is vertically split into

two rectangle 16×32 pixel blocks, and the right 16×32 pixel block is further horizontally split into two 16×16 pixel blocks (binary tree block splitting). The lower-right 32×32 pixel block is horizontally split into two 32×16 pixel blocks (binary tree block splitting). The upper-right square 32×32 pixel block is horizontally split into two rectangle 32×16 pixel blocks (binary tree block splitting). As a result, the lower-left square 64×64 pixel block is split into rectangle 16×32 pixel block **16**, two square 16×16 pixel blocks **17** and **18**, two square 32×32 pixel blocks **19** and **20**, and two rectangle 32×16 pixel blocks **21** and **22**.

(321) The lower-right 64×64 pixel block **23** is not split.

(322) As described above, in FIG. **10**, block **10** is split into thirteen variable-size blocks **11** through **23** based on recursive quadtree and binary tree block splitting. Such splitting is also referred to as quad-tree plus binary tree splitting (QTBT).

(323) It is to be noted that, in FIG. **10**, one block is split into four or two blocks (quadtree or binary tree block splitting), but splitting is not limited to these examples. For example, one block may be split into three blocks (ternary block splitting). Splitting including such ternary block splitting is also referred to as multi type tree (MBT) splitting.

(324) FIG. **11** is a diagram illustrating one example of a configuration of splitter **102**. As illustrated in FIG. **11**, splitter **102** may include block splitting determiner **102a**. Block splitting determiner **102a** may perform the following processes as examples.

(325) For example, block splitting determiner **102a** collects block information from either block memory **118** or frame memory **122**, and determines the above-described splitting pattern based on the block information. Splitter **102** splits the original image according to the splitting pattern, and outputs at least one block obtained by the splitting to subtractor **104**.

(326) In addition, for example, block splitting determiner **102a** outputs a parameter indicating the above-described splitting pattern to transformer **106**, inverse transformer **114**, intra predictor **124**, inter predictor **126**, and entropy encoder **110**. Transformer **106** may transform a prediction residual based on the parameter. Intra predictor **124** and inter predictor **126** may generate a prediction image based on the parameter. In addition, entropy encoder **110** may entropy encodes the parameter.

(327) The parameter related to the splitting pattern may be written in a stream as indicated below as one example.

(328) FIG. **12** is a diagram illustrating examples of splitting patterns. Examples of splitting patterns include: splitting into four regions (QT) in which a block is split into two regions both horizontally and vertically; splitting into three regions (HT or VT) in which a block is split in the same direction in a ratio of 1:2:1; splitting into two regions (HB or VB) in which a block is split in the same direction in a ratio of 1:1; and no splitting (NS).

(329) It is to be noted that the splitting pattern does not have any block splitting direction in the case of splitting into four regions and no splitting, and that the splitting pattern has splitting direction information in the case of splitting into two regions or three regions.

(330) FIGS. **13A** and **13B** are each a diagram illustrating one example of a syntax tree of a splitting pattern. In the example of FIG. **13A**, first, information indicating whether to perform splitting (S: Split flag) is present, and information indicating whether to perform splitting into four regions (QT: QT flag) is present next. Information indicating which one of splitting into three regions and two regions is to be performed (TT: TT flag or BT: BT flag) is present next, and lastly, information indicating a division direction (Ver: Vertical flag or Hor: Horizontal flag) is present. It is to be noted that each of at least one block obtained by splitting according to such a splitting pattern may be further split repeatedly in a similar process. In other words, as one example, whether splitting is performed, whether splitting into four regions is performed, which one of the horizontal direction and the vertical direction is the direction in which a splitting method is to be performed, which one of splitting into three regions and splitting into two regions is to be performed may be recursively determined, and the determination results may be encoded in a stream according to the encoding order disclosed by the syntax tree illustrated in FIG. **13A**.

(331) In addition, although information items respectively indicating S, QT, TT, and Ver are arranged in the listed order in the syntax tree illustrated in FIG. 13A, information items respectively indicating S, QT, Ver, and BT may be arranged in the listed order. In other words, in the example of FIG. 13B, first, information indicating whether to perform splitting (S: Split flag) is present, and information indicating whether to perform splitting into four regions (QT: QT flag) is present next. Information indicating the splitting direction (Ver: Vertical flag or Hor: Horizontal flag) is present next, and lastly, information indicating which one of splitting into two regions and splitting into three regions is to be performed (BT: BT flag or TT: TT flag) is present.

(332) It is to be noted that the splitting patterns described above are examples, and splitting patterns other than the described splitting patterns may be used, or part of the described splitting patterns may be used.

(333) [Subtractor]

(334) Subtractor **104** subtracts a prediction image (prediction image that is input from prediction controller **128**) from the original image in units of a block input from splitter **102** and split by splitter **102**. In other words, subtractor **104** calculates prediction residuals of a current block. Subtractor **104** then outputs the calculated prediction residuals to transformer **106**.

(335) The original signal is an input signal which has been input to encoder **100** and represents an image of each picture included in a video (for example, a luma signal and two chroma signals).

(336) [Transformer]

(337) Transformer **106** transforms prediction residuals in spatial domain into transform coefficients in frequency domain, and outputs the transform coefficients to quantizer **108**. More specifically, transformer **106** applies, for example, a predefined discrete cosine transform (DCT) or discrete sine transform (DST) to prediction residuals in spatial domain.

(338) It is to be noted that transformer **106** may adaptively select a transform type from among a plurality of transform types, and transform prediction residuals into transform coefficients by using a transform basis function corresponding to the selected transform type. This sort of transform is also referred to as explicit multiple core transform (EMT) or adaptive multiple transform (AMT). In addition, a transform basis function is also simply referred to as a basis.

(339) The transform types include, for example, DCT-II, DCT-V, DCT-VIII, DST-I, and DST-VII. It is to be noted that these transform types may be represented as DCT2, DCT5, DCT8, DST1, and DST7. FIG. 14 is a chart illustrating transform basis functions for each transform type. In FIG. 14, N indicates the number of input pixels. For example, selection of a transform type from among the plurality of transform types may depend on a prediction type (one of intra prediction and inter prediction), and may depend on an intra prediction mode.

(340) Information indicating whether to apply such EMT or AMT (referred to as, for example, an EMT flag or an AMT flag) and information indicating the selected transform type is normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, or CTU level).

(341) In addition, transformer **106** may re-transform the transform coefficients (which are transform results). Such re-transform is also referred to as adaptive secondary transform (AST) or non-separable secondary transform (NSST). For example, transformer **106** performs re-transform in units of a sub-block (for example, 4×4 pixel sub-block) included in a transform coefficient block corresponding to an intra prediction residual. Information indicating whether to apply NSST and information related to a transform matrix for use in NSST are normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, or CTU level).

(342) Transformer **106** may employ a separable transform and a non-separable transform. A separable transform is a method in which a transform is performed a plurality of times by

separately performing a transform for each of directions according to the number of dimensions of inputs. A non-separable transform is a method of performing a collective transform in which two or more dimensions in multidimensional inputs are collectively regarded as a single dimension.

(343) In one example of the non-separable transform, when an input is a 4×4 pixel block, the 4×4 pixel block is regarded as a single array including sixteen elements, and the transform applies a 16×16 transform matrix to the array.

(344) In another example of the non-separable transform, an input block of 4×4 pixels is regarded as a single array including sixteen elements, and then a transform (hypercube gives transform) in which gives revolution is performed on the array a plurality of times may be performed.

(345) In the transform in transformer **106**, the transform types of transform basis functions to be transformed into the frequency domain according to regions in a CU can be switched. Examples include a spatially varying transform (SVT).

(346) FIG. **15** is a diagram illustrating one example of SVT.

(347) In SVT, as illustrated in FIG. **15**, CUs are split into two equal regions horizontally or vertically, and only one of the regions is transformed into the frequency domain. A transform type can be set for each region. For example, DST7 and DST8 are used. For example, among the two regions obtained by splitting a CU vertically into two equal regions, DST7 and DCT8 may be used for the region at position 0. Alternatively, among the two regions, DST7 is used for the region at position 1. Likewise, among the two regions obtained by splitting a CU horizontally into two equal regions, DST7 and DCT8 are used for the region at position 0. Alternatively, among the two regions, DST7 is used for the region at position 1. Although only one of the two regions in a CU is transformed and the other is not transformed in the example illustrated in FIG. **15**, each of the two regions may be transformed. In addition, splitting method may include not only splitting into two regions but also splitting into four regions. In addition, the splitting method can be more flexible. For example, information indicating the splitting method may be encoded and may be signaled in the same manner as the CU splitting. It is to be noted that SVT is also referred to as sub-block transform (SBT).

(348) The AMT and EMT described above may be referred to as MTS (multiple transform selection). When MTS is applied, a transform type that is DST7, DCT8, or the like can be selected, and the information indicating the selected transform type may be encoded as index information for each CU. There is another process referred to as IMTS (implicit MTS) as a process for selecting, based on the shape of a CU, a transform type to be used for orthogonal transform performed without encoding index information. When IMTS is applied, for example, when a CU has a rectangle shape, orthogonal transform of the rectangle shape is performed using DST7 for the short side and DST2 for the long side. In addition, for example, when a CU has a square shape, orthogonal transform of the rectangle shape is performed using DCT2 when MTS is effective in a sequence and using DST7 when MTS is ineffective in the sequence. DCT2 and DST7 are mere examples. Other transform types may be used, and it is also possible to change the combination of transform types for use to a different combination of transform types. IMTS may be used only for intra prediction blocks, or may be used for both intra prediction blocks and inter prediction block.

(349) The three processes of MTS, SBT, and IMTS have been described above as selection processes for selectively switching transform types for use in orthogonal transform. However, all of the three selection processes may be made effective, or only part of the selection processes may be selectively made effective. Whether each of the selection processes is made effective can be identified based on flag information or the like in a header such as an SPS. For example, when all of the three selection processes are effective, one of the three selection processes is selected for each CU and orthogonal transform of the CU is performed. It is to be noted that the selection processes for selectively switching the transform types may be selection processes different from the above three selection processes, or each of the three selection processes may be replaced by another process as long as at least one of the following four functions [1] to [4] can be achieved.

Function [1] is a function for performing orthogonal transform of the entire CU and encoding information indicating the transform type used in the transform. Function [2] is a function for performing orthogonal transform of the entire CU and determining the transform type based on a predetermined rule without encoding information indicating the transform type. Function [3] is a function for performing orthogonal transform of a partial region of a CU and encoding information indicating the transform type used in the transform. Function [4] is a function for performing orthogonal transform of a partial region of a CU and determining the transform type based on a predetermined rule without encoding information indicating the transform type used in the transform.

(350) It is to be noted that whether each of MTS, IMTS, and SBT is applied may be determined for each processing unit. For example, whether each of MTS, IMTS, and SBT is applied may be determined for each sequence, picture, brick, slice, CTU, or CU.

(351) It is to be noted that a tool for selectively switching transform types in the present disclosure may be rephrased by a method for selectively selecting a basis for use in a transform process, a selection process, or a process for selecting a basis. In addition, the tool for selectively switching transform types may be rephrased by a mode for adaptively selecting a transform type.

(352) FIG. 16 is a flow chart illustrating one example of a process performed by transformer 106.

(353) For example, transformer 106 determines whether to perform orthogonal transform (Step St_1). Here, when determining to perform orthogonal transform (Yes in Step St_1), transformer 106 selects a transform type for use in orthogonal transform from a plurality of transform types (Step St_2). Next, transformer 106 performs orthogonal transform by applying the selected transform type to the prediction residual of a current block (Step St_3). Transformer 106 then outputs information indicating the selected transform type to entropy encoder 110, so as to allow entropy encoder 110 to encode the information (Step St_4). On the other hand, when determining not to perform orthogonal transform (No in Step St_1), transformer 106 outputs information indicating that no orthogonal transform is performed, so as to allow entropy encoder 110 to encode the information (Step St_5). It is to be noted that whether to perform orthogonal transform in Step St_1 may be determined based on, for example, the size of a transform block, a prediction mode applied to the CU, etc. Alternatively, orthogonal transform may be performed using a predefined transform type without encoding information indicating the transform type for use in orthogonal transform.

(354) FIG. 17 is a flow chart illustrating another example of a process performed by transformer 106. It is to be noted that the example illustrated in FIG. 17 is an example of orthogonal transform in the case where transform types for use in orthogonal transform are selectively switched as in the case of the example illustrated in FIG. 16.

(355) As one example, a first transform type group may include DCT2, DST7, and DCT8. As another example, a second transform type group may include DCT2. The transform types included in the first transform type group and the transform types included in the second transform type group may partly overlap with each other, or may be totally different from each other.

(356) More specifically, transformer 106 determines whether a transform size is smaller than or equal to a predetermined value (Step Su_1). Here, when determining that the transform size is smaller than or equal to the predetermined value (Yes in Step Su_1), transformer 106 performs orthogonal transform of the prediction residual of the current block using the transform type included in the first transform type group (Step Su_2). Next, transformer 106 outputs information indicating the transform type to be used among at least one transform type included in the first transform type group to entropy encoder 110, so as to allow entropy encoder 110 to encode the information (Step Su_3). On the other hand, when determining that the transform size is not smaller than or equal to the predetermined value (No in Step Su_1), transformer 106 performs orthogonal transform of the prediction residual of the current block using the second transform type group (Step Su_4).

(357) In Step Su_3, the information indicating the transform type for use in orthogonal transform may be information indicating a combination of the transform type to be applied vertically in the current block and the transform type to be applied horizontally in the current block. The first type group may include only one transform type, and the information indicating the transform type for use in orthogonal transform may not be encoded. The second transform type group may include a plurality of transform types, and information indicating the transform type for use in orthogonal transform among the one or more transform types included in the second transform type group may be encoded.

(358) Alternatively, a transform type may be determined based only on a transform size. It is to be noted that such determinations are not limited to the determination as to whether the transform size is smaller than or equal to the predetermined value, and other processes are also possible as long as the processes are for determining a transform type for use in orthogonal transform based on the transform size.

(359) [Quantizer]

(360) Quantizer **108** quantizes the transform coefficients output from transformer **106**. More specifically, quantizer **108** scans, in a determined scanning order, the transform coefficients of the current block, and quantizes the scanned transform coefficients based on quantization parameters (QP) corresponding to the transform coefficients. Quantizer **108** then outputs the quantized transform coefficients (hereinafter also referred to as quantized coefficients) of the current block to entropy encoder **110** and inverse quantizer **112**.

(361) A determined scanning order is an order for quantizing/inverse quantizing transform coefficients. For example, a determined scanning order is defined as ascending order of frequency (from low to high frequency) or descending order of frequency (from high to low frequency).

(362) A quantization parameter (QP) is a parameter defining a quantization step (quantization width). For example, when the value of the quantization parameter increases, the quantization step also increases. In other words, when the value of the quantization parameter increases, an error in quantized coefficients (quantization error) increases.

(363) In addition, a quantization matrix may be used for quantization. For example, several kinds of quantization matrices may be used correspondingly to frequency transform sizes such as 4×4 and 8×8 , prediction modes such as intra prediction and inter prediction, and pixel components such as luma and chroma pixel components. It is to be noted that quantization means digitalizing values sampled at predetermined intervals correspondingly to predetermined levels. In this technical field, quantization may be represented as other expressions such as rounding and scaling.

(364) Methods using quantization matrices include a method using a quantization matrix which has been set directly at the encoder **100** side and a method using a quantization matrix which has been set as a default (default matrix). At the encoder **100** side, a quantization matrix suitable for features of an image can be set by directly setting a quantization matrix. This case, however, has a disadvantage of increasing a coding amount for encoding the quantization matrix. It is to be noted that a quantization matrix to be used to quantize the current block may be generated based on a default quantization matrix or an encoded quantization matrix, instead of directly using the default quantization matrix or the encoded quantization matrix.

(365) There is a method for quantizing a high-frequency coefficient and a low-frequency coefficient in the same manner without using a quantization matrix. It is to be noted that this method is equivalent to a method using a quantization matrix (flat matrix) whose all coefficients have the same value.

(366) The quantization matrix may be encoded, for example, at the sequence level, picture level, slice level, brick level, or CTU level.

(367) When using a quantization matrix, quantizer **108** scales, for each transform coefficient, for example a quantization width which can be calculated based on a quantization parameter, etc., using the value of the quantization matrix. The quantization process performed without using any

quantization matrix may be a process of quantizing transform coefficients based on a quantization width calculated based on a quantization parameter, etc. It is to be noted that, in the quantization process performed without using any quantization matrix, the quantization width may be multiplied by a predetermined value which is common for all the transform coefficients in a block.

(368) FIG. **18** is a block diagram illustrating one example of a configuration of quantizer **108**.

(369) For example, quantizer **108** includes difference quantization parameter generator **108a**, predicted quantization parameter generator **108b**, quantization parameter generator **108c**, quantization parameter storage **108d**, and quantization executor **108e**.

(370) FIG. **19** is a flow chart illustrating one example of quantization performed by quantizer **108**.

(371) As one example, quantizer **108** may perform quantization for each CU based on the flow chart illustrated in FIG. **19**. More specifically, quantization parameter generator **108c** determines whether to perform quantization (Step Sv_1). Here, when determining to perform quantization (Yes in Step Sv_1), quantization parameter generator **108c** generates a quantization parameter for a current block (Step Sv_2), and stores the quantization parameter into quantization parameter storage **108d** (Step Sv_3).

(372) Next, quantization executor **108e** quantizes transform coefficients of the current block using the quantization parameter generated in Step Sv_2 (Step Sv_4). Predicted quantization parameter generator **108b** then obtains a quantization parameter for a processing unit different from the current block from quantization parameter storage **108d** (Step Sv_5). Predicted quantization parameter generator **108b** generates a predicted quantization parameter of the current block based on the obtained quantization parameter (Step Sv_6). Difference quantization parameter generator **108a** calculates the difference between the quantization parameter of the current block generated by quantization parameter generator **108c** and the predicted quantization parameter of the current block generated by predicted quantization parameter generator **108b** (Step Sv_7). The difference quantization parameter is generated by calculating the difference. Difference quantization parameter generator **108a** outputs the difference quantization parameter to entropy encoder **110**, so as to allow entropy encoder **110** to encode the difference quantization parameter (Step Sv_8).

(373) It is to be noted that the difference quantization parameter may be encoded, for example, at the sequence level, picture level, slice level, brick level, or CTU level. In addition, the initial value of the quantization parameter may be encoded at the sequence level, picture level, slice level, brick level, or CTU level. At this time, the quantization parameter may be generated using the initial value of the quantization parameter and the difference quantization parameter.

(374) It is to be noted that quantizer **108** may include a plurality of quantizers, and may apply dependent quantization in which transform coefficients are quantized using a quantization method selected from a plurality of quantization methods.

(375) [Entropy Encoder]

(376) FIG. **20** is a block diagram illustrating one example of a configuration of entropy encoder **110**.

(377) Entropy encoder **110** generates a stream by entropy encoding the quantized coefficients input from quantizer **108** and a prediction parameter input from prediction parameter generator **130**. For example, context-based adaptive binary arithmetic coding (CABAC) is used as the entropy encoding. More specifically, entropy encoder **110** includes binarizer **110a**, context controller **110b**, and binary arithmetic encoder **110c**. Binarizer **110a** performs binarization in which multi-level signals such as quantized coefficients and a prediction parameter are transformed into binary signals. Examples of binarization methods include truncated Rice binarization, exponential Golomb codes, and fixed length binarization. Context controller **110b** derives a context value according to a feature or a surrounding state of a syntax element, that is, an occurrence probability of a binary signal. Examples of methods for deriving a context value include bypass, referring to a syntax element, referring to an upper and left adjacent blocks, referring to hierarchical information, and others. Binary arithmetic encoder **110c** arithmetically encodes the binary signal using the derived

context value.

(378) FIG. 21 is a diagram illustrating a flow of CABAC in entropy encoder 110.

(379) First, initialization is performed in CABAC in entropy encoder 110. In the initialization, initialization in binary arithmetic encoder 110c and setting of an initial context value are performed. For example, binarizer 110a and binary arithmetic encoder 110c execute binarization and arithmetic encoding of a plurality of quantization coefficients in a CTU sequentially. At this time, context controller 110b updates the context value each time arithmetic encoding is performed. Context controller 110b then saves the context value as a post process. The saved context value is used, for example, to initialize the context value for the next CTU.

(380) [Inverse Quantizer]

(381) Inverse quantizer 112 inverse quantizes quantized coefficients which have been input from quantizer 108. More specifically, inverse quantizer 112 inverse quantizes, in a determined scanning order, quantized coefficients of the current block. Inverse quantizer 112 then outputs the inverse quantized transform coefficients of the current block to inverse transformer 114.

(382) [Inverse Transformer]

(383) Inverse transformer 114 restores prediction errors by inverse transforming the transform coefficients which have been input from inverse quantizer 112. More specifically, inverse transformer 114 restores the prediction residuals of the current block by performing an inverse transform corresponding to the transform applied to the transform coefficients by transformer 106. Inverse transformer 114 then outputs the restored prediction residuals to adder 116.

(384) It is to be noted that since information is normally lost in quantization, the restored prediction residuals do not match the prediction errors calculated by subtractor 104. In other words, the restored prediction residuals normally include quantization errors.

(385) [Adder]

(386) Adder 116 reconstructs the current block by adding the prediction residuals which have been input from inverse transformer 114 and prediction images which have been input from prediction controller 128. Consequently, a reconstructed image is generated. Adder 116 then outputs the reconstructed image to block memory 118 and loop filter 120.

(387) [Block Memory]

(388) Block memory 118 is storage for storing a block which is included in a current picture and is referred to in intra prediction. More specifically, block memory 118 stores a reconstructed image output from adder 116.

(389) [Frame Memory]

(390) Frame memory 122 is, for example, storage for storing reference pictures for use in inter prediction, and is also referred to as a frame buffer. More specifically, frame memory 122 stores a reconstructed image filtered by loop filter 120.

(391) [Loop Filter]

(392) Loop filter 120 applies a loop filter to a reconstructed image output by adder 116, and outputs the filtered reconstructed image to frame memory 122. A loop filter is a filter used in an encoding loop (in-loop filter). Examples of loop filters include, for example, an adaptive loop filter (ALF), a deblocking filter (DF or DBF), a sample adaptive offset (SAO), etc.

(393) FIG. 22 is a block diagram illustrating one example of a configuration of loop filter 120.

(394) For example, as illustrated in FIG. 22, loop filter 120 includes deblocking filter executor 120a, SAO executor 120b, and ALF executor 120c. Deblocking filter executor 120a performs a deblocking filter process of the reconstructed image. SAO executor 120b performs a SAO process of the reconstructed image after being subjected to the deblocking filter process. ALF executor 120c performs an ALF process of the reconstructed image after being subjected to the SAO process. The ALF and deblocking filter processes are described later in detail. The SAO process is a process for enhancing image quality by reducing ringing (a phenomenon in which pixel values are distorted like waves around an edge) and correcting deviation in pixel value. Examples of SAO

processes include an edge offset process and a band offset process. It is to be noted that loop filter **120** does not always need to include all the constituent elements disclosed in FIG. **22**, and may include only part of the constituent elements. In addition, loop filter **120** may be configured to perform the above processes in a processing order different from the one disclosed in FIG. **22**.

(395) [Loop Filter>Adaptive Loop Filter]

(396) In an ALF, a least square error filter for removing compression artifacts is applied. For example, one filter selected from among a plurality of filters based on the direction and activity of local gradients is applied for each of 2×2 pixel sub-blocks in the current block.

(397) More specifically, first, each sub-block (for example, each 2×2 pixel sub-block) is categorized into one out of a plurality of classes (for example, fifteen or twenty-five classes). The categorization of the sub-block is based on, for example, gradient directionality and activity. In a specific example, category index C (for example, $C=5D+A$) is calculated based on gradient directionality D (for example, 0 to 2 or 0 to 4) and gradient activity A (for example, 0 to 4). Then, based on category index C, each sub-block is categorized into one out of a plurality of classes.

(398) For example, gradient directionality D is calculated by comparing gradients of a plurality of directions (for example, the horizontal, vertical, and two diagonal directions). Moreover, for example, gradient activity A is calculated by adding gradients of a plurality of directions and quantizing the result of the addition.

(399) The filter to be used for each sub-block is determined from among the plurality of filters based on the result of such categorization.

(400) The filter shape to be used in an ALF is, for example, a circular symmetric filter shape. FIG. **23A** through FIG. **23C** illustrate examples of filter shapes used in ALFs. FIG. **23A** illustrates a 5×5 diamond shape filter, FIG. **23B** illustrates a 7×7 diamond shape filter, and FIG. **23C** illustrates a 9×9 diamond shape filter. Information indicating the filter shape is normally signaled at the picture level. It is to be noted that the signaling of such information indicating the filter shape does not necessarily need to be performed at the picture level, and may be performed at another level (for example, at the sequence level, slice level, brick level, CTU level, or CU level).

(401) The ON or OFF of the ALF is determined, for example, at the picture level or CU level. For example, the decision of whether to apply the ALF to luma may be made at the CU level, and the decision of whether to apply ALF to chroma may be made at the picture level. Information indicating ON or OFF of the ALF is normally signaled at the picture level or CU level. It is to be noted that the signaling of information indicating ON or OFF of the ALF does not necessarily need to be performed at the picture level or CU level, and may be performed at another level (for example, at the sequence level, slice level, brick level, or CTU level).

(402) In addition, as described above, one filter is selected from the plurality of filters, and an ALF process of a sub-block is performed. A coefficient set of coefficients to be used for each of the plurality of filters (for example, up to the fifteenth or twenty-fifth filter) is normally signaled at the picture level. It is to be noted that the coefficient set does not always need to be signaled at the picture level, and may be signaled at another level (for example, the sequence level, slice level, brick level, CTU level, CU level, or sub-block level).

(403) [Loop Filter>Cross Component Adaptive Loop Filter]

(404) FIG. **23D** is a diagram illustrating an example where Y samples (first component) are used for a cross component ALF (CCALF) for Cb and a CCALF for Cr (components different from the first component). FIG. **23E** is a diagram illustrating a diamond shaped filter.

(405) One example of CC-ALF operates by applying a linear, diamond shaped filter (FIGS. **23D**, **23E**) to a luma channel for each chroma component. The filter coefficients, for example, may be transmitted in the APS, scaled by a factor of $2^{\lceil \log_2(\text{range}) \rceil} / 10$, and rounded for fixed point representation. The application of the filters is controlled on a variable block size and signaled by a context-coded flag received for each block of samples. The block size along with a CC-ALF enabling flag is received at the slice-level for each chroma component. Syntax and

semantics for CC-ALF are provided in the Appendix. In the contribution, the following block sizes (in chroma samples) were supported: 16×16 , 32×32 , 64×64 , and 128×128 .

(406) [Loop Filter>Joint Chroma Cross Component Adaptive Loop Filter]

(407) FIG. 23F is a diagram illustrating an example for a joint chroma CCALF (JC-CCALF).

(408) One example of JC-CCALF, where only one CCALF filter will be used to generate one CCALF filtered output as a chroma refinement signal for one color component only, while a properly weighted version of the same chroma refinement signal will be applied to the other color component. In this way, the complexity of existing CCALF is reduced roughly by half.

(409) The weight value is coded into a sign flag and a weight index. The weight index (denoted as `weight_index`) is coded into 3 bits, and specifies the magnitude of the JC-CCALF weight `JcCcWeight`. It cannot be equal to 0. The magnitude of `JcCcWeight` is determined as follows. If `weight_index` is less than or equal to 4, `JcCcWeight` is equal to `weight_index >> 2`. Otherwise, `JcCcWeight` is equal to $4/(\text{weight_index}-4)$.

(410) The block-level on/off control of ALF filtering for Cb and Cr are separate. This is the same as in CCALF, and two separate sets of block-level on/off control flags will be coded. Different from CCALF, herein, the Cb, Cr on/off control block sizes are the same, and thus, only one block size variable is coded.

(411) [Loop Filter>Deblocking Filter]

(412) In a deblocking filter process, loop filter **120** performs a filter process on a block boundary in a reconstructed image so as to reduce distortion which occurs at the block boundary.

(413) FIG. 24 is a block diagram illustrating one example of a specific configuration of deblocking filter executor **120a**.

(414) For example, deblocking filter executor **120a** includes: boundary determiner **1201**; filter determiner **1203**; filter executor **1205**; process determiner **1208**; filter characteristic determiner **1207**; and switches **1202**, **1204**, and **1206**.

(415) Boundary determiner **1201** determines whether a pixel to be deblock filtered (that is, a current pixel) is present around a block boundary. Boundary determiner **1201** then outputs the determination result to switch **1202** and process determiner **1208**.

(416) In the case where boundary determiner **1201** has determined that a current pixel is present around a block boundary, switch **1202** outputs an unfiltered image to switch **1204**. In the opposite case where boundary determiner **1201** has determined that no current pixel is present around a block boundary, switch **1202** outputs an unfiltered image to switch **1206**. It is to be noted that the unfiltered image is an image configured with a current pixel and at least one surrounding pixel located around the current pixel.

(417) Filter determiner **1203** determines whether to perform deblocking filtering of the current pixel, based on the pixel value of at least one surrounding pixel located around the current pixel. Filter determiner **1203** then outputs the determination result to switch **1204** and process determiner **1208**.

(418) In the case where filter determiner **1203** has determined to perform deblocking filtering of the current pixel, switch **1204** outputs the unfiltered image obtained through switch **1202** to filter executor **1205**. In the opposite case where filter determiner **1203** has determined not to perform deblocking filtering of the current pixel, switch **1204** outputs the unfiltered image obtained through switch **1202** to switch **1206**.

(419) When obtaining the unfiltered image through switches **1202** and **1204**, filter executor **1205** executes, for the current pixel, deblocking filtering having the filter characteristic determined by filter characteristic determiner **1207**. Filter executor **1205** then outputs the filtered pixel to switch **1206**.

(420) Under control by process determiner **1208**, switch **1206** selectively outputs a pixel which has not been deblock filtered and a pixel which has been deblock filtered by filter executor **1205**.

(421) Process determiner **1208** controls switch **1206** based on the results of determinations made

by boundary determiner **1201** and filter determiner **1203**. In other words, process determiner **1208** causes switch **1206** to output the pixel which has been deblock filtered when boundary determiner **1201** has determined that the current pixel is present around the block boundary and filter determiner **1203** has determined to perform deblocking filtering of the current pixel. In addition, in a case other than the above case, process determiner **1208** causes switch **1206** to output the pixel which has not been deblock filtered. A filtered image is output from switch **1206** by repeating output of a pixel in this way. It is to be noted that the configuration illustrated in FIG. **24** is one example of a configuration in deblocking filter executor **120a**. Deblocking filter executor **120a** may have another configuration.

(422) FIG. **25** is a diagram illustrating an example of a deblocking filter having a symmetrical filtering characteristic with respect to a block boundary.

(423) In a deblocking filter process, one of two deblocking filters having different characteristics, that is, a strong filter and a weak filter is selected using pixel values and quantization parameters, for example. In the case of the strong filter, pixels p0 to p2 and pixels q0 to q2 are present across a block boundary as illustrated in FIG. **25**, the pixel values of the respective pixels q0 to q2 are changed to pixel values q'0 to q'2 by performing computations according to the expressions below.

$$q'0=(p1+2\times p0+2\times q0+2\times q1+q2+4)/8$$

$$q'1=(p0+q0+q1+q2+2)/4$$

$$q'2=(p0+q0+q1+3\times q2+2\times q3+4)/8$$

(424) It is to be noted that, in the above expressions, p0 to p2 and q0 to q2 are the pixel values of respective pixels p0 to p2 and pixels q0 to q2. In addition, q3 is the pixel value of neighboring pixel q3 located at the opposite side of pixel q2 with respect to the block boundary. In addition, in the right side of each of the expressions, coefficients which are multiplied with the respective pixel values of the pixels to be used for deblocking filtering are filter coefficients.

(425) Furthermore, in the deblocking filtering, clipping may be performed so that the calculated pixel values do not change over a threshold value. In the clipping process, the pixel values calculated according to the above expressions are clipped to a value obtained according to “a pre-computation pixel value $\pm$ 2 $\times$ a threshold value” using the threshold value determined based on a quantization parameter. In this way, it is possible to prevent excessive smoothing.

(426) FIG. **26** is a diagram for illustrating one example of a block boundary on which a deblocking filter process is performed. FIG. **27** is a diagram illustrating examples of Bs values.

(427) The block boundary on which the deblocking filter process is performed is, for example, a boundary between CUs, PUs, or TUs having 8 $\times$ 8 pixel blocks as illustrated in FIG. **26**. The deblocking filter process is performed, for example, in units of four rows or four columns. First, boundary strength (Bs) values are determined as indicated in FIG. **27** for block P and block Q illustrated in FIG. **26**.

(428) According to the Bs values in FIG. **27**, whether to perform deblocking filter processes of block boundaries belonging to the same image using different strengths may be determined. The deblocking filter process for a chroma signal is performed when a Bs value is 2. The deblocking filter process for a luma signal is performed when a Bs value is 1 or more and a determined condition is satisfied. It is to be noted that conditions for determining Bs values are not limited to those indicated in FIG. **27**, and a Bs value may be determined based on another parameter.

(429) [Predictor (Intra Predictor, Inter Predictor, Prediction Controller)]

(430) FIG. **28** is a flow chart illustrating one example of a process performed by a predictor of encoder **100**. It is to be noted that the predictor, as one example, includes all or part of the following constituent elements: intra predictor **124**; inter predictor **126**; and prediction controller **128**. The prediction executor includes, for example, intra predictor **124** and inter predictor **126**.

(431) The predictor generates a prediction image of a current block (Step Sb_1). It is to be noted that the prediction image is, for example, an intra prediction image (intra prediction signal) or an inter prediction image (inter prediction signal). More specifically, the predictor generates the

prediction image of the current block using a reconstructed image which has been already obtained for another block through generation of a prediction image, generation of a prediction residual, generation of quantized coefficients, restoring of a prediction residual, and addition of a prediction image.

(432) The reconstructed image may be, for example, an image in a reference picture or an image of an encoded block (that is, the other block described above) in a current picture which is the picture including the current block. The encoded block in the current picture is, for example, a neighboring block of the current block.

(433) FIG. 29 is a flow chart illustrating another example of a process performed by the predictor of encoder 100.

(434) The predictor generates a prediction image using a first method (Step Sc_1a), generates a prediction image using a second method (Step Sc_1b), and generates a prediction image using a third method (Step Sc_1c). The first method, the second method, and the third method may be mutually different methods for generating a prediction image. Each of the first to third methods may be an inter prediction method, an intra prediction method, or another prediction method. The above-described reconstructed image may be used in these prediction methods.

(435) Next, the predictor evaluates the prediction images generated in Steps Sc_1a, Sc_1b, and Sc_1c (Step Sc_2). For example, the predictor calculates costs C for the prediction images generated in Step Sc_1a, Sc_1b, and Sc_1c, and evaluates the prediction images by comparing the costs C of the prediction images. It is to be noted that cost C is calculated according to an expression of an R-D optimization model, for example, $C=D+\lambda \times R$. In this expression, D indicates compression artifacts of a prediction image, and is represented as, for example, a sum of absolute differences between the pixel value of a current block and the pixel value of a prediction image. In addition, R indicates a bit rate of a stream. In addition, λ indicates, for example, a multiplier according to the method of Lagrange multiplier.

(436) The predictor then selects one of the prediction images generated in Steps Sc_1a, Sc_1b, and Sc_1c (Step Sc_3). In other words, the predictor selects a method or a mode for obtaining a final prediction image. For example, the predictor selects the prediction image having the smallest cost C, based on costs C calculated for the prediction images. Alternatively, the evaluation in Step Sc_2 and the selection of the prediction image in Step Sc_3 may be made based on a parameter which is used in an encoding process. Encoder 100 may transform information for identifying the selected prediction image, the method, or the mode into a stream. The information may be, for example, a flag or the like. In this way, decoder 200 is capable of generating a prediction image according to the method or the mode selected by encoder 100, based on the information. It is to be noted that, in the example illustrated in FIG. 29, the predictor selects any of the prediction images after the prediction images are generated using the respective methods. However, the predictor may select a method or a mode based on a parameter for use in the above-described encoding process before generating prediction images, and may generate a prediction image according to the method or mode selected.

(437) For example, the first method and the second method may be intra prediction and inter prediction, respectively, and the predictor may select a final prediction image for a current block from prediction images generated according to the prediction methods.

(438) FIG. 30 is a flow chart illustrating another example of a process performed by the predictor of encoder 100.

(439) First, the predictor generates a prediction image using intra prediction (Step Sd_1a), and generates a prediction image using inter prediction (Step Sd_1b). It is to be noted that the prediction image generated by intra prediction is also referred to as an intra prediction image, and the prediction image generated by inter prediction is also referred to as an inter prediction image.

(440) Next, the predictor evaluates each of the intra prediction image and the inter prediction image (Step Sd_2). Cost C described above may be used in the evaluation. The predictor may then select

the prediction image for which the smallest cost C has been calculated among the intra prediction image and the inter prediction image, as the final prediction image for the current block (Step Sd_3). In other words, the prediction method or the mode for generating the prediction image for the current block is selected.

(441) [Intra Predictor]

(442) Intra predictor **124** generates a prediction image (that is, intra prediction image) of a current block by performing intra prediction (also referred to as intra frame prediction) of the current block by referring to a block or blocks in the current picture which is or are stored in block memory **118**. More specifically, intra predictor **124** generates an intra prediction image by performing intra prediction by referring to pixel values (for example, luma and/or chroma values) of a block or blocks neighboring the current block, and then outputs the intra prediction image to prediction controller **128**.

(443) For example, intra predictor **124** performs intra prediction by using one mode from among a plurality of intra prediction modes which have been predefined. The intra prediction modes normally include one or more non-directional prediction modes and a plurality of directional prediction modes.

(444) The one or more non-directional prediction modes include, for example, planar prediction mode and DC prediction mode defined in the H.265/HEVC standard.

(445) The plurality of directional prediction modes include, for example, the thirty-three directional prediction modes defined in the H.265/HEVC standard. It is to be noted that the plurality of directional prediction modes may further include thirty-two directional prediction modes in addition to the thirty-three directional prediction modes (for a total of sixty-five directional prediction modes). FIG. **31** is a diagram illustrating sixty-seven intra prediction modes in total used in intra prediction (two non-directional prediction modes and sixty-five directional prediction modes). The solid arrows represent the thirty-three directions defined in the H.265/HEVC standard, and the dashed arrows represent the additional thirty-two directions (the two non-directional prediction modes are not illustrated in FIG. **31**).

(446) In various kinds of mounting examples, a luma block may be referred to in intra prediction of a chroma block. In other words, a chroma component of the current block may be predicted based on a luma component of the current block. Such intra prediction is also referred to as cross-component linear model (CCLM). The intra prediction mode for a chroma block in which such a luma block is referred to (also referred to as, for example, a CCLM mode) may be added as one of the intra prediction modes for chroma blocks.

(447) Intra predictor **124** may correct intra-predicted pixel values based on horizontal/vertical reference pixel gradients. The intra prediction which accompanies this sort of correcting is also referred to as position dependent intra prediction combination (PDPC). Information indicating whether to apply PDPC (referred to as, for example, a PDPC flag) is normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, or CTU level).

(448) FIG. **32** is a flow chart illustrating one example of a process performed by intra predictor **124**.

(449) Intra predictor **124** selects one intra prediction mode from a plurality of intra prediction modes (Step Sw_1). Intra predictor **124** then generates a prediction image according to the selected intra prediction mode (Step Sw_2). Next, intra predictor **124** determines most probable modes (MPMs) (Step Sw_3). MPMs include, for example, six intra prediction modes. Two modes among the six intra prediction modes may be planar mode and DC prediction mode, and the other four modes may be directional prediction modes. Intra predictor **124** determines whether the intra prediction mode selected in Step Sw_1 is included in the MPMs (Step Sw_4).

(450) Here, when determining that the intra prediction mode selected in Step Sw_1 is included in

the MPMs (Yes in Step Sw_4), intra predictor **124** sets an MPM flag to 1 (Step Sw_5), and generates information indicating the selected intra prediction mode among the MPMs (Step Sw_6). It is to be noted that the MPM flag set to 1 and the information indicating the intra prediction mode are encoded as prediction parameters by entropy encoder **110**.

(451) When determining that the selected intra prediction mode is not included in the MPMs (No in Step Sw_4), intra predictor **124** sets the MPM flag to 0 (Step Sw_7). Alternatively, intra predictor **124** does not set any MPM flag. Intra predictor **124** then generates information indicating the selected intra prediction mode among at least one intra prediction mode which is not included in the MPMs (Step Sw_8). It is to be noted that the MPM flag set to 0 and the information indicating the intra prediction mode are encoded as prediction parameters by entropy encoder **110**. The information indicating the intra prediction mode indicates, for example, any one of 0 to 60.

(452) [Inter Predictor]

(453) Inter predictor **126** generates a prediction image (inter prediction image) by performing inter prediction (also referred to as inter frame prediction) of the current block by referring to a block or blocks in a reference picture which is different from the current picture and is stored in frame memory **122**. Inter prediction is performed in units of a current block or a current sub-block in the current block. The sub-block is included in the block and is a unit smaller than the block. The size of the sub-block may be 4×4 pixels, 8×8 pixels, or another size. The size of the sub-block may be switched for a unit such as slice, brick, picture, etc.

(454) For example, inter predictor **126** performs motion estimation in a reference picture for a current block or a current sub-block, and finds out a reference block or a reference sub-block which best matches the current block or current sub-block. Inter predictor **126** then obtains motion information (for example, a motion vector) which compensates a motion or a change from the reference block or the reference sub-block to the current block or the current sub-block. Inter predictor **126** generates an inter prediction image of the current block or the current sub-block by performing motion compensation (or motion prediction) based on the motion information. Inter predictor **126** outputs the generated inter prediction image to prediction controller **128**.

(455) The motion information used in motion compensation may be signaled as inter prediction images in various forms. For example, a motion vector may be signaled. As another example, the difference between a motion vector and a motion vector predictor may be signaled.

(456) [Reference Picture List]

(457) FIG. **33** is a diagram illustrating examples of reference pictures. FIG. **34** is a conceptual diagram illustrating examples of reference picture lists. Each reference picture list is a list indicating at least one reference picture stored in frame memory **122**. It is to be noted that, in FIG. **33**, each of rectangles indicates a picture, each of arrows indicates a picture reference relationship, the horizontal axis indicates time, I, P, and B in the rectangles indicate an intra prediction picture, a uni-prediction picture, and a bi-prediction picture, respectively, and numerals in the rectangles indicate a decoding order. As illustrated in FIG. **33**, the decoding order of the pictures is an order of I0, P1, B2, B3, and B4, and the display order of the pictures is an order of I0, B3, B2, B4, and P1. As illustrated in FIG. **34**, the reference picture list is a list representing reference picture candidates. For example, one picture (or a slice) may include at least one reference picture list. For example, one reference picture list is used when a current picture is a uni-prediction picture, and two reference picture lists are used when a current picture is a bi-prediction picture. In the examples of FIGS. **33** and **34**, picture B3 which is current picture currPic has two reference picture lists which are the L0 list and the L1 list. When current picture currPic is picture B3, reference picture candidates for current picture currPic are I0, P1, and B2, and the reference picture lists (which are the L0 list and the L1 list) indicate these pictures. Inter predictor **126** or prediction controller **128** specifies which picture in each reference picture list is to be actually referred to in form of a reference picture index refIdxLx. In FIG. **34**, reference pictures P1 and B2 are specified by reference picture indices refIdxL0 and refIdxL1.

(458) Such a reference picture list may be generated for each unit such as a sequence, picture, slice, brick, CTU, or CU. In addition, among reference pictures indicated in reference picture lists, a reference picture index indicating a reference picture to be referred to in inter prediction may be signaled at the sequence level, picture level, slice level, brick level, CTU level, or CU level. In addition, a common reference picture list may be used in a plurality of inter prediction modes.

(459) [Basic Flow of Inter Prediction]

(460) FIG. 35 is a flow chart illustrating a basic processing flow of inter prediction.

(461) First, inter predictor **126** generates a prediction signal (Steps Se_1 to Se_3). Next, subtractor **104** generates the difference between a current block and a prediction image as a prediction residual (Step Se_4).

(462) Here, in the generation of the prediction image, inter predictor **126** generates the prediction image through, for example, determination of a motion vector (MV) of the current block (Steps Se_1 and Se_2) and motion compensation (Step Se_3). Furthermore, in determination of an MV, inter predictor **126** determines the MV through, for example, selection of a motion vector candidate (MV candidate) (Step Se_1) and derivation of an MV (Step Se_2). The selection of the MV candidate is made by means of, for example, inter predictor **126** generating an MV candidate list and selecting at least one MV candidate from the MV candidate list. It is to be noted that MVs derived in the past may be added to the MV candidate list. Alternatively, in derivation of an MV, inter predictor **126** may further select at least one MV candidate from the at least one MV candidate, and determine the selected at least one MV candidate as the MV for the current block. Alternatively, inter predictor **126** may determine the MV for the current block by performing estimation in a reference picture region specified by each of the selected at least one MV candidate. It is to be noted that the estimation in the reference picture region may be referred to as motion estimation.

(463) In addition, although Steps Se_1 to Se_3 are performed by inter predictor **126** in the above-described example, a process that is, for example, Step Se_1, Step Se_2, or the like may be performed by another constituent element included in encoder **100**.

(464) It is to be noted that an MV candidate list may be generated for each process in inter prediction mode, or a common MV candidate list may be used in a plurality of inter prediction modes. The processes in Steps Se_3 and Se_4 correspond to Steps Sa_3 and Sa_4 illustrated in FIG. 9, respectively. The process in Step Se_3 corresponds to the process in Step Sd_1b in FIG. 30.

(465) [MV Derivation Flow]

(466) FIG. 36 is a flow chart illustrating one example of MV derivation.

(467) Inter predictor **126** may derive an MV for a current block in a mode for encoding motion information (for example, an MV). In this case, for example, the motion information may be encoded as a prediction parameter, and may be signaled. In other words, the encoded motion information is included in a stream.

(468) Alternatively, inter predictor **126** may derive an MV in a mode in which motion information is not encoded. In this case, no motion information is included in the stream.

(469) Here, MV derivation modes include a normal inter mode, a normal merge mode, a FRUC mode, an affine mode, etc. which are described later. Modes in which motion information is encoded among the modes include the normal inter mode, the normal merge mode, the affine mode (specifically, an affine inter mode and an affine merge mode), etc. It is to be noted that motion information may include not only an MV but also MV predictor selection information which is described later. Modes in which no motion information is encoded include the FRUC mode, etc. Inter predictor **126** selects a mode for deriving an MV of the current block from the plurality of modes, and derives the MV of the current block using the selected mode.

(470) FIG. 37 is a flow chart illustrating another example of MV derivation.

(471) Inter predictor **126** may derive an MV for a current block in a mode in which an MV difference is encoded. In this case, for example, the MV difference is encoded as a prediction

parameter, and is signaled. In other words, the encoded MV difference is included in a stream. The MV difference is the difference between the MV of the current block and the MV predictor. It is to be noted that the MV predictor is a motion vector predictor.

(472) Alternatively, inter predictor **126** may derive an MV in a mode in which no MV difference is encoded. In this case, no encoded MV difference is included in the stream.

(473) Here, as described above, the MV derivation modes include the normal inter mode, the normal merge mode, the FRUC mode, the affine mode, etc. which are described later. Modes in which an MV difference is encoded among the modes include the normal inter mode, the affine mode (specifically, the affine inter mode), etc. Modes in which no MV difference is encoded include the FRUC mode, the normal merge mode, the affine mode (specifically, the affine merge mode), etc. Inter predictor **126** selects a mode for deriving an MV of the current block from the plurality of modes, and derives the MV for the current block using the selected mode.

(474) [MV Derivation Modes]

(475) FIGS. **38A** and **38B** are each a diagram illustrating one example of categorization of modes for MV derivation. For example, as illustrated in FIG. **38A**, MV derivation modes are roughly categorized into three modes according to whether to encode motion information and whether to encode MV differences. The three modes are inter mode, merge mode, and frame rate up-conversion (FRUC) mode. The inter mode is a mode in which motion estimation is performed, and in which motion information and an MV difference are encoded. For example, as illustrated in FIG. **38B**, the inter mode includes affine inter mode and normal inter mode. The merge mode is a mode in which no motion estimation is performed, and in which an MV is selected from an encoded surrounding block and an MV for the current block is derived using the MV. The merge mode is a mode in which, basically, motion information is encoded and no MV difference is encoded. For example, as illustrated in FIG. **38B**, the merge modes include normal merge mode (also referred to as normal merge mode or regular merge mode), merge with motion vector difference (MMVD) mode, combined inter merge/intra prediction (CIIP) mode, triangle mode, ATMVP mode, and affine merge mode. Here, an MV difference is encoded exceptionally in the MMVD mode among the modes included in the merge modes. It is to be noted that the affine merge mode and the affine inter mode are modes included in the affine modes. The affine mode is a mode for deriving, as an MV of a current block, an MV of each of a plurality of sub-blocks included in the current block, assuming affine transform. The FRUC mode is a mode which is for deriving an MV of the current block by performing estimation between encoded regions, and in which neither motion information nor any MV difference is encoded. It is to be noted that the respective modes will be described later in detail.

(476) It is to be noted that the categorization of the modes illustrated in FIGS. **38A** and **38B** are examples, and categorization is not limited thereto. For example, when an MV difference is encoded in CIIP mode, the CIIP mode is categorized into inter modes.

(477) [MV Derivation>Normal Inter Mode]

(478) The normal inter mode is an inter prediction mode for deriving an MV of a current block by finding out a block similar to the image of the current block from a reference picture region specified by an MV candidate. In this normal inter mode, an MV difference is encoded.

(479) FIG. **39** is a flow chart illustrating an example of inter prediction by normal inter mode.

(480) First, inter predictor **126** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of encoded blocks temporally or spatially surrounding the current block (Step Sg_1). In other words, inter predictor **126** generates an MV candidate list.

(481) Next, inter predictor **126** extracts N (an integer of 2 or larger) MV candidates from the plurality of MV candidates obtained in Step Sg_1, as motion vector predictor candidates according to a predetermined priority order (Step Sg_2). It is to be noted that the priority order is determined in advance for each of the N MV candidates.

(482) Next, inter predictor **126** selects one MV predictor candidate from the N MV predictor

candidates as the MV predictor for the current block (Step Sg_3). At this time, inter predictor **126** encodes, in a stream, MV predictor selection information for identifying the selected MV predictor. In other words, inter predictor **126** outputs the MV predictor selection information as a prediction parameter to entropy encoder **110** through prediction parameter generator **130**.

(483) Next, inter predictor **126** derives an MV of a current block by referring to an encoded reference picture (Step Sg_4). At this time, inter predictor **126** further encodes, in the stream, the difference value between the derived MV and the MV predictor as an MV difference. In other words, inter predictor **126** outputs the MV difference as a prediction parameter to entropy encoder **110** through prediction parameter generator **130**. It is to be noted that the encoded reference picture is a picture including a plurality of blocks which have been reconstructed after being encoded.

(484) Lastly, inter predictor **126** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Sg_5). The processes in Steps Sg_1 to Sg_5 are executed on each block. For example, when the processes in Steps Sg_1 to Sg_5 are executed on each of all the blocks in the slice, inter prediction of the slice using the normal inter mode finishes. For example, when the processes in Steps Sg_1 to Sg_5 are executed on each of all the blocks in the picture, inter prediction of the picture using the normal inter mode finishes. It is to be noted that not all the blocks included in the slice may be subjected to the processes in Steps Sg_1 to Sg_5, and inter prediction of the slice using the normal inter mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the normal inter mode may finish when the processes in Steps Sg_1 to Sg_5 are executed on part of the blocks in the picture.

(485) It is to be noted that the prediction image is an inter prediction signal as described above. In addition, information indicating the inter prediction mode (normal inter mode in the above example) used to generate the prediction image is, for example, encoded as a prediction parameter in an encoded signal.

(486) It is to be noted that the MV candidate list may be also used as a list for use in another mode. In addition, the processes related to the MV candidate list may be applied to processes related to the list for use in another mode. The processes related to the MV candidate list include, for example, extraction or selection of an MV candidate from the MV candidate list, reordering of MV candidates, or deletion of an MV candidate.

(487) [MV Derivation>Normal Merge Mode]

(488) The normal merge mode is an inter prediction mode for selecting an MV candidate from an MV candidate list as an MV for a current block, thereby deriving the MV. It is to be noted that the normal merge mode is a merge mode in a narrow meaning and is also simply referred to as a merge mode. In this embodiment, the normal merge mode and the merge mode are distinguished, and the merge mode is used in a broad meaning.

(489) FIG. **40** is a flow chart illustrating an example of inter prediction by normal merge mode.

(490) First, inter predictor **126** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of encoded blocks temporally or spatially surrounding the current block (Step Sh_1). In other words, inter predictor **126** generates an MV candidate list.

(491) Next, inter predictor **126** selects one MV candidate from the plurality of MV candidates obtained in Step Sh_1, thereby deriving an MV for the current block (Step Sh_2). At this time, inter predictor **126** encodes, in a stream, MV selection information for identifying the selected MV candidate. In other words, inter predictor **126** outputs the MV selection information as a prediction parameter to entropy encoder **110** through prediction parameter generator **130**.

(492) Lastly, inter predictor **126** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Sh_3). The processes in Steps Sh_1 to Sh_3 are executed, for example, on each block. For example, when the processes in Steps Sh_1 to Sh_3 are executed on each of all the blocks in the slice, inter prediction of the slice using the normal merge mode finishes. In addition, when the

processes in Steps Sh_1 to Sh_3 are executed on each of all the blocks in the picture, inter prediction of the picture using the normal merge mode finishes. It is to be noted that not all the blocks included in the slice may be subjected to the processes in Steps Sh_1 to Sh_3, and inter prediction of the slice using the normal merge mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the normal merge mode may finish when the processes in Steps Sh_1 to Sh_3 are executed on part of the blocks in the picture.

(493) In addition, information indicating the inter prediction mode (normal merge mode in the above example) used to generate the prediction image is, for example, encoded as a prediction parameter in a stream.

(494) FIG. 41 is a diagram for illustrating one example of an MV derivation process for a current picture by normal merge mode.

(495) First, inter predictor 126 generates an MV candidate list in which MV candidates are registered. Examples of MV candidates include: spatially neighboring MV candidates which are MVs of a plurality of encoded blocks located spatially surrounding a current block; temporally neighboring MV candidates which are MVs of surrounding blocks on which the position of a current block in an encoded reference picture is projected; combined MV candidates which are MVs generated by combining the MV value of a spatially neighboring MV predictor and the MV value of a temporally neighboring MV predictor; and a zero MV candidate which is an MV having a zero value.

(496) Next, inter predictor 126 selects one MV candidate from a plurality of MV candidates registered in an MV candidate list, and determines the MV candidate as the MV of the current block.

(497) Furthermore, entropy encoder 110 writes and encodes, in a stream, merge_idx which is a signal indicating which MV candidate has been selected.

(498) It is to be noted that the MV candidates registered in the MV candidate list described in FIG. 41 are examples. The number of MV candidates may be different from the number of MV candidates in the diagram, the MV candidate list may be configured in such a manner that some of the kinds of the MV candidates in the diagram may not be included, or that one or more MV candidates other than the kinds of MV candidates in the diagram are included.

(499) A final MV may be determined by performing a dynamic motion vector refreshing (DMVR) to be described later using the MV of the current block derived by normal merge mode. It is to be noted that, in normal merge mode, no MV difference is encoded, but an MV difference is encoded. In MMVD mode, one MV candidate is selected from an MV candidate list as in the case of normal merge mode, an MV difference is encoded. As illustrated in FIG. 38B, MMVD may be categorized into merge modes together with normal merge mode. It is to be noted that the MV difference in MMVD mode does not always need to be the same as the MV difference for use in inter mode. For example, MV difference derivation in MMVD mode may be a process that requires a smaller amount of processing than the amount of processing required for MV difference derivation in inter mode.

(500) In addition, a combined inter merge/intra prediction (CIIP) mode may be performed. The mode is for overlapping a prediction image generated in inter prediction and a prediction image generated in intra prediction to generate a prediction image for a current block.

(501) It is to be noted that the MV candidate list may be referred to as a candidate list. In addition, merge_idx is MV selection information.

(502) [MV Derivation>HMVP Mode]

(503) FIG. 42 is a diagram for illustrating one example of an MV derivation process for a current picture by HMVP merge mode.

(504) In normal merge mode, an MV for, for example, a CU which is a current block is determined by selecting one MV candidate from an MV candidate list generated by referring to an encoded

block (for example, a CU). Here, another MV candidate may be registered in the MV candidate list. The mode in which such another MV candidate is registered is referred to as HMVP mode.

(505) In HMVP mode, MV candidates are managed using a first-in first-out (FIFO) buffer for HMVP, separately from the MV candidate list for normal merge mode.

(506) In FIFO buffer, motion information such as MVs of blocks processed in the past are stored newest first. In the management of the FIFO buffer, each time when one block is processed, the MV for the newest block (that is the CU processed immediately before) is stored in the FIFO buffer, and the MV of the oldest CU (that is, the CU processed earliest) is deleted from the FIFO buffer. In the example illustrated in FIG. 42, HMVP1 is the MV for the newest block, and HMVP5 is the MV for the oldest MV.

(507) Inter predictor 126 then, for example, checks whether each MV managed in the FIFO buffer is an MV different from all the MV candidates which have been already registered in the MV candidate list for normal merge mode starting from HMVP1. When determining that the MV is different from all the MV candidates, inter predictor 126 may add the MV managed in the FIFO buffer in the MV candidate list for normal merge mode as an MV candidate. At this time, the MV candidate registered from the FIFO buffer may be one or more.

(508) By using the HMVP mode in this way, it is possible to add not only the MV of a block which neighbors the current block spatially or temporally but also an MV for a block processed in the past. As a result, the variation of MV candidates for normal merge mode is expanded, which increases the probability that coding efficiency can be increased.

(509) It is to be noted that the MV may be motion information. In other words, information stored in the MV candidate list and the FIFO buffer may include not only MV values but also reference picture information, reference directions, the numbers of pictures, etc. In addition, the block is, for example, a CU.

(510) It is to be noted that the MV candidate list and the FIFO buffer illustrated in FIG. 42 are examples. The MV candidate list and FIFO buffer may be different in size from those in FIG. 42, or may be configured to register MV candidates in an order different from the one in FIG. 42. In addition, the process described here is common between encoder 100 and decoder 200.

(511) It is to be noted that the HMVP mode can be applied for modes other than the normal merge mode. For example, it is also excellent that motion information such as MVs of blocks processed in affine mode in the past may be stored newest first, and may be used as MV candidates. The mode obtained by applying HMVP mode to affine mode may be referred to as history affine mode.

(512) [MV Derivation>FRUC Mode]

(513) Motion information may be derived at the decoder 200 side without being signaled from the encoder 100 side. For example, motion information may be derived by performing motion estimation at the decoder 200 side. At this time, at the decoder 200 side, motion estimation is performed without using any pixel value in a current block. Modes in which motion estimation is performed at the decoder 200 side in this way include a frame rate up-conversion (FRUC) mode, a pattern matched motion vector derivation (PMMVD) mode, etc.

(514) One example of a FRUC process is illustrated in FIG. 43. First, a list which indicates, as MV candidates, MVs for encoded blocks each of which neighbors the current block spatially or temporally is generated by referring to the MVs (the list may be an MV candidate list, and be also used as the MV candidate list for normal merge mode) (Step Si_1). Next, a best MV candidate is selected from the plurality of MV candidates registered in the MV candidate list (Step Si_2). For example, the evaluation values of the respective MV candidates included in the MV candidate list are calculated, and one MV candidate is selected as the best MV candidate based on the evaluation values. Based on the selected best MV candidate, a motion vector for the current block is then derived (Step Si_4). More specifically, for example, the selected best MV candidate is directly derived as the MV for the current block. In addition, for example, the MV for the current block may be derived using pattern matching in a surrounding region of a position which is included in a

reference picture and corresponds to the selected best MV candidate. In other words, estimation using the pattern matching in a reference picture and the evaluation values may be performed in the surrounding region of the best MV candidate, and when there is an MV that yields a better evaluation value, the best MV candidate may be updated to the MV that yields the better evaluation value, and the updated MV may be determined as the final MV for the current block. Update to the MV that yields the better evaluation value may not be performed.

(515) Lastly, inter predictor **126** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Si_5). The processes in Steps Si_1 to Si_5 are executed, for example, on each block. For example, when the processes in Steps Si_1 to Si_5 are executed on each of all the blocks in the slice, inter prediction of the slice using the FRUC mode finishes. For example, when the processes in Steps Si_1 to Si_5 are executed on each of all the blocks in the picture, inter prediction of the picture using the FRUC mode finishes. It is to be noted that not all the blocks included in the slice may be subjected to the processes in Steps Si_1 to Si_5, and inter prediction of the slice using the FRUC mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the FRUC mode may finish when the processes in Steps Si_1 to Si_5 are executed on part of the blocks included in the picture.

(516) Each sub-block may be processed similarly to the above-described case of processing each block.

(517) Evaluation values may be calculated according to various kinds of methods. For example, a comparison is made between a reconstructed image in a region in a reference picture corresponding to an MV and a reconstructed image in a determined region (the region may be, for example, a region in another reference picture or a region in a neighboring block of a current picture, as indicated below). The difference between the pixel values of the two reconstructed images may be used for an evaluation value of the MV. It is to be noted that an evaluation value may be calculated using information other than the value of the difference.

(518) Next, pattern matching is described in detail. First, one MV candidate included in an MV candidate list (also referred to as a merge list) is selected as a starting point for estimation by pattern matching. As the pattern matching, either a first pattern matching or a second pattern matching may be used. The first pattern matching and the second pattern matching may be referred to as bilateral matching and template matching, respectively.

(519) [MV Derivation>FRUC>Bilateral Matching]

(520) In the first pattern matching, the pattern matching is performed between two blocks which are located along a motion trajectory of a current block and included in two different reference pictures. Accordingly, in the first pattern matching, a region in another reference picture located along the motion trajectory of the current block is used as a determined region for calculating the evaluation value of the above-described MV candidate.

(521) FIG. **44** is a diagram for illustrating one example of the first pattern matching (bilateral matching) between the two blocks in the two reference pictures located along the motion trajectory. As illustrated in FIG. **44**, in the first pattern matching, two motion vectors (MV0, MV1) are derived by estimating a pair which best matches among pairs of two blocks which are included in the two different reference pictures (Ref0, Ref1) and located along the motion trajectory of the current block (Cur block). More specifically, a difference between the reconstructed image at a specified position in the first encoded reference picture (Ref0) specified by an MV candidate and the reconstructed image at a specified position in the second encoded reference picture (Ref1) specified by a symmetrical MV obtained by scaling the MV candidate at a display time interval is derived for the current block, and an evaluation value is calculated using the value of the obtained difference. It is excellent to select, as the best MV, the MV candidate which yields the best evaluation value among the plurality of MV candidates.

(522) In the assumption of a continuous motion trajectory, the motion vectors (MV0, MV1)

specifying the two reference blocks are proportional to temporal distances (TD0, TD1) between the current picture (Cur Pic) and the two reference pictures (Ref0, Ref1). For example, when the current picture is temporally located between the two reference pictures and the temporal distances from the current picture to the respective two reference pictures are equal to each other, mirror-symmetrical bi-directional MVs are derived in the first pattern matching.

(523) [MV Derivation>FRUC>Template Matching]

(524) In the second pattern matching (template matching), pattern matching is performed between a block in a reference picture and a template in the current picture (the template is a block neighboring the current block in the current picture (the neighboring block is, for example, an upper and/or left neighboring block(s))). Accordingly, in the second pattern matching, the block neighboring the current block in the current picture is used as the determined region for calculating the evaluation value of the above-described MV candidate.

(525) FIG. 45 is a diagram for illustrating one example of pattern matching (template matching) between a template in a current picture and a block in a reference picture. As illustrated in FIG. 45, in the second pattern matching, the MV for the current block (Cur block) is derived by estimating, in the reference picture (Ref0), the block which best matches the block neighboring the current block in the current picture (Cur Pic). More specifically, the difference between a reconstructed image in an encoded region which neighbors both left and above or either left or above and a reconstructed image which is in a corresponding region in the encoded reference picture (Ref0) and is specified by an MV candidate is derived, and an evaluation value is calculated using the value of the obtained difference. It is excellent to select, as the best MV candidate, the MV candidate which yields the best evaluation value among the plurality of MV candidates.

(526) Such information indicating whether to apply the FRUC mode (referred to as, for example, a FRUC flag) may be signaled at the CU level. In addition, when the FRUC mode is applied (for example, when a FRUC flag is true), information indicating an applicable pattern matching method (either the first pattern matching or the second pattern matching) may be signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, CTU level, or sub-block level).

(527) [MV Derivation>Affine Mode]

(528) The affine mode is a mode for generating an MV using affine transform. For example, an MV may be derived in units of a sub-block based on motion vectors of a plurality of neighboring blocks. This mode is also referred to as an affine motion compensation prediction mode.

(529) FIG. 46A is a diagram for illustrating one example of MV derivation in units of a sub-block based on MVs of a plurality of neighboring blocks. In FIG. 46A, the current block includes sixteen 4×4 pixel sub-blocks. Here, motion vector v.sub.0 at an upper-left corner control point in the current block is derived based on an MV of a neighboring block, and likewise, motion vector v.sub.1 at an upper-right corner control point in the current block is derived based on an MV of a neighboring sub-block. Two motion vectors v.sub.0 and v.sub.1 are projected according to an expression (1A) indicated below, and motion vectors (v.sub.x, v.sub.y) for the respective sub-blocks in the current block are derived.

[MATH . 1]

$$(530) \quad \begin{cases} v_x = \frac{(v_{1x} - v_{0x})}{w}x - \frac{(v_{1y} - v_{0y})}{w}y + v_{0x} \\ v_y = \frac{(v_{1y} - v_{0y})}{w}x + \frac{(v_{1x} - v_{0x})}{w}y + v_{0y} \end{cases} \quad (1A)$$

(531) Here, x and y indicate the horizontal position and the vertical position of the sub-block, respectively, and w indicates a predetermined weighting coefficient.

(532) Such information indicating the affine mode (for example, referred to as an affine flag) may be signaled at the CU level. It is to be noted that the signaling of such information does not

necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, CTU level, or sub-block level). (533) In addition, the affine mode may include several modes for different methods for deriving MVs at the upper-left and upper-right corner control points. For example, the affine modes include two modes which are the affine inter mode (also referred to as an affine normal inter mode) and the affine merge mode.

(534) FIG. 46B is a diagram for illustrating one example of MV derivation in units of a sub-block in affine mode in which three control points are used. In FIG. 46B, the current block includes, for example, sixteen 4×4 pixel sub-blocks. Here, motion vector v.sub.0 at an upper-left corner control point in the current block is derived based on an MV of a neighboring block. Here, motion vector v.sub.1 at an upper-right corner control point in the current block is derived based on an MV of a neighboring block, and likewise, motion vector v.sub.2 at a lower-left corner control point for the current block is derived based on an MV of a neighboring block. Three motion vectors v.sub.0, v.sub.1, and v.sub.2 are projected according to an expression (1B) indicated below, and motion vectors (v.sub.x, v.sub.y) for the respective sub-blocks in the current block are derived.

[MATH . 2]

$$(535) \quad \begin{cases} v_x = \frac{(v_{1x} - v_{0x})}{w}x + \frac{(v_{2x} - v_{0x})}{h}y + v_{0x} \\ v_y = \frac{(v_{1y} - v_{0y})}{w}x + \frac{(v_{2y} - v_{0y})}{h}y + v_{0y} \end{cases} \quad (1B)$$

(536) Here, x and y indicate the horizontal position and the vertical position of the sub-block, respectively, and each of w and h indicates a predetermined weighting coefficient. Here, w may indicate the width of a current block, and h may indicate the height of the current block.

(537) Affine modes in which different numbers of control points (for example, two and three control points) are used may be switched and signaled at the CU level. It is to be noted that information indicating the number of control points in affine mode used at the CU level may be signaled at another level (for example, the sequence level, picture level, slice level, brick level, CTU level, or sub-block level).

(538) In addition, such an affine mode in which three control points are used may include different methods for deriving MVs at the upper-left, upper-right, and lower-left corner control points. For example, the affine modes in which three control points are used include two modes which are affine inter mode and affine merge mode, as in the case of affine modes in which two control points are used.

(539) It is to be noted that, in the affine modes, the size of each sub-block included in the current block may not be limited to 4×4 pixels, and may be another size. For example, the size of each sub-block may be 8×8 pixels.

(540) [MV Derivation>Affine Mode>Control Point]

(541) FIGS. 47A, 47B, and 47C are each a conceptual diagram for illustrating one example of MV derivation at control points in an affine mode.

(542) As illustrated in FIG. 47A, in the affine mode, for example, MV predictors at respective control points for a current block are calculated based on a plurality of MVs corresponding to blocks encoded according to the affine mode among encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) which neighbor the current block. More specifically, encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) are checked in the listed order, and the first effective block encoded according to the affine mode is identified. The MV at each control point for the current block is calculated based on the plurality of MVs corresponding to the identified block.

(543) For example, as illustrated in FIG. 47B, when block A which neighbors to the left of the current block has been encoded according to an affine mode in which two control points are used, motion vectors v.sub.3 and v.sub.4 projected at the upper-left corner position and the upper-right

corner position of the encoded block including block A are derived. Motion vector $v_{\text{sub}.0}$ at the upper-left control point and motion vector $v_{\text{sub}.1}$ at the upper-right control point for the current block are then calculated from derived motion vectors $v_{\text{sub}.3}$ and $v_{\text{sub}.4}$.

(544) For example, as illustrated in FIG. 47C, when block A which neighbors to the left of the current block has been encoded according to an affine mode in which three control points are used, motion vectors $v_{\text{sub}.3}$, $v_{\text{sub}.4}$, and $v_{\text{sub}.5}$ projected at the upper-left corner position, the upper-right corner position, and the lower-left corner position of the encoded block including block A are derived. Motion vector $v_{\text{sub}.0}$ at the upper-left control point for the current block, motion vector $v_{\text{sub}.1}$ at the upper-right control point for the current block, and motion vector v_2 at the lower-left control point for the current block are then calculated from derived motion vectors $v_{\text{sub}.3}$, $v_{\text{sub}.4}$, and $v_{\text{sub}.5}$.

(545) The MV derivation methods illustrated in FIGS. 47A to 47C may be used in the MV derivation at each control point for the current block in Step S_{k_1} illustrated in FIG. 50 described later, or may be used for MV predictor derivation at each control point for the current block in Step S_{j_1} illustrated in FIG. 51 described later.

(546) FIGS. 48A and 48B are each a conceptual diagram for illustrating another example of MV derivation at control points in affine mode.

(547) FIG. 48A is a diagram for illustrating an affine mode in which two control points are used.

(548) In the affine mode, as illustrated in FIG. 48A, an MV selected from MVs at encoded block A, block B, and block C which neighbor the current block is used as motion vector $v_{\text{sub}.0}$ at the upper-left corner control point for the current block. Likewise, an MV selected from MVs of encoded block D and block E which neighbor the current block is used as motion vector $v_{\text{sub}.1}$ at the upper-right corner control point for the current block.

(549) FIG. 48B is a diagram for illustrating an affine mode in which three control points are used.

(550) In the affine mode, as illustrated in FIG. 48B, an MV selected from MVs at encoded block A, block B, and block C which neighbor the current block is used as motion vector $v_{\text{sub}.0}$ at the upper-left corner control point for the current block. Likewise, an MV selected from MVs of encoded block D and block E which neighbor the current block is used as motion vector $v_{\text{sub}.1}$ at the upper-right corner control point for the current block. Furthermore, an MV selected from MVs of encoded block F and block G which neighbor the current block is used as motion vector $v_{\text{sub}.2}$ at the lower-left corner control point for the current block.

(551) It is to be noted that the MV derivation methods illustrated in FIGS. 48A and 48B may be used in the MV derivation at each control point for the current block in Step S_{k_1} illustrated in FIG. 50 described later, or may be used for MV predictor derivation at each control point for the current block in Step S_{j_1} illustrated in FIG. 51 described later.

(552) Here, when affine modes in which different numbers of control points (for example, two and three control points) are used may be switched and signaled at the CU level, the number of control points for an encoded block and the number of control points for a current block may be different from each other.

(553) FIGS. 49A and 49B are each a conceptual diagram for illustrating one example of a method for MV derivation at control points when the number of control points for an encoded block and the number of control points for a current block are different from each other.

(554) For example, as illustrated in FIG. 49A, a current block has three control points at the upper-left corner, the upper-right corner, and the lower-left corner, and block A which neighbors to the left of the current block has been encoded according to an affine mode in which two control points are used. In this case, motion vectors $v_{\text{sub}.3}$ and $v_{\text{sub}.4}$ projected at the upper-left corner position and the upper-right corner position in the encoded block including block A are derived. Motion vector $v_{\text{sub}.0}$ at the upper-left corner control point and motion vector $v_{\text{sub}.1}$ at the upper-right corner control point for the current block are then calculated from derived motion vectors $v_{\text{sub}.3}$ and $v_{\text{sub}.4}$. Furthermore, motion vector $v_{\text{sub}.2}$ at the lower-left corner control point is calculated from

derived motion vectors v.sub.0 and v.sub.1.

(555) For example, as illustrated in FIG. 49B, a current block has two control points at the upper-left corner and the upper-right corner, and block A which neighbors to the left of the current block has been encoded according to an affine mode in which three control points are used. In this case, motion vectors v.sub.3, v.sub.4, and v.sub.5 projected at the upper-left corner position in the encoded block including block A, the upper-right corner position in the encoded block, and the lower-left corner position in the encoded block are derived. Motion vector v.sub.0 at the upper-left corner control point for the current block and motion vector v.sub.1 at the upper-right corner control point for the current block are then calculated from derived motion vectors v.sub.3, v.sub.4, and v.sub.5.

(556) It is to be noted that the MV derivation methods illustrated in FIGS. 49A and 49B may be used in the MV derivation at each control point for the current block in Step Sk_1 illustrated in FIG. 50 described later, or may be used for MV predictor derivation at each control point for the current block in Step Sj_1 illustrated in FIG. 51 described later.

(557) [MV Derivation>Affine Mode>Affine Merge Mode]

(558) FIG. 50 is a flow chart illustrating one example of the affine merge mode.

(559) In the affine merge mode, first, inter predictor 126 derives MVs at respective control points for a current block (Step Sk_1). The control points are an upper-left corner point of the current block and an upper-right corner point of the current block as illustrated in FIG. 46A, or an upper-left corner point of the current block, an upper-right corner point of the current block, and a lower-left corner point of the current block as illustrated in FIG. 46B. At this time, inter predictor 126 may encode MV selection information for identifying two or three derived MVs in a stream.

(560) For example, when MV derivation methods illustrated in FIGS. 47A to 47C are used, as illustrated in FIG. 47A, inter predictor 126 checks encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) in the listed order, and identifies the first effective block encoded according to the affine mode.

(561) Inter predictor 126 derives the MV at the control point using the identified first effective block encoded according to the identified affine mode. For example, when block A is identified and block A has two control points, as illustrated in FIG. 47B, inter predictor 126 calculates motion vector v.sub.0 at the upper-left corner control point of the current block and motion vector v.sub.1 at the upper-right corner control point of the current block from motion vectors v 3 and v.sub.4 at the upper-left corner of the encoded block including block A and the upper-right corner of the encoded block. For example, inter predictor 126 calculates motion vector v.sub.0 at the upper-left corner control point of the current block and motion vector v.sub.1 at the upper-right corner control point of the current block by projecting motion vectors v.sub.3 and v.sub.4 at the upper-left corner and the upper-right corner of the encoded block onto the current block.

(562) Alternatively, when block A is identified and block A has three control points, as illustrated in FIG. 47C, inter predictor 126 calculates motion vector v.sub.0 at the upper-left corner control point of the current block, motion vector v.sub.1 at the upper-right corner control point of the current block, and motion vector v.sub.2 at the lower-left corner control point of the current block from motion vectors v.sub.3, v.sub.4, and v.sub.5 at the upper-left corner of the encoded block including block A, the upper-right corner of the encoded block, and the lower-left corner of the encoded block. For example, inter predictor 126 calculates motion vector v.sub.0 at the upper-left corner control point of the current block, motion vector v.sub.1 at the upper-right corner control point of the current block, and motion vector v.sub.2 at the lower-left corner control point of the current block by projecting motion vectors v.sub.3, v.sub.4, and v.sub.5 at the upper-left corner, the upper-right corner, and the lower-left corner of the encoded block onto the current block.

(563) It is to be noted that, as illustrated in FIG. 49A described above, MVs at three control points may be calculated when block A is identified and block A has two control points, and that, as illustrated in FIG. 49B described above, MVs at two control points may be calculated when block

A is identified and block A has three control points.

(564) Next, inter predictor **126** performs motion compensation of each of a plurality of sub-blocks included in the current block. In other words, inter predictor **126** calculates an MV for each of the plurality of sub-blocks as an affine MV, using either two motion vectors $v_{sub.0}$ and $v_{sub.1}$ and the above expression (1A) or three motion vectors $v_{sub.0}$, $v_{sub.1}$, and v_2 and the above expression (1B) (Step Sk_2). Inter predictor **126** then performs motion compensation of the sub-blocks using these affine MVs and encoded reference pictures (Step Sk_3). When the processes in Steps Sk_2 and Sk_3 are executed for each of all the sub-blocks included in the current block, the process for generating a prediction image using the affine merge mode for the current block finishes. In other words, motion compensation of the current block is performed to generate a prediction image of the current block.

(565) It is to be noted that the above-described MV candidate list may be generated in Step Sk_1. The MV candidate list may be, for example, a list including MV candidates derived using a plurality of MV derivation methods for each control point. The plurality of MV derivation methods may be any combination of the MV derivation methods illustrated in FIGS. 47A to 47C, the MV derivation methods illustrated in FIGS. 48A and 48B, the MV derivation methods illustrated in FIGS. 49A and 49B, and other MV derivation methods.

(566) It is to be noted that MV candidate lists may include MV candidates in a mode in which prediction is performed in units of a sub-block, other than the affine mode.

(567) It is to be noted that, for example, an MV candidate list including MV candidates in an affine merge mode in which two control points are used and an affine merge mode in which three control points are used may be generated as an MV candidate list. Alternatively, an MV candidate list including MV candidates in the affine merge mode in which two control points are used and an MV candidate list including MV candidates in the affine merge mode in which three control points are used may be generated separately. Alternatively, an MV candidate list including MV candidates in one of the affine merge mode in which two control points are used and the affine merge mode in which three control points are used may be generated. The MV candidate(s) may be, for example, MVs for encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left), or an MV for an effective block among the blocks.

(568) It is to be noted that index indicating one of the MVs in an MV candidate list may be transmitted as MV selection information.

(569) [MV Derivation>Affine Mode>Affine Inter Mode]

(570) FIG. 51 is a flow chart illustrating one example of an affine inter mode. In the affine inter mode, first, inter predictor **126** derives MV predictors ($v_{sub.0}$, $v_{sub.1}$) or ($v_{sub.0}$, $v_{sub.1}$, $v_{sub.2}$) of respective two or three control points for a current block (Step Sj_1). The control points are an upper-left corner point for the current block, an upper-right corner point of the current block, and a lower-left corner point for the current block as illustrated in FIG. 46A or FIG. 46B.

(571) For example, when the MV derivation methods illustrated in FIGS. 48A and 48B are used, inter predictor **126** derives the MV predictors ($v_{sub.0}$, $v_{sub.1}$) or ($v_{sub.0}$, $v_{sub.1}$, $v_{sub.2}$) at respective two or three control points for the current block by selecting MVs of any of the blocks among encoded blocks in the vicinity of the respective control points for the current block illustrated in either FIG. 48A or FIG. 48B. At this time, inter predictor **126** encodes, in a stream, MV predictor selection information for identifying the selected two or three MV predictors.

(572) For example, inter predictor **126** may determine, using a cost evaluation or the like, the block from which an MV as an MV predictor at a control point is selected from among encoded blocks neighboring the current block, and may write, in a bitstream, a flag indicating which MV predictor has been selected. In other words, inter predictor **126** outputs, as a prediction parameter, the MV predictor selection information such as a flag to entropy encoder **110** through prediction parameter generator **130**.

(573) Next, inter predictor **126** performs motion estimation (Steps Sj_3 and Sj_4) while updating

the MV predictor selected or derived in Step Sj_1 (Step Sj_2). In other words, inter predictor **126** calculates, as an affine MV, an MV of each of sub-blocks which corresponds to an updated MV predictor, using either the expression (1A) or expression (1B) described above (Step Sj_3). Inter predictor **126** then performs motion compensation of the sub-blocks using these affine MVs and encoded reference pictures (Step Sj_4). The processes in Steps Sj_3 and Sj_4 are executed on all the blocks in the current block each time an MV predictor is updated in Step Sj_2. As a result, for example, inter predictor **126** determines the MV predictor which yields the smallest cost as the MV at a control point in a motion estimation loop (Step Sj_5). At this time, inter predictor **126** further encodes, in the stream, the difference value between the determined MV and the MV predictor as an MV difference. In other words, inter predictor **126** outputs the MV difference as a prediction parameter to entropy encoder **110** through prediction parameter generator **130**.

(574) Lastly, inter predictor **126** generates a prediction image for the current block by performing motion compensation of the current block using the determined MV and the encoded reference picture (Step Sj_6).

(575) It is to be noted that the above-described MV candidate list may be generated in Step Sj_1. The MV candidate list may be, for example, a list including MV candidates derived using a plurality of MV derivation methods for each control point. The plurality of MV derivation methods may be any combination of the MV derivation methods illustrated in FIGS. 47A to 47C, the MV derivation methods illustrated in FIGS. 48A and 48B, the MV derivation methods illustrated in FIGS. 49A and 49B, and other MV derivation methods.

(576) It is to be noted that the MV candidate list may include MV candidates in a mode in which prediction is performed in units of a sub-block, other than the affine mode.

(577) It is to be noted that, for example, an MV candidate list including MV candidates in an affine inter mode in which two control points are used and an affine inter mode in which three control points are used may be generated as an MV candidate list. Alternatively, an MV candidate list including MV candidates in the affine inter mode in which two control points are used and an MV candidate list including MV candidates in the affine inter mode in which three control points are used may be generated separately. Alternatively, an MV candidate list including MV candidates in one of the affine inter mode in which two control points are used and the affine inter mode in which three control points are used may be generated. The MV candidate(s) may be, for example, MVs for encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left), or an MV for an effective block among the blocks.

(578) It is to be noted that index indicating one of the MV candidates in an MV candidate list may be transmitted as MV predictor selection information.

(579) [MV Derivation>Triangle Mode]

(580) Inter predictor **126** generates one rectangular prediction image for a rectangular current block in the above example. However, inter predictor **126** may generate a plurality of prediction images each having a shape different from a rectangle for the rectangular current block, and may combine the plurality of prediction images to generate the final rectangular prediction image. The shape different from a rectangle may be, for example, a triangle.

(581) FIG. 52A is a diagram for illustrating generation of two triangular prediction images.

(582) Inter predictor **126** generates a triangular prediction image by performing motion compensation of a first partition having a triangular shape in a current block by using a first MV of the first partition, to generate a triangular prediction image. Likewise, inter predictor **126** generates a triangular prediction image by performing motion compensation of a second partition having a triangular shape in a current block by using a second MV of the second partition, to generate a triangular prediction image. Inter predictor **126** then generates a prediction image having the same rectangular shape as the rectangular shape of the current block by combining these prediction images.

(583) It is to be noted that a first prediction image having a rectangular shape corresponding to a

current block may be generated as a prediction image for a first partition, using a first MV. In addition, a second prediction image having a rectangular shape corresponding to a current block may be generated as a prediction image for a second partition, using a second MV. A prediction image for the current block may be generated by performing a weighted addition of the first prediction image and the second prediction image. It is to be noted that the part which is subjected to the weighted addition may be a partial region across the boundary between the first partition and the second partition.

(584) FIG. 52B is a conceptual diagram for illustrating examples of a first portion of a first partition which overlaps with a second partition, and first and second sets of samples which may be weighted as part of a correction process. The first portion may be, for example, one fourth of the width or height of the first partition. In another example, the first portion may have a width corresponding to N samples adjacent to an edge of the first partition, where N is an integer greater than zero, and N may be, for example, the integer 2. As illustrated, the left example of FIG. 52B shows a rectangular partition having a rectangular portion with a width which is one fourth of the width of the first partition, with the first set of samples including samples outside of the first portion and samples inside of the first portion, and the second set of samples including samples within the first portion. The center example of FIG. 52B shows a rectangular partition having a rectangular portion with a height which is one fourth of the height of the first partition, with the first set of samples including samples outside of the first portion and samples inside of the first portion, and the second set of samples including samples within the first portion. The right example of FIG. 52B shows a triangular partition having a polygonal portion with a height which corresponds to two samples, with the first set of samples including samples outside of the first portion and samples inside of the first portion, and the second set of samples including samples within the first portion.

(585) The first portion may be a portion of the first partition which overlaps with an adjacent partition. FIG. 52C is a conceptual diagram for illustrating a first portion of a first partition, which is a portion of the first partition that overlaps with a portion of an adjacent partition. For ease of illustration, a rectangular partition having an overlapping portion with a spatially adjacent rectangular partition is shown. Partitions having other shapes, such as triangular partitions, may be employed, and the overlapping portions may overlap with a spatially or temporally adjacent partition.

(586) In addition, although an example is given in which a prediction image is generated for each of two partitions using inter prediction, a prediction image may be generated for at least one partition using intra prediction.

(587) FIG. 53 is a flow chart illustrating one example of a triangle mode.

(588) In the triangle mode, first, inter predictor **126** splits the current block into the first partition and the second partition (Step Sx_1). At this time, inter predictor **126** may encode, in a stream, partition information which is information related to the splitting into the partitions as a prediction parameter. In other words, inter predictor **126** may output the partition information as the prediction parameter to entropy encoder **110** through prediction parameter generator **130**.

(589) First, inter predictor **126** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of encoded blocks temporally or spatially surrounding the current block (Step Sx_2). In other words, inter predictor **126** generates an MV candidate list.

(590) Inter predictor **126** then selects the MV candidate for the first partition and the MV candidate for the second partition as a first MV and a second MV, respectively, from the plurality of MV candidates obtained in Step Sx_2 (Step Sx_3). At this time, inter predictor **126** encodes, in a stream, MV selection information for identifying the selected MV candidate, as a prediction parameter. In other words, inter predictor **126** outputs the MV selection information as a prediction parameter to entropy encoder **110** through prediction parameter generator **130**.

(591) Next, inter predictor **126** generates a first prediction image by performing motion

compensation using the selected first MV and an encoded reference picture (Step Sx_4). Likewise, inter predictor **126** generates a second prediction image by performing motion compensation using the selected second MV and an encoded reference picture (Step Sx_5).

(592) Lastly, inter predictor **126** generates a prediction image for the current block by performing a weighted addition of the first prediction image and the second prediction image (Step Sx_6).

(593) It is to be noted that, although the first partition and the second partition are triangles in the example illustrated in FIG. 52A, the first partition and the second partition may be trapezoids, or other shapes different from each other. Furthermore, although the current block includes two partitions in the example illustrated in FIG. 52A, the current block may include three or more partitions.

(594) In addition, the first partition and the second partition may overlap with each other. In other words, the first partition and the second partition may include the same pixel region. In this case, a prediction image for a current block may be generated using a prediction image in the first partition and a prediction image in the second partition.

(595) In addition, although the example in which the prediction image is generated for each of the two partitions using inter prediction has been illustrated, a prediction image may be generated for at least one partition using intra prediction.

(596) It is to be noted that the MV candidate list for selecting the first MV and the MV candidate list for selecting the second MV may be different from each other, or the MV candidate list for selecting the first MV may be also used as the MV candidate list for selecting the second MV.

(597) It is to be noted that partition information may include an index indicating the splitting direction in which at least a current block is split into a plurality of partitions. The MV selection information may include an index indicating the selected first MV and an index indicating the selected second MV. One index may indicate a plurality of pieces of information. For example, one index collectively indicating a part or the entirety of partition information and a part or the entirety of MV selection information may be encoded.

(598) [MV Derivation>ATMVP Mode]

(599) FIG. 54 is a diagram illustrating one example of an ATMVP mode in which an MV is derived in units of a sub-block.

(600) The ATMVP mode is a mode categorized into the merge mode. For example, in the ATMVP mode, an MV candidate for each sub-block is registered in an MV candidate list for use in normal merge mode.

(601) More specifically, in the ATMVP mode, first, as illustrated in FIG. 54, a temporal MV reference block associated with a current block is identified in an encoded reference picture specified by an MV (MV0) of a neighboring block located at the lower-left position with respect to the current block. Next, in each sub-block in the current block, the MV used to encode the region corresponding to the sub-block in the temporal MV reference block is identified. The MV identified in this way is included in an MV candidate list as an MV candidate for the sub-block in the current block. When the MV candidate for each sub-block is selected from the MV candidate list, the sub-block is subjected to motion compensation in which the MV candidate is used as the MV for the sub-block. In this way, a prediction image for each sub-block is generated.

(602) Although the block located at the lower-left position with respect to the current block is used as a surrounding MV reference block in the example illustrated in FIG. 54, it is to be noted that another block may be used. In addition, the size of the sub-block may be 4×4 pixels, 8×8 pixels, or another size. The size of the sub-block may be switched for a unit such as a slice, brick, picture, etc.

(603) [Motion Estimation>DMVR]

(604) FIG. 55 is a diagram illustrating a relationship between a merge mode and DMVR.

(605) Inter predictor **126** derives an MV for a current block according to the merge mode (Step Sl_1). Next, inter predictor **126** determines whether to perform estimation of an MV that is motion

estimation (Step Sl_2). Here, when determining not to perform motion estimation (No in Step Sl_2), inter predictor **126** determines the MV derived in Step Sl_1 as the final MV for the current block (Step Sl_4). In other words, in this case, the MV for the current block is determined according to the merge mode.

(606) When determining to perform motion estimation in Step Sl_1 (Yes in Step Sl_2), inter predictor **126** derives the final MV for the current block by estimating a surrounding region of the reference picture specified by the MV derived in Step Sl_1 (Step Sl_3). In other words, in this case, the MV for the current block is determined according to the DMVR.

(607) FIG. 56 is a conceptual diagram for illustrating another example of DMVR for determining an MV.

(608) First, in the merge mode for example, MV candidates (L0 and L1) are selected for the current block. A reference pixel is identified from a first reference picture (L0) which is an encoded picture in the L0 list according to the MV candidate (L0). Likewise, a reference pixel is identified from a second reference picture (L1) which is an encoded picture in the L1 list according to the MV candidate (L1). A template is generated by calculating an average of these reference pixels.

(609) Next, each of the surrounding regions of MV candidates of the first reference picture (L0) and the second reference picture (L1) are estimated using the template, and the MV which yields the smallest cost is determined to be the final MV. It is to be noted that the cost may be calculated, for example, using a difference value between each of the pixel values in the template and a corresponding one of the pixel values in the estimation region, the values of MV candidates, etc.

(610) Exactly the same processes described here do not always need to be performed. Any process for enabling derivation of the final MV by estimation in surrounding regions of MV candidates may be used.

(611) FIG. 57 is a conceptual diagram for illustrating another example of DMVR for determining an MV. Unlike the example of DMVR illustrated in FIG. 56, in the example illustrated in FIG. 57, costs are calculated without generating any template.

(612) First, inter predictor **126** estimates a surrounding region of a reference block included in each of reference pictures in the L0 list and L1 list, based on an initial MV which is an MV candidate obtained from each MV candidate list. For example, as illustrated in FIG. 57, the initial MV corresponding to the reference block in the L0 list is InitMV_L0, and the initial MV corresponding to the reference block in the L1 list is InitMV_L1. In motion estimation, inter predictor **126** firstly sets a search position for the reference picture in the L0 list. Based on the position indicated by the vector difference indicating the search position to be set, specifically, the initial MV (that is, InitMV_L0), the vector difference to the search position is MVd_L0. Inter predictor **126** then determines the estimation position in the reference picture in the L1 list. This search position is indicated by the vector difference to the search position from the position indicated by the initial MV (that is, InitMV_L1). More specifically, inter predictor **126** determines the vector difference as MVd_L1 by mirroring of MVd_L0. In other words, inter predictor **126** determines the position which is symmetrical with respect to the position indicated by the initial MV to be the search position in each reference picture in the L0 list and the L1 list. Inter predictor **126** calculates, for each search position, the total sum of the absolute differences (SADs) between values of pixels at search positions in blocks as a cost, and finds out the search position that yields the smallest cost.

(613) FIG. 58A is a diagram illustrating one example of motion estimation in DMVR, and FIG. 58B is a flow chart illustrating one example of the motion estimation.

(614) First, in Step 1, inter predictor **126** calculates the cost between the search position (also referred to as a starting point) indicated by the initial MV and eight surrounding search positions. Inter predictor **126** then determines whether the cost at each of the search positions other than the starting point is the smallest. Here, when determining that the cost at the search position other than the starting point is the smallest, inter predictor **126** changes a target to the search position at which the smallest cost is obtained, and performs the process in Step 2. When the cost at the starting point

is the smallest, inter predictor **126** skips the process in Step 2 and performs the process in Step 3.
(615) In Step 2, inter predictor **126** performs the search similar to the process in Step 1, regarding, as a new starting point, the search position after the target change according to the result of the process in Step 1. Inter predictor **126** then determines whether the cost at each of the search positions other than the starting point is the smallest. Here, when determining that the cost at the search position other than the starting point is the smallest, inter predictor **126** performs the process in Step 4. When the cost at the starting point is the smallest, inter predictor **126** performs the process in Step 3.

(616) In Step 4, inter predictor **126** regards the search position at the starting point as the final search position, and determines the difference between the position indicated by the initial MV and the final search position to be a vector difference.

(617) In Step 3, inter predictor **126** determines the pixel position at sub-pixel accuracy at which the smallest cost is obtained, based on the costs at the four points located at upper, lower, left, and right positions with respect to the starting point in Step 1 or Step 2, and regards the pixel position as the final search position. The pixel position at the sub-pixel accuracy is determined by performing weighted addition of each of the four upper, lower, left, and right vectors ((0, 1), (0, -1), (-1, 0), and (1, 0)), using, as a weight, the cost at a corresponding one of the four search positions. Inter predictor **126** then determines the difference between the position indicated by the initial MV and the final search position to be the vector difference.

(618) [Motion Compensation>BIO/OBMC/LIC]

(619) Motion compensation involves a mode for generating a prediction image, and correcting the prediction image. The mode is, for example, BIO, OBMC, and LIC to be described later.

(620) FIG. 59 is a flow chart illustrating one example of generation of a prediction image.

(621) Inter predictor **126** generates a prediction image (Step Sm_1), and corrects the prediction image according to any of the modes described above (Step Sm_2).

(622) FIG. 60 is a flow chart illustrating another example of generation of a prediction image.

(623) Inter predictor **126** derives an MV of a current block (Step Sn_1). Next, inter predictor **126** generates a prediction image using the MV (Step Sn_2), and determines whether to perform a correction process (Step Sn_3). Here, when determining to perform a correction process (Yes in Step Sn_3), inter predictor **126** generates the final prediction image by correcting the prediction image (Step Sn_4). It is to be noted that, in LIC described later, luminance and chrominance may be corrected in Step Sn_4. When determining not to perform a correction process (No in Step Sn_3), inter predictor **126** outputs the prediction image as the final prediction image without correcting the prediction image (Step Sn_5).

(624) [Motion Compensation>OBMC]

(625) It is to be noted that an inter prediction image may be generated using motion information for a neighboring block in addition to motion information for the current block obtained by motion estimation. More specifically, an inter prediction image may be generated for each sub-block in a current block by performing weighted addition of a prediction image based on the motion information obtained by motion estimation (in a reference picture) and a prediction image based on the motion information of the neighboring block (in the current picture). Such inter prediction (motion compensation) is also referred to as overlapped block motion compensation (OBMC) or an OBMC mode.

(626) In OBMC mode, information indicating a sub-block size for OBMC (referred to as, for example, an OBMC block size) may be signaled at the sequence level. Moreover, information indicating whether to apply the OBMC mode (referred to as, for example, an OBMC flag) may be signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the sequence level and CU level, and may be performed at another level (for example, at the picture level, slice level, brick level, CTU level, or sub-block level).

(627) The OBMC mode will be described in further detail. FIGS. 61 and 62 are a flow chart and a

conceptual diagram for illustrating an outline of a prediction image correction process performed by OBMC.

(628) First, as illustrated in FIG. 62, a prediction image (Pred) by normal motion compensation is obtained using an MV assigned to a current block. In FIG. 62, the arrow “MV” points a reference picture, and indicates what the current block of the current picture refers to in order to obtain the prediction image.

(629) Next, a prediction image (Pred_L) is obtained by applying a motion vector (MV_L) which has been already derived for the encoded block neighboring to the left of the current block to the current block (re-using the motion vector for the current block). The motion vector (MV_L) is indicated by an arrow “MV_L” indicating a reference picture from a current block. A first correction of a prediction image is performed by overlapping two prediction images Pred and Pred_L. This provides an effect of blending the boundary between neighboring blocks.

(630) Likewise, a prediction image (Pred_U) is obtained by applying an MV (MV_U) which has been already derived for the encoded block neighboring above the current block to the current block (re-using the MV for the current block). The MV (MV_U) is indicated by an arrow “MV_U” indicating a reference picture from a current block. A second correction of a prediction image is performed by overlapping the prediction image Pred_U to the prediction images (for example, Pred and Pred_L) on which the first correction has been performed. This provides an effect of blending the boundary between neighboring blocks. The prediction image obtained by the second correction is the one in which the boundary between the neighboring blocks has been blended (smoothed), and thus is the final prediction image of the current block.

(631) Although the above example is a two-path correction method using left and upper neighboring blocks, it is to be noted that the correction method may be three- or more-path correction method using also the right neighboring block and/or the lower neighboring block.

(632) It is to be noted that the region in which such overlapping is performed may be only part of a region near a block boundary instead of the pixel region of the entire block.

(633) It is to be noted that the prediction image correction process according to OBMC for obtaining one prediction image Pred from one reference picture by overlapping additional prediction images Pred_L and Pred_U has been described above. However, when a prediction image is corrected based on a plurality of reference images, a similar process may be applied to each of the plurality of reference pictures. In such a case, after corrected prediction images are obtained from the respective reference pictures by performing OBMC image correction based on the plurality of reference pictures, the obtained corrected prediction images are further overlapped to obtain the final prediction image.

(634) It is to be noted that, in OBMC, a current block unit may be a PU or a sub-block unit obtained by further splitting the PU.

(635) One example of a method for determining whether to apply OBMC is a method for using an obmc_flag which is a signal indicating whether to apply OBMC. As one specific example, encoder **100** may determine whether the current block belongs to a region having complicated motion. Encoder **100** sets the obmc_flag to a value of “1” when the block belongs to a region having complicated motion and applies OBMC when encoding, and sets the obmc_flag to a value of “0” when the block does not belong to a region having complicated motion and encodes the block without applying OBMC. Decoder **200** switches between application and non-application of OBMC by decoding the obmc_flag written in a stream.

(636) [Motion Compensation>BIO]

(637) Next, an MV derivation method is described. First, a mode for deriving an MV based on a model assuming uniform linear motion is described. This mode is also referred to as a bi-directional optical flow (BIO) mode. In addition, this bi-directional optical flow may be written as BDOF instead of BIO.

(638) FIG. 63 is a diagram for illustrating a model assuming uniform linear motion. In FIG. 63,

(vx, vy) indicates a velocity vector, and τ_0 and τ_1 indicate temporal distances between a current picture (Cur Pic) and two reference pictures (Ref.sub.0, Ref.sub.1). (MVx.sub.0, MVy.sub.0) indicates an MV corresponding to reference picture Ref.sub.0, and (MVx.sub.1, MVy.sub.1) indicates an MV corresponding to reference picture Ref.sub.1.

(639) Here, under the assumption of uniform linear motion exhibited by a velocity vector (vx, vy), (MVx.sub.0, MVy.sub.0) and (MVx.sub.1, MVy.sub.1) are represented as (vx τ_0 , vy τ_0) and (-vx τ_1 , -vy τ_1), respectively, and the following optical flow equation (2) is given.

[MATH. 3]

$$\partial I.\text{sup.}(k)/\partial t + v.\text{sub.}x \partial I.\text{sup.}(k)/\partial x + v.\text{sub.}y \partial I.\text{sup.}(k)/\partial y = 0 \quad (2)$$

(640) Here, I(k) denotes a luma value from reference image k (k=0, 1) after motion compensation. This optical flow equation shows that the sum of (i) the time derivative of the luma value, (ii) the product of the horizontal velocity and the horizontal component of the spatial gradient of a reference image, and (iii) the product of the vertical velocity and the vertical component of the spatial gradient of a reference image is equal to zero. A motion vector of each block obtained from, for example, an MV candidate list may be corrected in units of a pixel, based on a combination of the optical flow equation and Hermite interpolation.

(641) It is to be noted that a motion vector may be derived on the decoder **200** side using a method other than deriving a motion vector based on a model assuming uniform linear motion. For example, a motion vector may be derived in units of a sub-block based on MVs of a plurality of neighboring blocks.

(642) FIG. **64** is a flow chart illustrating one example of inter prediction according to BIO. FIG. **65** is a diagram illustrating one example of a configuration of inter predictor **126** which performs inter prediction according to BIO.

(643) As illustrated in FIG. **65**, inter predictor **126** includes, for example, memory **126a**, interpolated image deriver **126b**, gradient image deriver **126c**, optical flow deriver **126d**, correction value deriver **126e**, and prediction image corrector **126f**. It is to be noted that memory **126a** may be frame memory **122**.

(644) Inter predictor **126** derives two motion vectors (M0, M1), using two reference pictures (Ref.sub.0, Ref.sub.1) different from the picture (Cur Pic) including a current block. Inter predictor **126** then derives a prediction image for the current block using the two motion vectors (M0, M1) (Step Sy_1). It is to be noted that motion vector M0 is motion vector (MVx.sub.0, MVy.sub.0) corresponding to reference picture Ref.sub.0, and motion vector M1 is motion vector (MVx.sub.1, MVy.sub.1) corresponding to reference picture Ref.sub.1.

(645) Next, interpolated image deriver **126b** derives interpolated image I.sup.0 for the current block, using motion vector M0 and reference picture L0 by referring to memory **126a**. Next, interpolated image deriver **126b** derives interpolated image I.sup.1 for the current block, using motion vector M1 and reference picture L1 by referring to memory **126a** (Step Sy_2). Here, interpolated image I.sup.0 is an image included in reference picture Ref.sub.0 and to be derived for the current block, and interpolated image I.sup.1 is an image included in reference picture Ref.sub.1 and to be derived for the current block. Each of interpolated image I.sup.0 and interpolated image I.sup.1 may be the same in size as the current block. Alternatively, each of interpolated image I.sup.0 and interpolated image I.sup.1 may be an image larger than the current block. Furthermore, interpolated image I.sup.0 and interpolated image I.sup.1 may include a prediction image obtained by using motion vectors (M0, M1) and reference pictures (L0, L1) and applying a motion compensation filter.

(646) In addition, gradient image deriver **126c** derives gradient images (Ix.sup.0, Ix.sup.1, Iy.sup.0, Iy.sup.1) of the current block, from interpolated image I.sup.0 and interpolated image I.sup.1. It is to be noted that the gradient images in the horizontal direction are (Ix.sup.0, Ix.sup.1), and the gradient images in the vertical direction are (Iy.sup.0, Iy.sup.1). Gradient image deriver **126c** may derive each gradient image by, for example, applying a gradient filter to the interpolated images. It

is only necessary that a gradient image indicate the amount of spatial change in pixel value along the horizontal direction or the vertical direction.

(647) Next, optical flow deriver **126d** derives, for each sub-block of the current block, an optical flow (vx, vy) which is a velocity vector, using the interpolated images (I.sup.0, I.sup.1) and the gradient images (Ix.sup.0, Ix.sup.1, Iy.sup.0, Iy.sup.1). The optical flow indicates coefficients for correcting the amount of spatial pixel movement, and may be referred to as a local motion estimation value, a corrected motion vector, or a corrected weighting vector. As one example, a sub-block may be 4×4 pixel sub-CU. It is to be noted that the optical flow derivation may be performed for each pixel unit, or the like, instead of being performed for each sub-block.

(648) Next, inter predictor **126** corrects a prediction image for the current block using the optical flow (vx, vy). For example, correction value deriver **126e** derives a correction value for the value of a pixel included in a current block, using the optical flow (vx, vy) (Step Sy_5). Prediction image corrector **126f** may then correct the prediction image for the current block using the correction value (Step Sy_6). It is to be noted that the correction value may be derived in units of a pixel, or may be derived in units of a plurality of pixels or in units of a sub-block.

(649) It is to be noted that the BIO process flow is not limited to the process disclosed in FIG. 64. Only part of the processes disclosed in FIG. 64 may be performed, or a different process may be added or used as a replacement, or the processes may be executed in a different processing order.

(650) [Motion Compensation>LIC]

(651) Next, one example of a mode for generating a prediction image (prediction) using a local illumination compensation (LIC) is described.

(652) FIG. 66A is a diagram for illustrating one example of a prediction image generation method using a luminance correction process performed by LIC. FIG. 66B is a flow chart illustrating one example of a prediction image generation method using the LIC.

(653) First, inter predictor **126** derives an MV from an encoded reference picture, and obtains a reference image corresponding to the current block (Step Sz_1).

(654) Next, inter predictor **126** extracts, for the current block, information indicating how the luma value has changed between the current block and the reference picture (Step Sz_2). This extraction is performed based on the luma pixel values of the encoded left neighboring reference region (surrounding reference region) and the encoded upper neighboring reference region (surrounding reference region) in the current picture, and the luma pixel values at the corresponding positions in the reference picture specified by the derived MVs. Inter predictor **126** calculates a luminance correction parameter, using the information indicating how the luma value has changed (Step Sz_3).

(655) Inter predictor **126** generates a prediction image for the current block by performing a luminance correction process in which the luminance correction parameter is applied to the reference image in the reference picture specified by the MV (Step Sz_4). In other words, the prediction image which is the reference image in the reference picture specified by the MV is subjected to the correction based on the luminance correction parameter. In this correction, luminance may be corrected, or chrominance may be corrected. In other words, a chrominance correction parameter may be calculated using information indicating how chrominance has changed, and a chrominance correction process may be performed.

(656) It is to be noted that the shape of the surrounding reference region illustrated in FIG. 66A is one example; another shape may be used.

(657) Moreover, although the process in which a prediction image is generated from a single reference picture has been described here, cases in which a prediction image is generated from a plurality of reference pictures can be described in the same manner. The prediction image may be generated after performing a luminance correction process of the reference images obtained from the reference pictures in the same manner as described above.

(658) One example of a method for determining whether to apply LIC is a method for using a

lic_flag which is a signal indicating whether to apply the LIC. As one specific example, encoder **100** determines whether the current block belongs to a region having a luminance change. Encoder **100** sets the lic_flag to a value of “1” when the block belongs to a region having a luminance change and applies LIC when encoding, and sets the lic_flag to a value of “0” when the block does not belong to a region having a luminance change and performs encoding without applying LIC. Decoder **200** may decode the lic_flag written in the stream and decode the current block by switching between application and non-application of LIC in accordance with the flag value.

(659) One example of a different method of determining whether to apply a LIC process is a determining method in accordance with whether a LIC process has been applied to a surrounding block. As one specific example, when a current block has been processed in merge mode, inter predictor **126** determines whether an encoded surrounding block selected in MV derivation in merge mode has been encoded using LIC. Inter predictor **126** performs encoding by switching between application and non-application of LIC according to the result. It is to be noted that, also in this example, the same processes are applied to processes at the decoder **200** side.

(660) The luminance correction (LIC) process has been described with reference to FIGS. **66A** and **66B**, and is further described below.

(661) First, inter predictor **126** derives an MV for obtaining a reference image corresponding to a current block from a reference picture which is an encoded picture.

(662) Next, inter predictor **126** extracts information indicating how the luma value of the reference picture has been changed to the luma value of the current picture, using the luma pixel values of encoded surrounding reference regions which neighbor to the left of and above the current block and the luma pixel values in the corresponding positions in the reference pictures specified by MVs, and calculates a luminance correction parameter. For example, it is assumed that the luma pixel value of a given pixel in the surrounding reference region in the current picture is p_0 , and that the luma pixel value of the pixel corresponding to the given pixel in the surrounding reference region in the reference picture is p_1 . Inter predictor **126** calculates coefficients A and B for optimizing $A \times p_1 + B = p_0$ as the luminance correction parameter for a plurality of pixels in the surrounding reference region.

(663) Next, inter predictor **126** performs a luminance correction process using the luminance correction parameter for the reference image in the reference picture specified by the MV, to generate a prediction image for the current block. For example, it is assumed that the luma pixel value in the reference image is p_2 , and that the luminance-corrected luma pixel value of the prediction image is p_3 . Inter predictor **126** generates the prediction image after being subjected to the luminance correction process by calculating $A \times p_2 + B = p_3$ for each of the pixels in the reference image.

(664) It is to be noted that part of the surrounding reference regions illustrated in FIG. **66A** may be used. For example, a region having a determined number of pixels extracted from each of upper neighboring pixels and left neighboring pixels may be used as a surrounding reference region. In addition, the surrounding reference region is not limited to a region which neighbors the current block, and may be a region which does not neighbor the current block. In the example illustrated in FIG. **66A**, the surrounding reference region in the reference picture may be a region specified by another MV in a current picture, from a surrounding reference region in the current picture. For example, the other MV may be an MV in a surrounding reference region in the current picture.

(665) Although operations performed by encoder **100** have been described here, it is to be noted that decoder **200** performs similar operations.

(666) It is to be noted that LIC may be applied not only to luma but also to chroma. At this time, a correction parameter may be derived individually for each of Y, Cb, and Cr, or a common correction parameter may be used for any of Y, Cb, and Cr.

(667) In addition, the LIC process may be applied in units of a sub-block. For example, a correction parameter may be derived using a surrounding reference region in a current sub-block and a

surrounding reference region in a reference sub-block in a reference picture specified by an MV of the current sub-block.

(668) [Prediction Controller]

(669) Prediction controller **128** selects one of an intra prediction image (an image or a signal output from intra predictor **124**) and an inter prediction image (an image or a signal output from inter predictor **126**), and outputs the selected prediction image to subtractor **104** and adder **116**.

(670) [Prediction Parameter Generator]

(671) Prediction parameter generator **130** may output information related to intra prediction, inter prediction, selection of a prediction image in prediction controller **128**, etc. as a prediction parameter to entropy encoder **110**. Entropy encoder **110** may generate a stream, based on the prediction parameter which is input from prediction parameter generator **130** and quantized coefficients which are input from quantizer **108**. The prediction parameter may be used in decoder **200**. Decoder **200** may receive and decode the stream, and perform the same processes as the prediction processes performed by intra predictor **124**, inter predictor **126**, and prediction controller **128**. The prediction parameter may include (i) a selection prediction signal (for example, an MV, a prediction type, or a prediction mode used by intra predictor **124** or inter predictor **126**), or (ii) an optional index, a flag, or a value which is based on a prediction process performed in each of intra predictor **124**, inter predictor **126**, and prediction controller **128**, or which indicates the prediction process.

(672) [Decoder]

(673) Next, decoder **200** capable of decoding a stream output from encoder **100** described above is described. FIG. **67** is a block diagram illustrating a configuration of decoder **200** according to this embodiment. Decoder **200** is an apparatus which decodes a stream that is an encoded image in units of a block.

(674) As illustrated in FIG. **67**, decoder **200** includes entropy decoder **202**, inverse quantizer **204**, inverse transformer **206**, adder **208**, block memory **210**, loop filter **212**, frame memory **214**, intra predictor **216**, inter predictor **218**, prediction controller **220**, prediction parameter generator **222**, and splitting determiner **224**. It is to be noted that intra predictor **216** and inter predictor **218** are configured as part of a prediction executor.

(675) [Mounting Example of Decoder]

(676) FIG. **68** is a block diagram illustrating a mounting example of decoder **200**. Decoder **200** includes processor **b1** and memory **b2**. For example, the plurality of constituent elements of decoder **200** illustrated in FIG. **67** are mounted on processor **b1** and memory **b2** illustrated in FIG. **68**.

(677) Processor **b1** is circuitry which performs information processing and is accessible to memory **b2**. For example, processor **b1** is a dedicated or general electronic circuit which decodes a stream. Processor **b1** may be a processor such as a CPU. In addition, processor **b1** may be an aggregate of a plurality of electronic circuits. In addition, for example, processor **b1** may take the roles of two or more constituent elements other than a constituent element for storing information out of the plurality of constituent elements of decoder **200** illustrated in FIG. **67**, etc.

(678) Memory **b2** is dedicated or general memory for storing information that is used by processor **b1** to decode a stream. Memory **b2** may be electronic circuitry, and may be connected to processor **131**. In addition, memory **b2** may be included in processor **b1**. In addition, memory **b2** may be an aggregate of a plurality of electronic circuits. In addition, memory **b2** may be a magnetic disc, an optical disc, or the like, or may be represented as a storage, a medium, or the like. In addition, memory **b2** may be non-volatile memory, or volatile memory.

(679) For example, memory **b2** may store an image or a stream. In addition, memory **b2** may store a program for causing processor **b1** to decode a stream.

(680) In addition, for example, memory **b2** may take the roles of two or more constituent elements for storing information out of the plurality of constituent elements of decoder **200** illustrated in

FIG. 67, etc. More specifically, memory b2 may take the roles of block memory **210** and frame memory **214** illustrated in FIG. 67. More specifically, memory b2 may store a reconstructed image (specifically, a reconstructed block, a reconstructed picture, or the like).

(681) It is to be noted that, in decoder **200**, not all of the plurality of constituent elements illustrated in FIG. 67, etc. may be implemented, and not all the processes described above may be performed. Part of the constituent elements indicated in FIG. 67, etc. may be included in another device, or part of the processes described above may be performed by another device.

(682) Hereinafter, an overall flow of the processes performed by decoder **200** is described, and then each of the constituent elements included in decoder **200** is described. It is to be noted that, some of the constituent elements included in decoder **200** perform the same processes as performed by some of the constituent elements included in encoder **100**, and thus the same processes are not repeatedly described in detail. For example, inverse quantizer **204**, inverse transformer **206**, adder **208**, block memory **210**, frame memory **214**, intra predictor **216**, inter predictor **218**, prediction controller **220**, and loop filter **212** included in decoder **200** perform similar processes as performed by inverse quantizer **112**, inverse transformer **114**, adder **116**, block memory **118**, frame memory **122**, intra predictor **124**, inter predictor **126**, prediction controller **128**, and loop filter **120** included in encoder **100**, respectively.

(683) [Overall Flow of Decoding Process]

(684) FIG. 69 is a flow chart illustrating one example of an overall decoding process performed by decoder **200**.

(685) First, splitting determiner **224** in decoder **200** determines a splitting pattern of each of a plurality of fixed-size blocks (128×128 pixels) included in a picture, based on a parameter which is input from entropy decoder **202** (Step Sp_1). This splitting pattern is a splitting pattern selected by encoder **100**. Decoder **200** then performs processes of Steps Sp_2 to Sp_6 for each of a plurality of blocks of the splitting pattern.

(686) Entropy decoder **202** decodes (specifically, entropy decodes) encoded quantized coefficients and a prediction parameter of a current block (Step Sp_2).

(687) Next, inverse quantizer **204** performs inverse quantization of the plurality of quantized coefficients and inverse transformer **206** performs inverse transform of the result, to restore prediction residuals of the current block (Step Sp_3).

(688) Next, the prediction executor including all or part of intra predictor **216**, inter predictor **218**, and prediction controller **220** generates a prediction image of the current block (Step Sp_4).

(689) Next, adder **208** adds the prediction image to a prediction residual to generate a reconstructed image (also referred to as a decoded image block) of the current block (Step Sp_5).

(690) When the reconstructed image is generated, loop filter **212** performs filtering of the reconstructed image (Step Sp_6).

(691) Decoder **200** then determines whether decoding of the entire picture has been finished (Step Sp_7). When determining that the decoding has not yet been finished (No in Step Sp_7), decoder **200** repeatedly executes the processes starting with Step Sp_1.

(692) It is to be noted that the processes of these Steps Sp_1 to Sp_7 may be performed sequentially by decoder **200**, or two or more of the processes may be performed in parallel. The processing order of the two or more of the processes may be modified.

(693) [Splitting Determiner]

(694) FIG. 70 is a diagram illustrating a relationship between splitting determiner **224** and other constituent elements. Splitting determiner **224** may perform the following processes as examples.

(695) For example, splitting determiner **224** collects block information from block memory **210** or frame memory **214**, and furthermore obtains a parameter from entropy decoder **202**. Splitting determiner **224** may then determine the splitting pattern of a fixed-size block, based on the block information and the parameter. Splitting determiner **224** may then output information indicating the determined splitting pattern to inverse transformer **206**, intra predictor **216**, and inter predictor **218**.

Inverse transformer **206** may perform inverse transform of transform coefficients, based on the splitting pattern indicated by the information from splitting determiner **224**. Intra predictor **216** and inter predictor **218** may generate a prediction image, based on the splitting pattern indicated by the information from splitting determiner **224**.

(696) [Entropy Decoder]

(697) FIG. **71** is a block diagram illustrating one example of a configuration of entropy decoder **202**.

(698) Entropy decoder **202** generates quantized coefficients, a prediction parameter, and a parameter related to a splitting pattern, by entropy decoding the stream. For example, CABAC is used in the entropy decoding. More specifically, entropy decoder **202** includes, for example, binary arithmetic decoder **202a**, context controller **202b**, and debinarizer **202c**. Binary arithmetic decoder **202a** arithmetically decodes the stream using a context value derived by context controller **202b** to a binary signal. Context controller **202b** derives a context value according to a feature or a surrounding state of a syntax element, that is, an occurrence probability of a binary signal, in the same manner as performed by context controller **110b** of encoder **100**. Debinarizer **202c** performs debinarization for transforming the binary signal output from binary arithmetic decoder **202a** to a multi-level signal indicating quantized coefficients as described above. This binarization is performed according to the binarization method described above.

(699) With this, entropy decoder **202** outputs quantized coefficients of each block to inverse quantizer **204**. Entropy decoder **202** may output a prediction parameter included in a stream (see FIG. **1**) to intra predictor **216**, inter predictor **218**, and prediction controller **220**. Intra predictor **216**, inter predictor **218**, and prediction controller **220** are capable of executing the same prediction processes as those performed by intra predictor **124**, inter predictor **126**, and prediction controller **128** at the encoder **100** side.

(700) [Entropy Decoder]

(701) FIG. **72** is a diagram illustrating a flow of CABAC in entropy decoder **202**.

(702) First, initialization is performed in CABAC in entropy decoder **202**. In the initialization, initialization in binary arithmetic decoder **202a** and setting of an initial context value are performed. Binary arithmetic decoder **202a** and debinarizer **202c** then execute arithmetic decoding and debinarization of, for example, encoded data of a CTU. At this time, context controller **202b** updates the context value each time arithmetic decoding is performed. Context controller **202b** then saves the context value as a post process. The saved context value is used, for example, to initialize the context value for the next CTU.

(703) [Inverse Quantizer]

(704) Inverse quantizer **204** inverse quantizes quantized coefficients of a current block which are inputs from entropy decoder **202**. More specifically, inverse quantizer **204** inverse quantizes the quantized coefficients of the current block, based on quantization parameters corresponding to the quantized coefficients. Inverse quantizer **204** then outputs the inverse quantized transform coefficients (that are transform coefficients) of the current block to inverse transformer **206**.

(705) FIG. **73** is a block diagram illustrating one example of a configuration of inverse quantizer **204**.

(706) Inverse quantizer **204** includes, for example, quantization parameter generator **204a**, predicted quantization parameter generator **204b**, quantization parameter storage **204d**, and inverse quantization executor **204e**.

(707) FIG. **74** is a flow chart illustrating one example of inverse quantization performed by inverse quantizer **204**.

(708) Inverse quantizer **204** may perform an inverse quantization process as one example for each CU based on the flow illustrated in FIG. **74**. More specifically, quantization parameter generator **204a** determines whether to perform inverse quantization (Step Sv_11). Here, when determining to perform inverse quantization (Yes in Step Sv_11), quantization parameter generator **204a** obtains a

difference quantization parameter for the current block from entropy decoder **202** (Step Sv_12).
(709) Next, predicted quantization parameter generator **204b** then obtains a quantization parameter for a processing unit different from the current block from quantization parameter storage **204d** (Step Sv_13). Predicted quantization parameter generator **204b** generates a predicted quantization parameter of the current block based on the obtained quantization parameter (Step Sv_14).
(710) Quantization parameter generator **204a** then adds the difference quantization parameter for the current block obtained from entropy decoder **202** and the predicted quantization parameter for the current block generated by predicted quantization parameter generator **204b** (Step Sv_15). This addition generates a quantization parameter for the current block. In addition, quantization parameter generator **204a** stores the quantization parameter for the current block in quantization parameter storage **204d** (Step Sv_16).
(711) Next, inverse quantization executor **204e** inverse quantizes the quantized coefficients of the current block into transform coefficients, using the quantization parameter generated in Step Sv_15 (Step Sv_17).
(712) It is to be noted that the difference quantization parameter may be decoded at the bit sequence level, picture level, slice level, brick level, or CTU level. In addition, the initial value of the quantization parameter may be decoded at the sequence level, picture level, slice level, brick level, or CTU level. At this time, the quantization parameter may be generated using the initial value of the quantization parameter and the difference quantization parameter.
(713) It is to be noted that inverse quantizer **204** may include a plurality of inverse quantizers, and may inverse quantize the quantized coefficients using an inverse quantization method selected from a plurality of inverse quantization methods.
(714) [Inverse Transformer]
(715) Inverse transformer **206** restores prediction residuals by inverse transforming the transform coefficients which are inputs from inverse quantizer **204**.
(716) For example, when information parsed from a stream indicates that EMT or AMT is to be applied (for example, when an AMT flag is true), inverse transformer **206** inverse transforms the transform coefficients of the current block based on information indicating the parsed transform type.
(717) Moreover, for example, when information parsed from a stream indicates that NSST is to be applied, inverse transformer **206** applies a secondary inverse transform to the transform coefficients.
(718) FIG. 75 is a flow chart illustrating one example of a process performed by inverse transformer **206**.
(719) For example, inverse transformer **206** determines whether information indicating that no orthogonal transform is performed is present in a stream (Step St_11). Here, when determining that no such information is present (No in Step St_11), inverse transformer **206** obtains information indicating the transform type decoded by entropy decoder **202** (Step St_12). Next, based on the information, inverse transformer **206** determines the transform type used for the orthogonal transform in encoder **100** (Step St_13). Inverse transformer **206** then performs inverse orthogonal transform using the determined transform type (Step St_14).
(720) FIG. 76 is a flow chart illustrating another example of a process performed by inverse transformer **206**.
(721) For example, inverse transformer **206** determines whether a transform size is smaller than or equal to a predetermined value (Step Su_11). Here, when determining that the transform size is smaller than or equal to a predetermined value (Yes in Step Su_11), inverse transformer **206** obtains, from entropy decoder **202**, information indicating which transform type has been used by encoder **100** among at least one transform type included in the first transform type group (Step Su_12). It is to be noted that such information is decoded by entropy decoder **202** and output to inverse transformer **206**.

(722) Based on the information, inverse transformer **206** determines the transform type used for the orthogonal transform in encoder **100** (Step Su_13). Inverse transformer **206** then inverse orthogonal transforms the transform coefficients of the current block using the determined transform type (Step Su_14). When determining that a transform size is not smaller than or equal to the predetermined value (No in Step Su_11), inverse transformer **206** inverse transforms the transform coefficients of the current block using the second transform type group (Step Su_15).

(723) It is to be noted that the inverse orthogonal transform by inverse transformer **206** may be performed according to the flow illustrated in FIG. 75 or FIG. 76 for each TU as one example. In addition, inverse orthogonal transform may be performed by using a predefined transform type without decoding information indicating a transform type used for orthogonal transform. In addition, the transform type is specifically DST7, DCT8, or the like. In inverse orthogonal transform, an inverse transform basis function corresponding to the transform type is used.

(724) [Adder]

(725) Adder **208** reconstructs the current block by adding a prediction residual which is an input from inverse transformer **206** and a prediction image which is an input from prediction controller **220**. In other words, a reconstructed image of the current block is generated. Adder **208** then outputs the reconstructed image of the current block to block memory **210** and loop filter **212**.

(726) [Block Memory]

(727) Block memory **210** is storage for storing a block which is included in a current picture and is referred to in intra prediction. More specifically, block memory **210** stores a reconstructed image output from adder **208**.

(728) [Loop Filter]

(729) Loop filter **212** applies a loop filter to the reconstructed image generated by adder **208**, and outputs the filtered reconstructed image to frame memory **214** and a display device, etc.

(730) When information indicating ON or OFF of an ALF parsed from a stream indicates that an ALF is ON, one filter from among a plurality of filters is selected based on the direction and activity of local gradients, and the selected filter is applied to the reconstructed image.

(731) FIG. 77 is a block diagram illustrating one example of a configuration of loop filter **212**. It is to be noted that loop filter **212** has a configuration similar to the configuration of loop filter **120** of encoder **100**.

(732) For example, as illustrated in FIG. 77, loop filter **212** includes deblocking filter executor **212a**, SAO executor **212b**, and ALF executor **212c**. Deblocking filter executor **212a** performs a deblocking filter process of the reconstructed image. SAO executor **212b** performs a SAO process of the reconstructed image after being subjected to the deblocking filter process. ALF executor **212c** performs an ALF process of the reconstructed image after being subjected to the SAO process. It is to be noted that loop filter **212** does not always need to include all the constituent elements disclosed in FIG. 77, and may include only part of the constituent elements. In addition, loop filter **212** may be configured to perform the above processes in a processing order different from the one disclosed in FIG. 77.

(733) [Frame Memory]

(734) Frame memory **214** is, for example, storage for storing reference pictures for use in inter prediction, and is also referred to as a frame buffer. More specifically, frame memory **214** stores a reconstructed image filtered by loop filter **212**.

(735) [Predictor (Intra Predictor, Inter Predictor, Prediction Controller)]

(736) FIG. 78 is a flow chart illustrating one example of a process performed by a predictor of decoder **200**. It is to be noted that the prediction executor includes all or part of the following constituent elements: intra predictor **216**; inter predictor **218**; and prediction controller **220**. The prediction executor includes, for example, intra predictor **216** and inter predictor **218**.

(737) The predictor generates a prediction image of a current block (Step Sq_1). This prediction image is also referred to as a prediction signal or a prediction block. It is to be noted that the

prediction signal is, for example, an intra prediction signal or an inter prediction signal. More specifically, the predictor generates the prediction image of the current block using a reconstructed image which has been already obtained for another block through generation of a prediction image, restoration of a prediction residual, and addition of a prediction image. The predictor of decoder **200** generates the same prediction image as the prediction image generated by the predictor of encoder **100**. In other words, the prediction images are generated according to a method common between the predictors or mutually corresponding methods.

(738) The reconstructed image may be, for example, an image in a reference picture, or an image of a decoded block (that is, the other block described above) in a current picture which is the picture including the current block. The decoded block in the current picture is, for example, a neighboring block of the current block.

(739) FIG. **79** is a flow chart illustrating another example of a process performed by the predictor of decoder **200**.

(740) The predictor determines either a method or a mode for generating a prediction image (Step Sr_1). For example, the method or mode may be determined based on, for example, a prediction parameter, etc.

(741) When determining a first method as a mode for generating a prediction image, the predictor generates a prediction image according to the first method (Step Sr_2a). When determining a second method as a mode for generating a prediction image, the predictor generates a prediction image according to the second method (Step Sr_2b). When determining a third method as a mode for generating a prediction image, the predictor generates a prediction image according to the third method (Step Sr_2c).

(742) The first method, the second method, and the third method may be mutually different methods for generating a prediction image. Each of the first to third methods may be an inter prediction method, an intra prediction method, or another prediction method. The above-described reconstructed image may be used in these prediction methods.

(743) FIG. **80A** and FIG. **80B** illustrate a flow chart illustrating another example of a process performed by a predictor of decoder **200**.

(744) The predictor may perform a prediction process according to the flow illustrated in FIG. **80A** and FIG. **80B** as one example. It is to be noted that intra block copy illustrated in FIG. **80A** and FIG. **80B** is one mode which belongs to inter prediction, and in which a block included in a current picture is referred to as a reference image or a reference block. In other words, no picture different from the current picture is referred to in intra block copy. In addition, the PCM mode illustrated in FIG. **80A** is one mode which belongs to intra prediction, and in which no transform and quantization is performed.

(745) [Intra Predictor]

(746) Intra predictor **216** performs intra prediction by referring to a block in a current picture stored in block memory **210**, based on the intra prediction mode parsed from the stream, to generate a prediction image of a current block (that is, an intra prediction image). More specifically, intra predictor **216** performs intra prediction by referring to pixel values (for example, luma and/or chroma values) of a block or blocks neighboring the current block to generate an intra prediction image, and then outputs the intra prediction image to prediction controller **220**.

(747) It is to be noted that when an intra prediction mode in which a luma block is referred to in intra prediction of a chroma block is selected, intra predictor **216** may predict the chroma component of the current block based on the luma component of the current block.

(748) Moreover, when information parsed from a stream indicates that PDPC is to be applied, intra predictor **216** corrects intra predicted pixel values based on horizontal/vertical reference pixel gradients.

(749) FIG. **81** is a diagram illustrating one example of a process performed by intra predictor **216** of decoder **200**.

(750) Intra predictor **216** firstly determines whether an MPM flag indicating 1 is present in the stream (Step Sw_11). Here, when determining that the MPM flag indicating 1 is present (Yes in Step Sw_11), intra predictor **216** obtains, from entropy decoder **202**, information indicating the intra prediction mode selected in encoder **100** among MPMs (Step Sw_12). It is to be noted that such information is decoded by entropy decoder **202** and output to intra predictor **216**. Next, intra predictor **216** determines an MPM (Step Sw_13). MPMs include, for example, six intra prediction modes. Intra predictor **216** then determines the intra prediction mode which is included in a plurality of intra prediction modes included in the MPMs and is indicated by the information obtained in Step Sw_12 (Step Sw_14).

(751) When determining that no MPM flag indicating 1 is present (No in Step Sw_11), intra predictor **216** obtains information indicating the intra prediction mode selected in encoder **100** (Step Sw_15). In other words, intra predictor **216** obtains, from entropy decoder **202**, information indicating the intra prediction mode selected in encoder **100** from among at least one intra prediction mode which is not included in the MPMs. It is to be noted that such information is decoded by entropy decoder **202** and output to intra predictor **216**. Intra predictor **216** then determines the intra prediction mode which is not included in a plurality of intra prediction modes included in the MPMs and is indicated by the information obtained in Step Sw_15 (Step Sw_17).

(752) Intra predictor **216** generates a prediction image according to the intra prediction mode determined in Step Sw_14 or Step Sw_17 (Step Sw_18).

(753) [Inter Predictor]

(754) Inter predictor **218** predicts the current block by referring to a reference picture stored in frame memory **214**. Prediction is performed in units of a current block or a current sub-block in the current block. It is to be noted that the sub-block is included in the block and is a unit smaller than the block. The size of the sub-block may be 4×4 pixels, 8×8 pixels, or another size. The size of the sub-block may be switched for a unit such as a slice, brick, picture, etc.

(755) For example, inter predictor **218** generates an inter prediction image of a current block or a current sub-block by performing motion compensation using motion information (for example, an MV) parsed from a stream (for example, a prediction parameter output from entropy decoder **202**), and outputs the inter prediction image to prediction controller **220**.

(756) When the information parsed from the stream indicates that the OBMC mode is to be applied, inter predictor **218** generates the inter prediction image using motion information of a neighboring block in addition to motion information of the current block obtained through motion estimation.

(757) Moreover, when the information parsed from the stream indicates that the FRUC mode is to be applied, inter predictor **218** derives motion information by performing motion estimation in accordance with the pattern matching method (bilateral matching or template matching) parsed from the stream. Inter predictor **218** then performs motion compensation (prediction) using the derived motion information.

(758) Moreover, when the BIO mode is to be applied, inter predictor **218** derives an MV based on a model assuming uniform linear motion. In addition, when the information parsed from the stream indicates that the affine mode is to be applied, inter predictor **218** derives an MV for each sub-block, based on the MVs of a plurality of neighboring blocks.

(759) [MV Derivation Flow]

(760) FIG. **82** is a flow chart illustrating one example of MV derivation in decoder **200**.

(761) Inter predictor **218** determines, for example, whether to decode motion information (for example, an MV). For example, inter predictor **218** may make the determination according to the prediction mode included in the stream, or may make the determination based on other information included in the stream. Here, when determining to decode motion information, inter predictor **218** derives an MV for a current block in a mode in which the motion information is decoded. When determining not to decode motion information, inter predictor **218** derives an MV in a mode in

which no motion information is decoded.

(762) Here, MV derivation modes include a normal inter mode, a normal merge mode, a FRUC mode, an affine mode, etc. which are described later. Modes in which motion information is decoded among the modes include the normal inter mode, the normal merge mode, the affine mode (specifically, an affine inter mode and an affine merge mode), etc. It is to be noted that motion information may include not only an MV but also MV predictor selection information which is described later. Modes in which no motion information is decoded include the FRUC mode, etc. Inter predictor **218** selects a mode for deriving an MV for the current block from the plurality of modes, and derives the MV for the current block using the selected mode.

(763) FIG. **83** is a flow chart illustrating another example of MV derivation in decoder **200**.

(764) For example, inter predictor **218** may determine whether to decode an MV difference, that is for example, may make the determination according to the prediction mode included in the stream, or may make the determination based on other information included in the stream. Here, when determining to decode an MV difference, inter predictor **218** may derive an MV for a current block in a mode in which the MV difference is decoded. In this case, for example, the MV difference included in the stream is decoded as a prediction parameter.

(765) When determining not to decode any MV difference, inter predictor **218** derives an MV in a mode in which no MV difference is decoded. In this case, no encoded MV difference is included in the stream.

(766) Here, as described above, the MV derivation modes include the normal inter mode, the normal merge mode, the FRUC mode, the affine mode, etc. which are described later. Modes in which an MV difference is encoded among the modes include the normal inter mode and the affine mode (specifically, the affine inter mode), etc. Modes in which no MV difference is encoded include the FRUC mode, the normal merge mode, the affine mode (specifically, the affine merge mode), etc. Inter predictor **218** selects a mode for deriving an MV for the current block from the plurality of modes, and derives the MV for the current block using the selected mode.

(767) [MV Derivation>Normal Inter Mode]

(768) For example, when information parsed from a stream indicates that the normal inter mode is to be applied, inter predictor **218** derives an MV based on the information parsed from the stream and performs motion compensation (prediction) using the MV.

(769) FIG. **84** is a flow chart illustrating an example of inter prediction by normal inter mode in decoder **200**.

(770) Inter predictor **218** of decoder **200** performs motion compensation for each block. At this time, first, inter predictor **218** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of decoded blocks temporally or spatially surrounding the current block (Step Sg_11). In other words, inter predictor **218** generates an MV candidate list.

(771) Next, inter predictor **218** extracts N (an integer of 2 or larger) MV candidates from the plurality of MV candidates obtained in Step Sg_11, as motion vector predictor candidates (also referred to as MV predictor candidates) according to the predetermined ranks in priority order (Step Sg_12). It is to be noted that the ranks in priority order are determined in advance for the respective N MV predictor candidates.

(772) Next, inter predictor **218** decodes the MV predictor selection information from the input stream, and selects one MV predictor candidate from the N MV predictor candidates as the MV predictor for the current block using the decoded MV predictor selection information (Step Sg_13).

(773) Next, inter predictor **218** decodes an MV difference from the input stream, and derives an MV for the current block by adding a difference value which is the decoded MV difference and the selected MV predictor (Step Sg_14).

(774) Lastly, inter predictor **218** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the decoded reference picture (Step Sg_15). The processes in Steps Sg_11 to Sg_15 are executed on each block. For example,

when the processes in Steps Sg_11 to Sg_15 are executed on each of all the blocks in the slice, inter prediction of the slice using the normal inter mode finishes. For example, when the processes in Steps Sg_11 to Sg_15 are executed on each of all the blocks in the picture, inter prediction of the picture using the normal inter mode finishes. It is to be noted that not all the blocks included in the slice may be subjected to the processes in Steps Sg_11 to Sg_15, and inter prediction of the slice using the normal inter mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the normal inter mode may finish when the processes in Steps Sg_11 to Sg_15 are executed on part of the blocks in the picture.

(775) [MV Derivation>Normal Merge Mode]

(776) For example, when information parsed from a stream indicates that the normal merge mode is to be applied, inter predictor **218** derives an MV and performs motion compensation (prediction) using the MV.

(777) FIG. **85** is a flow chart illustrating an example of inter prediction by normal merge mode in decoder **200**.

(778) At this time, first, inter predictor **218** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of decoded blocks temporally or spatially surrounding the current block (Step Sh_11). In other words, inter predictor **218** generates an MV candidate list.

(779) Next, inter predictor **218** selects one MV candidate from the plurality of MV candidates obtained in Step Sh_11, thereby deriving an MV for the current block (Step Sh_12). More specifically, inter predictor **218** obtains MV selection information included as a prediction parameter in a stream, and selects the MV candidate identified by the MV selection information as the MV for the current block.

(780) Lastly, inter predictor **218** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the decoded reference picture (Step Sh_13). The processes in Steps Sh_11 to Sh_13 are executed, for example, on each block. For example, when the processes in Steps Sh_11 to Sh_13 are executed on each of all the blocks in the slice, inter prediction of the slice using the normal merge mode finishes. In addition, when the processes in Steps Sh_11 to Sh_13 are executed on each of all the blocks in the picture, inter prediction of the picture using the normal merge mode finishes. It is to be noted that not all the blocks included in the slice are subjected to the processes in Steps Sh_11 to Sh_13, and inter prediction of the slice using the normal merge mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the normal merge mode may finish when the processes in Steps Sh_11 to Sh_13 are executed on part of the blocks in the picture.

(781) [MV Derivation>FRUC Mode]

(782) For example, when information parsed from a stream indicates that the FRUC mode is to be applied, inter predictor **218** derives an MV in the FRUC mode and performs motion compensation (prediction) using the MV. In this case, the motion information is derived at the decoder **200** side without being signaled from the encoder **100** side. For example, decoder **200** may derive the motion information by performing motion estimation. In this case, decoder **200** performs motion estimation without using any pixel value in a current block.

(783) FIG. **86** is a flow chart illustrating an example of inter prediction by FRUC mode in decoder **200**.

(784) First, inter predictor **218** generates a list indicating MVs of decoded blocks spatially or temporally neighboring the current block by referring to the MVs as MV candidates (the list is an MV candidate list, and may be used also as an MV candidate list for normal merge mode (Step Si_11). Next, a best MV candidate is selected from the plurality of MV candidates registered in the MV candidate list (Step Si_12). For example, inter predictor **218** calculates the evaluation value of each MV candidate included in the MV candidate list, and selects one of the MV candidates as the

best MV candidate based on the evaluation values. Based on the selected best MV candidate, inter predictor **218** then derives an MV for the current block (Step Si_14). More specifically, for example, the selected best MV candidate is directly derived as the MV for the current block. In addition, for example, the MV for the current block may be derived using pattern matching in a surrounding region of a position which is included in a reference picture and corresponds to the selected best MV candidate. In other words, estimation using the pattern matching in a reference picture and the evaluation values may be performed in the surrounding region of the best MV candidate, and when there is an MV that yields a better evaluation value, the best MV candidate may be updated to the MV that yields the better evaluation value, and the updated MV may be determined as the final MV for the current block. Update to the MV that yields the better evaluation value may not be performed.

(785) Lastly, inter predictor **218** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the decoded reference picture (Step Si_15). The processes in Steps Si_11 to Si_15 are executed, for example, on each block. For example, when the processes in Steps Si_11 to Si_15 are executed on each of all the blocks in the slice, inter prediction of the slice using the FRUC mode finishes. For example, when the processes in Steps Si_11 to Si_15 are executed on each of all the blocks in the picture, inter prediction of the picture using the FRUC mode finishes. Each sub-block may be processed similarly to the above-described case of processing each block.

(786) [MV Derivation>Affine Merge Mode]

(787) For example, when information parsed from a stream indicates that the affine merge mode is to be applied, inter predictor **218** derives an MV in the affine merge mode and performs motion compensation (prediction) using the MV.

(788) FIG. **87** is a flow chart illustrating an example of inter prediction by the affine merge mode in decoder **200**.

(789) In the affine merge mode, first, inter predictor **218** derives MVs at respective control points for a current block (Step Sk_11). The control points are an upper-left corner point of the current block and an upper-right corner point of the current block as illustrated in FIG. **46A**, or an upper-left corner point of the current block, an upper-right corner point of the current block, and a lower-left corner point of the current block as illustrated in FIG. **46B**.

(790) For example, when the MV derivation methods illustrated in FIGS. **47A** to **47C** are used, as illustrated in FIG. **47A**, inter predictor **218** checks decoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) in this order, and identifies the first effective block decoded according to the affine mode.

(791) Inter predictor **218** derives the MV at the control point using the identified first effective block decoded according to the affine mode. For example, when block A is identified and block A has two control points, as illustrated in FIG. **47B**, inter predictor **218** calculates motion vector $v_{sub.0}$ at the upper-left corner control point of the current block and motion vector $v_{sub.1}$ at the upper-right corner control point of the current block by projecting motion vectors $v_{sub.3}$ and $v_{sub.4}$ at the upper-left corner and the upper-right corner of the decoded block including block A onto the current block. In this way, the MV at each control point is derived.

(792) It is to be noted that, as illustrated in FIG. **49A**, MVs at three control points may be calculated when block A is identified and block A has two control points, and that, as illustrated in FIG. **49B**, MVs at two control points may be calculated when block A is identified and when block A has three control points.

(793) In addition, when MV selection information is included as a prediction parameter in a stream, inter predictor **218** may derive the MV at each control point for the current block using the MV selection information.

(794) Next, inter predictor **218** performs motion compensation of each of a plurality of sub-blocks included in the current block. In other words, inter predictor **218** calculates an MV for each of the

plurality of sub-blocks as an affine MV, using either two motion vectors $v_{sub.0}$ and $v_{sub.1}$ and the above expression (1A) or three motion vectors $v_{sub.0}$, $v_{sub.1}$, and $v_{sub.2}$ and the above expression (1B) (Step Sk_12). Inter predictor **218** then performs motion compensation of the sub-blocks using these affine MVs and decoded reference pictures (Step Sk_13). When the processes in Steps Sk_12 and Sk_13 are executed for each of all the sub-blocks included in the current block, the inter prediction using the affine merge mode for the current block finishes. In other words, motion compensation of the current block is performed to generate a prediction image of the current block.

(795) It is to be noted that the above-described MV candidate list may be generated in Step Sk_11. The MV candidate list may be, for example, a list including MV candidates derived using a plurality of MV derivation methods for each control point. The plurality of MV derivation methods may be any combination of the MV derivation methods illustrated in FIGS. 47A to 47C, the MV derivation methods illustrated in FIGS. 48A and 48B, the MV derivation methods illustrated in FIGS. 49A and 49B, and other MV derivation methods.

(796) It is to be noted that an MV candidate list may include MV candidates in a mode in which prediction is performed in units of a sub-block, other than the affine mode.

(797) It is to be noted that, for example, an MV candidate list including MV candidates in an affine merge mode in which two control points are used and an affine merge mode in which three control points are used may be generated as an MV candidate list. Alternatively, an MV candidate list including MV candidates in the affine merge mode in which two control points are used and an MV candidate list including MV candidates in the affine merge mode in which three control points are used may be generated separately. Alternatively, an MV candidate list including MV candidates in one of the affine merge mode in which two control points are used and the affine merge mode in which three control points are used may be generated.

(798) [MV Derivation>Affine Inter Mode]

(799) For example, when information parsed from a stream indicates that the affine inter mode is to be applied, inter predictor **218** derives an MV in the affine inter mode and performs motion compensation (prediction) using the MV.

(800) FIG. 88 is a flow chart illustrating an example of inter prediction by the affine inter mode in decoder **200**.

(801) In the affine inter mode, first, inter predictor **218** derives MV predictors ($v_{sub.0}$, $v_{sub.1}$) or ($v_{sub.0}$, $v_{sub.1}$, $v_{sub.2}$) of respective two or three control points for a current block (Step Sj_11). The control points are an upper-left corner point of the current block, an upper-right corner point of the current block, and a lower-left corner point of the current block as illustrated in FIG. 46A or FIG. 46B.

(802) Inter predictor **218** obtains MV predictor selection information included as a prediction parameter in the stream, and derives the MV predictor at each control point for the current block using the MV identified by the MV predictor selection information. For example, when the MV derivation methods illustrated in FIGS. 48A and 48B are used, inter predictor **218** derives the motion vector predictors ($v_{sub.0}$, $v_{sub.1}$) or ($v_{sub.0}$, $v_{sub.1}$, $v_{sub.2}$) at control points for the current block by selecting the MV of the block identified by the MV predictor selection information among decoded blocks in the vicinity of the respective control points for the current block illustrated in either FIG. 48A or FIG. 48B.

(803) Next, inter predictor **218** obtains each MV difference included as a prediction parameter in the stream, and adds the MV predictor at each control point for the current block and the MV difference corresponding to the MV predictor (Step Sj_12). In this way, the MV at each control point for the current block is derived.

(804) Next, inter predictor **218** performs motion compensation of each of a plurality of sub-blocks included in the current block. In other words, inter predictor **218** calculates an MV for each of the plurality of sub-blocks as an affine MV, using either two motion vectors $v_{sub.0}$ and $v_{sub.1}$ and the

above expression (1A) or three motion vectors v.sub.0, v.sub.1, and v.sub.2 and the above expression (1B) (Step Sj_13). Inter predictor **218** then performs motion compensation of the sub-blocks using these affine MVs and decoded reference pictures (Step Sj_14). When the processes in Steps Sj_13 and Sj_14 are executed for each of all the sub-blocks included in the current block, the inter prediction using the affine merge mode for the current block finishes. In other words, motion compensation of the current block is performed to generate a prediction image of the current block. (805) It is to be noted that the above-described MV candidate list may be generated in Step Sj_11 as in Step Sk_11.

(806) [MV Derivation>Triangle Mode]

(807) For example, when information parsed from a stream indicates that the triangle mode is to be applied, inter predictor **218** derives an MV in the triangle mode and performs motion compensation (prediction) using the MV.

(808) FIG. **89** is a flow chart illustrating an example of inter prediction by the triangle mode in decoder **200**.

(809) In the triangle mode, first, inter predictor **218** splits the current block into a first partition and a second partition (Step Sx_11). At this time, inter predictor **218** may obtain, from the stream, partition information which is information related to the splitting as a prediction parameter. Inter predictor **218** may then split a current block into a first partition and a second partition according to the partition information.

(810) Next, first, inter predictor **218** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of decoded blocks temporally or spatially surrounding the current block (Step Sx_12). In other words, inter predictor **218** generates an MV candidate list.

(811) Inter predictor **218** then selects the MV candidate for the first partition and the MV candidate for the second partition as a first MV and a second MV, respectively, from the plurality of MV candidates obtained in Step Sx_11 (Step Sx_13). At this time, inter predictor **218** may obtain, from the stream, MV selection information for identifying each selected MV candidate, as a prediction parameter. Inter predictor **218** may then select the first MV and the second MV according to the MV selection information.

(812) Next, inter predictor **218** generates a first prediction image by performing motion compensation using the selected first MV and a decoded reference picture (Step Sx_14). Likewise, inter predictor **218** generates a second prediction image by performing motion compensation using the selected second MV and a decoded reference picture (Step Sx_15).

(813) Lastly, inter predictor **218** generates a prediction image for the current block by performing a weighted addition of the first prediction image and the second prediction image (Step Sx_16).

(814) [Motion Estimation>DMVR]

(815) For example, information parsed from a stream indicates that DMVR is to be applied, inter predictor **218** performs motion estimation using DMVR.

(816) FIG. **90** is a flow chart illustrating an example of motion estimation by DMVR in decoder **200**.

(817) Inter predictor **218** derives an MV for a current block according to the merge mode (Step Sl_11). Next, inter predictor **218** derives the final MV for the current block by searching the region surrounding the reference picture indicated by the MV derived in Sl_11 (Step Sl_12). In other words, the MV of the current block is determined according to the DMVR.

(818) FIG. **91** is a flow chart illustrating a specific example of motion estimation by DMVR in decoder **200**.

(819) First, in Step **1** illustrated in FIG. **58A**, inter predictor **218** calculates the cost between the search position (also referred to as a starting point) indicated by the initial MV and eight surrounding search positions. Inter predictor **218** then determines whether the cost at each of the search positions other than the starting point is the smallest. Here, when determining that the cost at one of the search positions other than the starting point is the smallest, inter predictor **218** changes

a target to the search position at which the smallest cost is obtained, and performs the process in Step 2 illustrated in FIG. 58A. When the cost at the starting point is the smallest, inter predictor 218 skips the process in Step 2 illustrated in FIG. 58A and performs the process in Step 3.

(820) In Step 2 illustrated in FIG. 58A, inter predictor 218 performs search similar to the process in Step 1, regarding the search position after the target change as a new starting point according to the result of the process in Step 1. Inter predictor 218 then determines whether the cost at each of the search positions other than the starting point is the smallest. Here, when determining that the cost at one of the search positions other than the starting point is the smallest, inter predictor 218 performs the process in Step 4. When the cost at the starting point is the smallest, inter predictor 218 performs the process in Step 3.

(821) In Step 4, inter predictor 218 regards the search position at the starting point as the final search position, and determines the difference between the position indicated by the initial MV and the final search position to be a vector difference.

(822) In Step 3 illustrated in FIG. 58A, inter predictor 218 determines the pixel position at sub-pixel accuracy at which the smallest cost is obtained, based on the costs at the four points located at upper, lower, left, and right positions with respect to the starting point in Step 1 or Step 2, and regards the pixel position as the final search position. The pixel position at the sub-pixel accuracy is determined by performing weighted addition of each of the four upper, lower, left, and right vectors ((0, 1), (0, -1), (-1, 0), and (1, 0)), using, as a weight, the cost at a corresponding one of the four search positions. Inter predictor 218 then determines the difference between the position indicated by the initial MV and the final search position to be the vector difference.

(823) [Motion Compensation>BIO/OBMC/LIC]

(824) For example, when information parsed from a stream indicates that correction of a prediction image is to be performed, upon generating a prediction image, inter predictor 218 corrects the prediction image based on the mode for the correction. The mode is, for example, one of BIO, OBMC, and LIC described above.

(825) FIG. 92 is a flow chart illustrating one example of generation of a prediction image in decoder 200.

(826) Inter predictor 218 generates a prediction image (Step Sm_11), and corrects the prediction image according to any of the modes described above (Step Sm_12).

(827) FIG. 93 is a flow chart illustrating another example of generation of a prediction image in decoder 200.

(828) Inter predictor 218 derives an MV for a current block (Step Sn_11). Next, inter predictor 218 generates a prediction image using the MV (Step Sn_12), and determines whether to perform a correction process (Step Sn_13). For example, inter predictor 218 obtains a prediction parameter included in the stream, and determines whether to perform a correction process based on the prediction parameter. This prediction parameter is, for example, a flag indicating whether each of the above-described modes is to be applied. Here, when determining to perform a correction process (Yes in Step Sn_13), inter predictor 218 generates the final prediction image by correcting the prediction image (Step Sn_14). It is to be noted that, in LIC, the luminance and chrominance of the prediction image may be corrected in Step Sn_14. When determining not to perform a correction process (No in Step Sn_13), inter predictor 218 outputs the final prediction image without correcting the prediction image (Step Sn_15).

(829) [Motion Compensation>OBMC]

(830) For example, when information parsed from a stream indicates that OBMC is to be performed, upon generating a prediction image, inter predictor 218 corrects the prediction image according to the OBMC.

(831) FIG. 94 is a flow chart illustrating an example of correction of a prediction image by OBMC in decoder 200. It is to be noted that the flow chart in FIG. 94 indicates the correction flow of a prediction image using the current picture and the reference picture illustrated in FIG. 62.

(832) First, as illustrated in FIG. 62, inter predictor **218** obtains a prediction image (Pred) by normal motion compensation using an MV assigned to the current block.

(833) Next, inter predictor **218** obtains a prediction image (Pred_L) by applying a motion vector (MV_L) which has been already derived for the decoded block neighboring to the left of the current block to the current block (re-using the motion vector for the current block). Inter predictor **218** then performs a first correction of a prediction image by overlapping two prediction images Pred and Pred_L. This provides an effect of blending the boundary between neighboring blocks.

(834) Likewise, inter predictor **218** obtains a prediction image (Pred_U) by applying an MV (MV_U) which has been already derived for the decoded block neighboring above the current block to the current block (re-using the motion vector for the current block). Inter predictor **218** then performs a second correction of the prediction image by overlapping the prediction image Pred_U to the prediction images (for example, Pred and Pred_L) on which the first correction has been performed. This provides an effect of blending the boundary between neighboring blocks. The prediction image obtained by the second correction is the one in which the boundary between the neighboring blocks has been blended (smoothed), and thus is the final prediction image of the current block.

(835) [Motion Compensation>BIO]

(836) For example, when information parsed from a stream indicates that BIO is to be performed, upon generating a prediction image, inter predictor **218** corrects the prediction image according to the BIO.

(837) FIG. 95 is a flow chart illustrating an example of correction of a prediction image by the BIO in decoder **200**.

(838) As illustrated in FIG. 63, inter predictor **218** derives two motion vectors (M0, M1), using two reference pictures (Ref.sub.0, Ref.sub.1) different from the picture (Cur Pic) including a current block. Inter predictor **218** then derives a prediction image for the current block using the two motion vectors (M0, M1) (Step Sy_11). It is to be noted that motion vector M0 is a motion vector (MVx.sub.0, MVy.sub.0) corresponding to reference picture Ref.sub.0, and motion vector M1 is a motion vector (MVx.sub.1, MVy.sub.1) corresponding to reference picture Ref.sub.1.

(839) Next, inter predictor **218** derives interpolated image I.sup.0 for the current block using motion vector M0 and reference picture L0. In addition, inter predictor **218** derives interpolated image I.sup.1 for the current block using motion vector M1 and reference picture L1 (Step Sy_12). Here, interpolated image I.sup.0 is an image included in reference picture Ref.sub.0 and to be derived for the current block, and interpolated image I.sup.1 is an image included in reference picture Ref.sub.1 and to be derived for the current block. Each of interpolated image I.sup.0 and interpolated image I.sup.1 may be the same in size as the current block. Alternatively, each of interpolated image I.sup.0 and interpolated image I.sup.1 may be an image larger than the current block. Furthermore, interpolated image I.sup.0 and interpolated image I.sup.1 may include a prediction image obtained by using motion vectors (M0, M1) and reference pictures (L0, L1) and applying a motion compensation filter.

(840) In addition, inter predictor **218** derives gradient images (Ix.sup.0, Ix.sup.1, Iy.sup.0, Iy.sup.1) of the current block, from interpolated image I.sup.0 and interpolated image I.sup.1 (Step Sy_13). It is to be noted that the gradient images in the horizontal direction are (Ix.sup.0, Ix.sup.1), and the gradient images in the vertical direction are (Iy.sup.0, Iy.sup.1). Inter predictor **218** may derive the gradient images by, for example, applying a gradient filter to the interpolated images. The gradient images may be the ones each of which indicates the amount of spatial change in pixel value along the horizontal direction or the amount of spatial change in pixel value along the vertical direction.

(841) Next, inter predictor **218** derives, for each sub-block of the current block, an optical flow (vx, vy) which is a velocity vector, using the interpolated images (I.sup.0, I.sup.1) and the gradient images (Ix.sup.0, Ix.sup.1, Iy.sup.0, Iy.sup.1). As one example, a sub-block may be 4×4 pixel sub-CU.

(842) Next, inter predictor **218** corrects a prediction image for the current block using the optical flow (vx, vy). For example, inter predictor **218** derives a correction value for the value of a pixel included in a current block, using the optical flow (vx, vy) (Step Sy_15). Inter predictor **218** may then correct the prediction image for the current block using the correction value (Step Sy_16). It is to be noted that the correction value may be derived in units of a pixel, or may be derived in units of a plurality of pixels or in units of a sub-block.

(843) It is to be noted that the BIO process flow is not limited to the process disclosed in FIG. 95. Only part of the processes disclosed in FIG. 95 may be performed, or a different process may be added or used as a replacement, or the processes may be executed in a different processing order.

(844) [Motion Compensation>LIC]

(845) For example, when information parsed from a stream indicates that LIC is to be performed, upon generating a prediction image, inter predictor **218** corrects the prediction image according to the LIC.

(846) FIG. 96 is a flow chart illustrating an example of correction of a prediction image by the LIC in decoder **200**.

(847) First, inter predictor **218** obtains a reference image corresponding to a current block from a decoded reference picture using an MV (Step Sz_11).

(848) Next, inter predictor **218** extracts, for the current block, information indicating how the luma value has changed between the current picture and the reference picture (Step Sz_12). This extraction is performed based on the luma pixel values for the decoded left neighboring reference region (surrounding reference region) and the decoded upper neighboring reference region (surrounding reference region), and the luma pixel values at the corresponding positions in the reference picture specified by the derived MVs. Inter predictor **218** calculates a luminance correction parameter, using the information indicating how the luma value changed (Step Sz_13).

(849) Inter predictor **218** generates a prediction image for the current block by performing a luminance correction process in which the luminance correction parameter is applied to the reference image in the reference picture specified by the MV (Step Sz_14). In other words, the prediction image which is the reference image in the reference picture specified by the MV is subjected to the correction based on the luminance correction parameter. In this correction, luminance may be corrected, or chrominance may be corrected.

(850) [Prediction Controller]

(851) Prediction controller **220** selects either an intra prediction image or an inter prediction image, and outputs the selected image to adder **208**. As a whole, the configurations, functions, and processes of prediction controller **220**, intra predictor **216**, and inter predictor **218** at the decoder **200** side may correspond to the configurations, functions, and processes of prediction controller **128**, intra predictor **124**, and inter predictor **126** at the encoder **100** side.

(852) [Constraints on Resolution]

(853) The constraints on resolution are described below by way of several aspects and examples.

(854) (Aspect 1)

(855) The frequent occurrence of reference picture resampling (RPR) causes switching between sizes of subsequent-stage processes including displaying to occur frequently, which may complicate control. The frequent occurrence of RPR causes switching between memory access methods for a current picture and a reference picture (switching between memory maps) to occur frequently, which may complicate control.

(856) In Aspect 1, in order to reduce the frequency of the switching, the resolution change is enabled only at random access pictures or their multiples (i.e., the k-th random access picture). In this manner, it is possible to more simplify control than a case without constraints.

(857) (Aspect 2)

(858) A large number of switching patterns of RPR lead to the corresponding number of patterns of matching sizes of subsequent-stage processes including displaying, which complicates mounting. A

large number of switching patterns of RPR also lead to the corresponding number of patterns of switching between memory access methods for a current picture and a reference picture (switching between memory maps), which complicates mounting.

(859) In Aspect 2, in order to reduce the number of switching patterns, the resolution change (e.g. the RPR) is enabled only between a set of predefined resolutions. In this manner, it is possible to more simplify mounting than a case without constraints.

(860) (Aspect 3)

(861) When a large size not actually used in a group of pictures (GOP) is described in an SPS, decoder **200** secures memory resources etc. in accordance with the size of the SPS, and it is necessary to secure uselessly large resources.

(862) In Aspect 3, in order to enable a more accurate resource scheduling, the SPS and PPS parameters regarding the picture resolution are signaled more consistently. For example, the resolution signaled in the SPS is not allowed to be larger than the maximum of the resolutions signaled in Picture Parameter Sets.

(863) The three aspects mentioned above may be employed to more efficiently use resources for encoding/decoding picture sequences. The three aspects may be employed independently of each other. However, they may also be combined (any two of them or all three aspects). Some specific examples of the present disclosure address these aspects as will be explained below in detail. In particular, some specific embodiments relate to constraining bitstreams using reference picture resampling to ease receivers' (decoders' **200**) processing. It is noted that a picture sequence is a generic term used to refer to pictures.

(864) Some embodiments relate to encoding and/or decoding a sequence of pictures while applying a constraint according to which a picture resolution change is allowed for pictures at every k-th random access point (k being a positive integer), but prohibited for pictures not at the every k-th random access point. The picture resolution change is a change from a first resolution of pictures preceding a picture at a random access point (inter-coded pictures) to a second resolution of the picture at the random access point (intra-coded picture) and pictures following the picture at the random access point (inter-coded pictures).

(865) [Change Picture Resolution]

(866) Versatile Video Coding (VVC) enables the possibility to change the picture resolution of a sequence of pictures at any picture within the sequence, not necessarily at a random access intra-predicted picture. To enable such a feature, it is allowed for any picture to have reference pictures that are not of the same resolution as illustrated in FIG. **97**.

(867) FIG. **97** illustrates reference picture resampling among inter-predicted pictures. In particular, in FIG. **97**, a part of picture sequence **300** is shown, including six pictures. Pictures **310** have a first resolution and pictures **320** have a second resolution lower than the first resolution. Pictures **310** as well as pictures **320** are inter-predicted (i.e. predicted by a prediction mode including temporal prediction) and ordered in sequence of their temporal dependency.

(868) In this example, an arrow from a first picture to a second picture means that the second picture uses the first picture as a reference (correspondingly, the first picture predicts the second picture). As can be seen in FIG. **97**, the resolution change occurs after three pictures **310**, at the first picture among pictures **320**.

(869) Such resolution change among the inter-predicted pictures requires resampling the reference pictures, for example using the interpolation filter of the motion compensation process also as a resampling filter. VVC provides such a tool, which is called Reference Picture Resampling (RPR) and which was introduced to ease picture resolution changes for applications such as video conferencing that typically do not use many random-access pictures, and may have different speakers with different resolution video streams depending e.g. on their equipment. This may lead to often changing resolution depending on who actively speaks.

(870) Although the concept of VVC RPR was developed for video conferencing applications, it

may also be used for broadcast or streaming applications that would typically use random-access pictures regularly (typically at least every couple of seconds).

(871) In some particular broadcast applications, so-called open Group Of Pictures (open GOP) structures are employed, in which pictures belonging to the previous GOP may be used to predict pictures that are located after a random-access picture in encoding order and before the random-access picture in display order, whereby a GOP is defined as the pictures located between two random-access pictures in display order, including the random-access picture starting the GOP. Alternatively, a GOP may be defined as pictures located between two random-access pictures in encoding order.

(872) After the random-access picture in display order, all following pictures in the same GOP are predicted using only the random-access picture or pictures following it. In this manner, random-access is enabled at that point.

(873) Regarding the terminology, random access intra-predicted picture is a picture which is predicted spatially (intra-prediction) and which enables random access, as its decoding and the decoding of the following pictures in display order does not depend on previous decoding of other pictures of the bitstream. Typically, random access intra-predicted picture does not include any inter-predicted parts and the entropy coding/decoding is restarted at the beginning of the picture.

(874) The random access intra-predicted picture may be also referred to as an intra random access point picture (IRAP). The random access intra-predicted picture may be further referred to as a random access point picture, a random access picture, a random access point, or an IRAP. The GOP corresponds to a unit of random access.

(875) The term “display order” refers to the order in which the pictures decoded from the bitstream are to be displayed for reproduction. In more general terms, the display order corresponds to the order in which pictures are input to encoder **100** for picture encoding (input order) and to the order in which pictures are output from decoder **200** (output order).

(876) The encoding order is the order in which pictures are encoded, and the decoding order is the order in which pictures are decoded. Basically, the encoding order is the same as the decoding order. In contrast, the encoding order is often different from the display order.

(877) [Receiver Friendly Constraints on Reference Picture Resampling (Aspect 1)]

(878) The RPR is a useful feature, but the frequent occurrence of the RPR may cause some implementation issues, especially at the receiver side. In order to address these issues, in the following, some constraints in connection with the resolution change are introduced.

(879) [Change of Resolution at Random-Access Point]

(880) According to an aspect of the present disclosure, an apparatus (encoder **100**) is provided for encoding a sequence of pictures into a bitstream. More specifically, the apparatus comprises a processing circuitry which is configured to encode the sequence of pictures into the bitstream. Moreover, the processing circuitry is configured to apply a constraint (in the encoding) according to which a picture resolution change is: allowed for pictures at every k-th random access point (k being a positive integer); and prohibited for pictures not at the every k-th random access point.

(881) When referring to “every k-th random access point”, what is meant is, for instance every one, or every second or every third, etc. random access point.

(882) In other words, “every k-th random access point” corresponds to random access points each identified by interposing k-1 random access points, and corresponds to one of k random access points in each group.

(883) For example, in the case of k=1, “every k-th random access point” corresponds to each random access point. Moreover, in the case of k=2, “every k-th random access point” corresponds to one of two random access points in each group, and in the case of k=3, “every k-th random access point” corresponds to one of three random access points in each group.

(884) It is noted that an initial offset may be given, which means that the first random access point at which the resolution change is allowed is not necessarily k-th random access point when counted

from the start of the sequence, i.e., it may actually be a random access point corresponding to the order larger than or smaller than k.

(885) The picture resolution change is a change from a first resolution of pictures preceding a picture. “Preceding” indicates that the order is earlier in the encoding and/or the input order. More specifically, the picture resolution change is a change from a first resolution of (one or more) pictures preceding a picture at a random access point in the encoding and/or input order to a second resolution of the picture at a random access point and (one or more) pictures following said picture at the random access point in the encoding and/or input order.

(886) With the above configuration, reference picture resampling can be enabled also for example in the particular case of broadcast and broadband delivery applications, which frequently use random-access pictures and open-GOP encoding structure. As a particular constraint, it is allowed for the picture resolution to change only at a random-access picture. The change may be effectuated between typical broadcast video resolutions, as illustrated in three exemplary implementations on FIG. 98A, FIG. 98B, and FIG. 98C, where each rectangle represents a picture, for a change of resolution from a HD resolution to a 4K resolution.

(887) The term “4K resolution” generally refers to a picture resolution with the number of about 8 megapixels, roughly corresponding to a vertical or horizontal size of 4000 pixels (e.g. luma samples). The particular vertical and horizontal sizes of 4K pictures may differ, e.g. 4K pictures may still have various different aspect ratios, such as 4096×2160, 3996×2160, 3996×1716, 3840×2160, 3840×2400, or the like. These numbers specify number of columns times number of rows, i.e. horizontal times vertical resolution in units of pixels (samples).

(888) The term “HD resolution” or “full HD resolution” typically refers to a picture resolution of 1920×1080, resulting in approximately 2 megapixels per picture. However, the term may also be applied for other aspect ratios of approximately 2 megapixels. For the purpose of the example shown in FIG. 98A, FIG. 98B, and FIG. 98C, the HD resolution may be defined as 1920×1080 pixels and the 4K resolution may be defined as 3840×2160 pixels. However, the present disclosure is not limited to any specific resolutions and can be applied to any resolution change.

(889) FIG. 98A, FIG. 98B, and FIG. 98C illustrate three scenarios. Each scenario shows pictures ordered in display order (which corresponds to the input order for encoder 100 and to the output order for decoder 200). The pictures in this example are encoded in four temporal sub-layers (tId: 0 to 3) to support temporal scalability. The temporal scalability may be implemented, e.g. as in the VVC. The term “tId” in FIG. 98A, FIG. 98B, and FIG. 98C denotes a temporal ID, i.e. an ID of a temporal sub-layer. However, the temporal sub-layer coding is shown only as an example. The present disclosure is applicable to streams with or without temporal scalability.

(890) In an exemplary implementation, the encoding is performed using VVC codec, and the resolution change is indicated in the bitstream by both or one of: 1)

pps_pic_width_in_luma_samples and/or pps_pic_height_in_luma_samples, and 2) one or more of pps_scaling_win_left_offset, pps_scaling_win_right_offset, pps_scaling_win_top_offset, and pps_scaling_win_bottom_offset.

(891) For example, in case of a VVC based implementation, it is possible to use reference picture resampling within the same VVC layer by setting the syntax elements sps_ref_pic_resampling_enabled_flag and sps_res_change_in_clvs_allowed_flag of the Sequence Parameter Set (SPS) both equal to 1. The parameter sps_ref_pic_resampling_enabled_flag can take one of two values which indicate enabling and disabling of the reference picture resampling for the sequence referring to the SPS in which this parameter is carried.

(892) The sps_ref_pic_resampling_enabled_flag equal to 1 specifies that reference picture resampling is enabled. In this case, a current picture referring to the SPS may have slices that refer to a reference picture that has one or more of the following seven parameters different than that of the current picture: 1) pps_pic_width_in_luma_samples, 2) pps_pic_height_in_luma_samples, 3) pps_scaling_win_left_offset, 4) pps_scaling_win_right_offset, 5) pps_scaling_win_top_offset, 6)

pps_scaling_win_bottom_offset, and 7) sps_num_subpics_minus1.

(893) The sps_ref_pic_resampling_enabled_flag equal to 0 specifies that reference picture resampling is disabled. In this case, no current picture referring to the SPS has slices that refer to a reference picture that has one or more of these seven parameters different than that of the current picture.

(894) The parameter sps_res_change_in_clvs_allowed_flag indicates enabling and disabling of the reference picture resampling for a particular coded layer video sequence (CLVS) referring to the SPS.

(895) In particular, sps_res_change_in_clvs_allowed_flag equal to 1 specifies that the picture spatial resolution may change within a CLVS referring to the SPS, thereby enabling resolution change within a single-layer bitstream. The sps_res_change_in_clvs_allowed_flag equal to 0 specifies that the picture spatial resolution does not change within any CLVS referring to the SPS, but may change between layers, thereby enabling spatial scalability between different layers.

(896) The resolution change itself is then signaled e.g. by changing the values of pps_pic_width_in_luma_samples and pps_pic_height_in_luma_samples of the Picture Parameter Set (PPS) from the resolution of the previous GOP to the resolution of the new GOP. PPS parameters pps_pic_width_in_luma_samples and pps_pic_height_in_luma_samples indicate respectively the width and the height of pictures referring to said PPS.

(897) It is noted that if the resolutions are recognizable only based on the width or only based on the height, only one of them may be signaled without signaling the other one in some conceivable implementations.

(898) In the present embodiment, changes to the values of the two syntax elements pps_pic_width_in_luma_samples and pps_pic_height_in_luma_samples in the referred PPSs are only allowed at a random-access point, typically a VVC Intra Random Access Point IRAP picture.

(899) In case the VVC scaling window is used, the following syntax elements specified as offsets for the scaling window in the referred PPSs are also only allowed to change at a random-access point: 1) pps_scaling_win_left_offset, 2) pps_scaling_win_right_offset, 3) pps_scaling_win_top_offset, and 4) pps_scaling_win_bottom_offset.

(900) The above parameters specify the offsets that are applied to the picture size for scaling ratio calculation. When not present, the values of these parameters are inferred to be equal to the following conformance window parameters, respectively: 1) pps_conf_win_left_offset, 2) pps_conf_win_right_offset, 3) pps_conf_win_top_offset, and 4) pps_conf_win_bottom_offset.

(901) FIG. 99 is a schematic drawing illustrating a relationship between the parameters. The picture width in a whole range is denoted by pps_pic_width_in_luma_samples, and the picture height in a whole range is denoted by pps_pic_height_in_luma_samples.

(902) The picture width in a reference range can be obtained by removing pps_scaling_win_left_offset and pps_scaling_win_right_offset from pps_pic_width_in_luma_samples. The picture height in a reference range can be obtained by removing pps_scaling_win_left_offset and pps_scaling_win_right_offset from pps_pic_height_in_luma_samples.

(903) The picture resolution may be represented by a picture size in a whole range or by a picture size in a reference range. In encoding or decoding a current picture, when the resolution is different between a current picture and a reference picture, the reference picture is resampled so that the current picture and the reference picture are substantially the same in the resolution.

(904) More specifically, a reference image in the reference picture is resampled so that the reference image in the reference picture is scaled according to the ratio of the resolution of the current picture to the resolution of the reference picture. For example, when the resolution of the current picture is larger than the resolution of the reference picture, the reference image in the reference picture is up-sampled using the motion compensation filter. When the resolution of the current picture is smaller than the resolution of the reference picture, the reference image in the

reference picture is down-sampled using the motion compensation filter.

(905) Even when the size of the current picture in a whole range is the same as the size of the reference picture in a whole range, the reference image in a reference range of the reference picture can be resampled using an offset for determining the reference range of the reference picture. More specifically, the image in a reference range of the reference picture may be resampled so that the reference picture in the reference range and the current picture in the whole range are substantially the same in the resolution. For example, the offset is useful when a part of the reference picture is applied to the whole prediction image of the current picture according to zoom or the like.

(906) When the offset is constant, the picture resolution change may be controlled by changing the picture size in the whole range. Alternatively, when the whole range of the picture is constant, the picture resolution change may be controlled by changing the picture size in the reference range. In particular, in some examples of the present disclosure, the picture resolution is changed only at a random access point.

(907) The constraint of changing the resolution of the pictures only at a random-access point covers the three examples from FIG. **98A**, FIG. **98B**, and FIG. **98C**.

(908) In the example of FIG. **98A**, the processing circuitry of encoder **100** is further configured to encode an intra-coded random-access picture (IRAP—4K in the figure) of a second group of pictures (with a 4K resolution), followed by encoding leading pictures (leading inter-predicted pictures) **330** which belong to a first group of pictures (previous GOP) preceding the second group of pictures in input order, followed by encoding of trailing pictures (trailing inter-predicted pictures) belonging to the second group of pictures.

(909) The resolution of leading pictures **330** does not match the resolution of the IRAP and the trailing pictures. This can be seen in FIG. **98A** in that leading pictures **330** have a HD resolution wherein the IRAP and the trailing pictures following the IRAP in the input order have a 4K resolution. Leading pictures **330** may use one or more pictures (typically the IRAP picture) from the second group of pictures as reference pictures.

(910) The term “group of pictures (GOP)” herein may be, but is not necessarily the (closed) GOP as is known from some early codecs. As is exemplified above in FIG. **98A**, the open GOP may also be supported.

(911) The present disclosure employs the term “group of pictures” more broadly as a set of consecutive pictures including an intra-predicted (or not predicted at all) random access picture and one or more inter-predicted pictures. In general, groups of pictures in a sequence of pictures (video) may have the same size or may have different sizes (in term of the amount of pictures). The present disclosure is not limited to any particular structure of the group of pictures.

(912) It is noted that inter-predicted pictures may still comprise some regions or blocks (coding unit) encoded in intra mode. However, intra-predicted pictures typically do not comprise any parts or blocks which would be inter-predicted. IRAP pictures are typically intra-predicted pictures with additional rules on emptying the reference picture buffer at some point at or after the IRAP in coding order to enable random-access from the IRAP.

(913) In FIG. **98A**, an open-GOP with leading pictures **330** is used, but leading pictures **330** use a different resolution from the IRAP and its trailing pictures. The reference picture resampling is applied in this example, and the resolution change happens at the IRAP in the display order (input order), but not in the coding order, since leading pictures **330** are coded after the IRAP picture. In this example, a display controller can easily synchronize the timing of the IRAP display with the timing of the resolution change.

(914) Decoder **200** decodes the bitstream and may use the constraints to plan its resources. Trailing inter-predicted pictures may not use leading pictures **330** as reference in the open GOP scenario, otherwise it would not be possible to random-access the video at the IRAP, because leading pictures **330** typically use the pictures from the previous GOP also as reference pictures in the open GOP scenario.

(915) In order to enable random access, the reference picture buffer is emptied of all pictures other than the IRAP at the IRAP (closed GOP case) or when starting to decode the trailing pictures (open GOP case). In FIG. **98A**, the reference picture resampling is also used in that case because leading pictures **330** use the IRAP picture of a different resolution as a reference picture.

(916) FIG. **98B** illustrates another exemplary implementation of the constraint that the resolution change may only occur at random access points. Accordingly, the processing circuitry is further configured to encode an intra-coded random-access picture (IRAP—4K in the figure) of a second group of pictures, followed by encoding leading pictures (leading inter predicted pictures) **340** which belong to a first group of pictures (previous GOP) preceding the second group of pictures in input order, followed by encoding of trailing pictures (trailing inter-predicted pictures) belonging to the second group of pictures.

(917) Unlike the implementation of FIG. **98A**, the implementation of FIG. **98B** shows leading pictures **340** with the same resolution (4K) as the IRAP. Nevertheless, these leading pictures **340** still follow the IRAP in the encoding order but not in the input order. Leading pictures **340** use one or more pictures of the first group of pictures (previous GOP) as reference pictures in the open GOP scenario.

(918) The resolution of leading pictures **340** is the same as the resolution of the IRAP (4K in this example) and the resolution of the trailing pictures of the second group of pictures, but not the same as the resolution (HD in this example) of the remaining pictures in the first group of pictures.

(919) Since the IRAP and leading pictures **340** have the same resolution, no resampling is necessary at encoder **100** (and decoder **200**) when using the IRAP as reference picture, but reference picture resampling may be necessary when using any of the remaining pictures in the first group of pictures (the trailing pictures in the previous GOP) as reference pictures during the encoding/decoding. In particular, in FIG. **98B**, the reference picture resampling may be used if leading pictures **340** are using pictures from the previous GOP as references.

(920) However, reference picture resampling is not necessary if leading pictures **340** use only the IRAP as reference. In FIG. **98B**, the resolution change happens at the IRAP picture in coding order, but at first leading picture **340** in the display order (input order in encoder **100** view, output order in decoder **200** view). In display order, except first leading picture **340**, encoder **100** and decoder **200** do not need to change the resolution within a GOP.

(921) FIG. **98C** illustrates another example in which the bitstream includes said sequence of pictures encoded in a closed group of pictures. A second closed group of pictures comprises an intra-coded random-access picture, IRAP, and one or more inter-predicted pictures which follow the IRAP in both the input and coding orders. These inter predicted pictures are referred to in the figure as trailing pictures—4K. A closed group of pictures is a group of pictures in which the one or more inter-predicted pictures in the group of pictures use, as reference pictures, pictures only from said group of pictures.

(922) Accordingly, in the example of FIG. **98C**, no leading picture is used; a first group of pictures has only a not shown IRAP picture and trailing pictures denoted in the figure as trailing pictures of previous GOP-HD. The resolution change happens at the IRAP picture in both coding and display order, and reference picture resampling is not necessary. The example of FIG. **98C**, shares the benefits of both examples in FIG. **98A** and FIG. **98B**, but may be less efficient than the open-GOP structure in terms of coding efficiency.

(923) In summary, in FIG. **98A**, the resolution change is at the IRAP in output order (term used for display order in video coding), but not in the decoding order, since the leading pictures having a resolution different from that of the IRAP picture are decoded after the IRAP. In FIG. **98B**, the resolution change is at the IRAP in the decoding order, but not in the output order, since the leading pictures having the same resolution as the IRAP picture are output before the IRAP picture. In FIG. **98C**, the resolution change is at the IRAP in both decoded and output order.

(924) According to the above constraints, only when the current picture is a leading picture, the

resolution may be different between the current picture and the reference picture. Accordingly, only when the current picture is a leading picture, the reference picture may be allowed to be resampled. (925) It is noted that the leading picture can be defined as a picture preceding the IRAP picture in display order and following the IRAP picture in encoding order. The trailing picture can be defined as a picture following the IRAP picture in display order and following the IRAP picture in encoding order.

(926) [Change of Resolution with Limited Frequency]

(927) According to the above constraints, the resolution change is allowed to occur only at IRAP (in encoding/decoding or input/output order), for pictures at every k -th (k being an integer equal to or greater than 1) random access point and not for other pictures. According to an exemplary embodiment, said k is a preconfigured integer larger than one. In other words, setting k larger than 1 may limit the frequency of resolution change. In this manner, the frequent resolution change is prevented, and an issue especially at receivers when post-processing and displaying the picture in broadcast and broadband applications is reduced.

(928) For example, in one aspect of the present disclosure, two successive changes of resolutions is caused by two successive changes of `pps_pic_width_in_luma_samples` and `pps_pic_height_in_luma_samples` or two successive changes to the scaling window offsets if used in the referred PPSs. The time interval between two successive changes of resolutions is not allowed to be less than a value in seconds, which should correspond to several random-access periods.

(929) A suitable value may be 5 seconds, which corresponds to 4 to 5 random-access pictures when random-access pictures are inserted approximately every second of the video. However, it is noted that the present disclosure is not limited to these numbers. In general, the frequency of the random access pictures may differ and be adapted to the application such as video conferencing, streaming, broadcasting, or the like, and thus the time interval size for limiting the resolution change for broadcast and broadband applications may be defined. The time interval size may be defined in terms of the number of IRAPs (k) or in terms of seconds.

(930) According to this constraint, an example shown in FIG. 100A would be a bitstream conforming to the constraint, for instance, for broadcast and broadband applications (e.g. a VVC bitstream following the constraint), whereas an example in FIG. 100B would not be a bitstream conforming to the constraint, for instance, for the broadcast and broadband applications. FIG. 100A and FIG. 100B shows the two examples of resolution changes within a bitstream.

(931) In 100A, the first GOP has a HD resolution, followed by five GOPs 350 with 4K resolution and another GOP with HD resolution again. Thus, FIG. 100A is conform with possible requirements $k=5$, $k=3$, and/or $k=2$. In 100B, the first GOP has a HD resolution, followed by a single GOP with 4K resolution, then a single GOP with HD resolution, three GOPs with 4K resolution and finally a GOP with HD resolution. This stream does not comply with any $k>1$.

(932) In one exemplary implementation, the minimum allowed time interval between two successive changes in resolution may be a multiple value of a fixed random-access period when IRAP pictures are inserted in the bitstream using a fixed period. In this case, the minimum allowed time interval and/or the fixed random-access period may be defined in terms of time units (seconds). Other implementations are conceivable, in which the minimum allowed time interval between two successive changes in resolution may be a multiple value of a variable random-access period. Such implementations would still reduce the frequency of the resolution change.

(933) It is noted that the random-access period corresponds to a period from a random access point to the next random access point, and is also referred to as the IRAP period. Moreover, as described above, the minimum allowed time interval need not necessarily be defined in terms of time units such as seconds, and may be defined in terms of the number of pictures, etc. Accordingly, the minimum allowed time interval may be referred to as the minimum allowed interval or the minimum interval.

(934) The constraints discussed above (e.g. k being equal to or larger than 1) may be implemented at encoder **100** as a part of generating the picture sequence to be encoded or pre-processing of the picture sequence before the encoding.

(935) An exemplary implementation of an encoding method is provided in FIG. **101**. A sequence of pictures to be encoded is obtained at first (**S410**). Such obtaining may be from content producer, from a camera (such as a webcam) operated by a streaming or video conferencing application or the like.

(936) The sequence may already have the resolution change in the desired positions (defined by k and the desired IRAP period). For example, it may be ensured that the picture sequence to be encoded has resolution change only for each k times the random access period. As a matter of pre-processing (**S420**), if this is not the case, some of the pictures may be resealed to match this limitation. Rescaling may be upscaling (e.g. up-sampling) or downscaling (e.g. down-sampling). Rescaling approaches are known in the art and any of the known approaches may be readily applied.

(937) Then, encoder **100** applies the constraint by employing the random access period and k as described above to encode the input sequence during the encoding (**S430**). The encoding may comply with VVC or other standardized codec to produce a bitstream.

(938) The bitstream may be decoded by a decoding method illustrated on the right hand side of FIG. **101**. In particular, the bitstream is obtained, e.g. from a channel such as broadcast broadband channel or via streaming or video conferencing or the like (**S415**).

(939) The bitstream or additional control information may indicate to decoder **200** the application of the constraints (k and/or the IRAP period). Decoder **200** configures its resources accordingly (e.g. allocates memory and processing power depending on the resolution for the constrained time periods) (**S425**). It is noted that the configuration (**S425**) may be performed at some decoders and does not need to be performed at other decoders. In other words, the configuration (**S425**) may be performed, or need not be performed.

(940) Then the sequence of pictures is decoded from the bitstream in step **S435**. The decoding (**S435**) itself does not necessarily need the explicit information of the constraints.

(941) A more detailed exemplary implementation of a method for a broadcast encoder corresponding to encoder **100** is shown in FIG. **102**. FIG. **102** is a flowchart illustrating an example process for a broadcast encoder that can take as an input a video signal that may be coming from several cameras using different resolutions, and create a VVC bitstream out of it.

(942) First, when starting encoding of a new piece of content, variable $tResChange$ representing the time (time instant) of the last resolution change is set equal to $-x$ seconds in step **S505**, where x is set to the value of the minimum allowed time interval between successive resolution changes, for example $x=5$ seconds in the example of FIG. **102**. As described above, the time interval may be measured and indicated in terms of time units such as seconds or in terms of the number of pictures or the number of GOPs or the like. As an alternative, $tResChange$ may also be initialized to zero.

(943) Then, in step **S510**, the picture from camera $currC$ (e.g. a current camera out of a plurality of cameras capturing content) with a resolution of width $currW$ (current width) and height $currH$ (current height) is encoded.

(944) Next, in step **S520**, it is decided whether there are pictures to be encoded from camera $altC$ with a resolution of width $newW$ and height $newH$. Decision step **S520** represents checking which camera is active. For example, in case the camera delivering the input video signal suddenly changes, more pictures arrive from a camera different from $currC$ (denoted thus as alternative camera $altC$) with a resolution of width $newW$ and height $newH$. In this case (“yes” in step **S520**), in step **S530**, it is checked whether the values of $currW$ and $newW$ are equal and whether the values of $currH$ and $newH$ are equal.

(945) When this is the case (“yes” in step **S530**), the encoding can continue without resolution change and variable $currC$ denoting the current camera is simply set to $altC$ in step **S540**.

Otherwise (“no” in step S530), variable t representing the current media time is determined in step S550. Typically, t would be calculated as the number of frames in display order (input order at encoder 100) before the first picture coming from camera altC divided by the frame rate of the content.

(946) Then, in step S560, it is checked how big the time interval (corresponding to t minus $t_{\text{ResChange}}$) since the last change in resolution is. When the value of t minus $t_{\text{ResChange}}$ is less than x seconds (x being equal to 5 in the example of FIG. 102) (“no” in step S560), a resolution change is not allowed, so that the picture from camera altC is resampled in step S580 to a resolution of width currW and height currH , before continuing the encoding (S510) with currC set to altC in step S540.

(947) When the value of t minus $t_{\text{ResChange}}$ is greater than or equal to x seconds (x being equal to 5 in the example of FIG. 102) (“yes” in step S560), an IRAP picture with a resolution of width newW and height newH is inserted into the bitstream. Any remaining pictures from camera currC not yet encoded are encoded with the previous resolution of width currW and height currH as leading pictures of the IRAP picture using reference picture resampling (S570).

(948) Then, variables currW , currH and $t_{\text{ResChange}}$ are updated to the new values of newW , newH and t , respectively in step S590, before continuing the encoding (S510) with currC set to altC in step S540.

(949) When there are no pictures to be encoded from another camera altC, i.e., no more pictures from any camera arrive to encoder 100 (“no” in step S520), the process ends with the encoded bitstream as output.

(950) It is noted that the example of FIG. 102 is not to limit the present disclosure. It is a detailed example of encoder 100 suitable for broadcasting or other applications. However, the particular implementation may differ in many ways. For example, instead of testing whether t minus $t_{\text{ResChange}}$ exceeds x , $t_{\text{ResChange}} - t$ may be tested. The example of FIG. 102 generates a bitstream similar to FIG. 98A. As mentioned above, the constraints may be implemented in various different ways, and bitstreams as exemplified in FIG. 98B and FIG. 98C may be generated.

(951) [Exemplary Implementation Based on DASH]

(952) Dynamic Adaptive Streaming over HTTP (DASH) is also known as MPEG-DASH and is an adaptive bitrate streaming technique that enables streaming of media content over the Internet delivered from conventional HTTP web servers. MPEG-DASH works by breaking the content (e.g. picture sequence) into a sequence of small segments, which are served over HTTP.

(953) Each segment contains a short interval of playback time of the content (e.g. a movie or a live broadcast or the like). The content may be made available at a plurality of different bit rates, i.e., alternative segments encoded at different bit rates covering aligned short intervals of playback time. While the content is being played back by an MPEG-DASH client, the client may use a bit rate adaptation algorithm to automatically select the segment with the highest bit rate possible that can be downloaded in time for playback without causing stalls or re-buffering events in the playback. In other words, DASH already implements a possibility of an adaptive bitrate.

(954) According to an embodiment which may be based on any of the above mentioned embodiments implementing constraints on the resolution change, the minimum time interval between two successive changes in resolution may also be a value equal to a segment size or a multiple of a segment size in a DASH based delivery. In the DASH based delivery of content, as mentioned above, a bitstream is split into multiple segments, and each of the segments starts with an IRAP picture to enable switching between segments of different DASH representations (e.g. with different bitrates mentioned above).

(955) For example, an apparatus (encoder 100) is provided with a processing circuitry as mentioned above. The processing circuitry may be further configured to encode a first bitstream that is a first representation of the sequence of pictures and encode a second bitstream that is a second representation of the same sequence of pictures with a different quality or resolution. The

processing circuitry may be further configured to receive, via DASH protocol, a request for the first representation or the second representation from a receiver and to transmit, via DASH protocol, to the receiver the first bitstream or the second bitstream, depending on the request.

(956) In other words, encoder **100** of the present disclosure may also support DASH. The encoded first bitstream and the encoded second bitstream may include the same number (amount) of DASH segments. In order to support the preceding embodiments, the minimum allowed time interval between two successive changes in resolution of the first encoded bitstream and/or the second encoded bitstream is equal to the segment size or a multiple of the segment size. In other words, the k GOPs mentioned above may be constrained to include an integer number of segments, the integer number being 1 or more than one.

(957) DASH based services typically already offer different representations of the same content. Each representation has a different bitrate or even a different resolution. A DASH receiver has the choice of which DASH segments to download, decode and display. While a DASH service may not change the resolution at all within a DASH representation or the interval between two successive changes of resolution within a DASH representation may be constrained as in the present disclosure. However, due to the switching between DASH representations, a DASH receiver is free to change the resolution of the downloaded bitstream more often.

(958) For example, even when a DASH service uses DASH segments of 0.5 seconds in duration and the interval between two successive changes of resolution within a DASH representation is constrained to at least 5 seconds, a DASH receiver may still decide to switch between DASH representations with different resolutions more often. In other words, a DASH receiver may download a bitstream such as in FIG. **100B** by switching between DASH representations often.

(959) On the other hand, a DASH receiver that would not be able to deal with such frequent changes in resolutions should stick to downloading only the segments of one DASH representation during the minimum allowed time interval, since the constraint ensures that there are no frequent changes of resolutions within a single DASH Representation.

(960) As usual for DASH based delivery, DASH receivers have the choice to ignore a constraint when the DASH receiver has better capabilities than the minimum required. Thus, even though DASH already provides flexibility, it may be advantageous to impose a constraint on the DASH encoder corresponding to encoder **100** to allow resolution change within one representation after a minimum allowed time interval corresponding to multiples (1, 2, 3, or more) of DASH segments. The constraint in some exemplary implementation also may correspond to the constraints mentioned above, namely changing the resolution only at multiples (k) of IRAPs (after multiples of groups of pictures).

(961) FIG. **103** is a schematic drawing illustrating a server and a receiver compliant with DASH protocol. In this example, server **1001** corresponding to encoder **100** holds representation A for a normal bit rate and representation B for a low bit rate. Representation A includes segments A-1, A-2, A-3, A-4, . . . , A- m . Representation B includes segments B-1, B-2, B-3, B-4, . . . , B- m .

(962) Server **1001** receives a request from receiver **1002** corresponding to decoder **200**, and transmits a segment sequence to receiver **1002** according to the request. In this example, server **1001** transmits segments A-1, A-2, B-3, B-4, . . . , A- m to receiver **1002**.

(963) Each segment includes one or more GOPs. Each of the GOPs belongs to only a single segment. For example, the resolution change is not allowed in each segment, and the resolution change is allowed between segments.

(964) Two representations A and B are shown here, but the number of representations is not limited to 2. A different resolution may be set for each representation. In each representation, the resolution change may be allowed only at the segment interval of j segments or more. When the resolution change is not allowed in each segment, in each representation, the resolution change may be allowed in units of segments only at the segment interval of j segments or more.

(965) For example, when receiver **1002** requests multiple segments of the same representation, the

resolution can be changed only at the segment interval of j segments or more. On the other hand, when receiver **1002** requests multiple segments while switching between representations different from each other, the resolution can be changed more often.

(966) When having capabilities to handle high frequent resolution change, receiver **1002** may request multiple segments while frequently switching between representations. In contrast, when having no capabilities to handle high frequent resolution change, receiver **1002** may request multiple segments without frequently switching between representations.

(967) [Resolution Change for Rate Control]

(968) According to an exemplary implementation, resolution change may be employed as means to perform rate control. For example, in the above mentioned encoder **100** side apparatus, the processing circuitry may be further configured to monitor an amount of bits in the encoded bitstream. Then, based on the monitored amount of bits, the processing circuitry may select the first resolution or the second resolution, and encode the sequence of pictures with the selected resolution. Here, the monitoring the amount of bits corresponds to monitoring of rate.

(969) There may be some requirements defined for the rate to be achieved which have to be checked and according to which it is decided to change or not the resolution. The requirements may be based on the number of bits per unit of time (such as second) over some predetermined time interval not exceeding certain value or the like. A detailed and non-limiting example of such rate control mechanism is provided below. The rate control mechanism may still obey the constraints defined in any of the preceding embodiments regarding the resolution changes at IRAPs or multiples of IRAPs (every k -th IRAP).

(970) FIG. **104A** and FIG. **104B** are a flowchart illustrating an example process for a broadcast encoder (encoder **100**) using rate control and taking as an input a video signal (sequence of pictures) with a resolution of width baseW and height baseH . First, in step **S610**, variables currW and currH representing the current resolution width and height, respectively, are set equal to baseW and baseH , respectively.

(971) In step **S615**, variable tResChange representing the time (time instant) of the last resolution change is set equal to $-x$ seconds, where x is set to the value of the minimum allowed time interval between successive resolution changes (for example $x=5$ seconds). As alternative, tResChange may also be initialised to zero. Variable BufferFullness representing the number of bits in the Coded Picture Buffer is set equal to zero (also in step **S615**).

(972) After these initializations, one Group Of Pictures (GOP) of the signal (picture sequence) is encoded with the resolution of width currW and height currH (**S620**) and the number of bits in the Coded Picture Buffer is tracked in variable BufferFullness .

(973) It is determined whether a picture to be encoded remains in step **S625**. When a picture to be encoded remains (“yes” in step **S625**), it is checked in step **S628** whether the value of BufferFullness is greater than a lower threshold denoted as LowThresBFullness and lower than or equal to a standard threshold denoted as StdThresBFullness , to check that the current bitstream rate is compatible with keeping the number of bits in the Coded Picture Buffer within a range determined by the application.

(974) When this is the case (“yes” in step **S628**), the encoding (**S620**) can continue without resolution change and the next GOP is encoded with resolution of width currW and height currH . Otherwise (“no” in step **S628**), variable t representing the current media time is determined to find out whether a resolution change for increasing or decreasing the bitstream bitrate is appropriate. Typically, t would be calculated as the number of frames in display order obtained so far divided by the frame rate of the content (**S630**) similarly as already mentioned with reference to FIG. **102**.

(975) Then, it is checked in step **S635**, how big the time interval since the last change in resolution is. When the value of t minus tResChange is less than x seconds (“no” in step **S635**), a resolution change is not allowed, so that the encoding (**S620**) has to continue without resolution change and the next GOP is encoded with resolution of width currW and height currH .

(976) When the value of t minus $tResChange$ is greater than or equal to x seconds (“yes” in step **S635**), it is checked more in details whether the current bitstream rate is appropriate by comparing $BufferFullness$ to the lower threshold $LowThresBFullness$ in step **S638**. When $BufferFullness$ is lower than the threshold $LowThresBFullness$ (“yes” in step **S638**), the bitstream rate is actually too low, so that there is no need to decrease the content resolution for the purpose of the rate control. (977) In that case, variables $newW$ and $newH$ representing the new width and height, respectively, are reset to the original input values of $baseW$ and $baseH$, respectively in step **S640**. An IRAP picture with resolution width $newW$ and height $newH$ is inserted into the bitstream (**S650**) and the encoding starts a new GOP with the new resolution using reference picture resampling. Variables $currW$ and $currH$ are updated to the new resolution $newW$ and $newH$, respectively, and variable $tResChange$ is updated to new value t at which the resolution change occurs in step **S660**, before continuing the encoding (**S620**) with resolution of width $currW$ and height $currH$.

(978) When $BufferFullness$ is greater than $LowThresBFullness$ (“no” in step **S638**), it means that $BufferFullness$ is also greater than the standard threshold $StdThresBFullness$, so that the resolution of the content should be decreased to decrease the bitrate of the bitstream. One possible way to decrease the resolution is by dividing the current width and/or height by a predefined factor, such as 2 for example, as in step **S670**.

(979) Then, it is tested in step **S675**, whether the bitrate is too high. When the bitrate is really very high, $BufferFullness$ is even greater than a second higher threshold denoted as $HighThresBFullness$. In this case (“no” in step **S675**), it may even be beneficial to reduce the current resolution even more by dividing the current width and/or height by twice (or more times) the predefined factor as shown in step **S680**. Additional decision units may be added to even further decrease the current resolution based on predetermined buffer fullness ranges and factors. In other words, the present disclosure may include selection of the predefined factor or its multiples according to the size of the $BufferFullness$.

(980) $BufferFullness$ is greater than standard threshold $StdThresBFullness$ and lower than or equal to the second higher threshold $HighThresBFullness$ (“Yes” in step **S675**), the above process (step **S680**) is skipped.

(981) Once a new resolution of width $newW$ and height $newH$ has been determined (in step **S670** or **S680**), the input video signal is resampled from its original resolution of width $baseW$ and height $baseH$ to the new calculated resolution of width $newW$ and height $newH$ (**S690**), before inserting an IRAP picture of resolution width $newW$ and height $newH$ into the bitstream and starting a new GOP with the new resolution using reference picture resampling in step **S650**.

(982) When no more pictures to be encoded remains (“no” in step **S625**), the process ends with the encoded bitstream as output.

(983) It is noted that the above detailed example may be modified. For example, three thresholds have been discussed: $LowThresBFullness$, $HighThresBFullness$, and $StdThresBFullness$. However, it is conceivable only having two thresholds ($StdThresBFullness$ and one of the $LowThresBFullness$ and $HighThresBFullness$) or even one threshold only ($StdThresBFullness$). Having more thresholds to account for various ranges of bitrates may also be possible.

(984) Moreover, also in decoder **200**, the monitoring the amount of bits may be performed in the same manner as the above-mentioned process of encoder **100**. However, decoder **200** can omit the monitoring the amount of bits by performing processes according to the resolution change performed by encoder **100**.

(985) [Operation at Bitstream Receiver]

(986) Operation of a receiver corresponding to decoder **200** has been already briefly described with reference to FIG. **101**. A receiver of the bitstream may simply obtain the bitstream and decode it according to the control information provided in the bitstream or as a supplementary side information. However, the above mentioned constraints applied by encoder **100** may be applied by the receiver corresponding to decoder **200**. For example, decoding may be configured based on the

knowledge of the constraints.

(987) For example, decoder **200** for decoding a bitstream into a sequence of pictures (corresponding to the bitstream receiver) may be provided. Decoder **200** comprises a processing circuitry which is configured to decode the bitstream into the sequence of pictures including one or more picture resolution changes. In the decoding, a constraint is applied, according to which a picture resolution change is: allowed for pictures at every k-th random access point (k being a positive integer); and prohibited for pictures not at the every k-th random access point.

(988) As also mentioned with reference to encoder **100** above, the picture resolution change is a change from a first resolution of pictures preceding a picture at a random access point in the decoding and/or output order to a second resolution of the picture at a random access point and pictures following the picture at a random access point in the decoding and/or output order.

(989) FIG. **105** is a flowchart illustrating an example in which the processing circuitry is configured to apply the constraint in that it allocates computational resources and/or memory resources for decoding based on the constraint. FIG. **105** is a flow chart of a process for receiving a bitstream using reference picture resampling and conforming to the above mentioned constraints on resolution change at IRAPs or multiples of IRAPs only (and possibly any further constraints exemplified above).

(990) First the maximum resolution for the bitstream is read in step **S710** from the SPS by reading syntax elements `sps_pic_width_max_in_luma_samples` and `sps_pic_height_max_in_luma_samples`. Syntax elements `sps_pic_width_max_in_luma_samples` and `sps_pic_height_max_in_luma_samples` indicate the maximum width and height respectively, in units of luma samples, of each decoded picture referring to the SPS.

(991) Then, still in step **S710**, variable MaxW representing the maximum width of pictures in the bitstream is set equal to `sps_pic_width_max_in_luma_samples`, variable MaxH representing the maximum height of pictures in the bitstream is set equal to `sps_pic_height_max_in_luma_samples`, and decoding resources are allocated based on MaxW and MaxH. Variable currW representing the current width is set equal to MaxW, variable currH representing the current height is set equal to MaxH, and post-processing resources are allocated based on currW, currH and the display (output) resolution.

(992) Then decoding of the bitstream starts at an IRAP picture and the resolution is read in step **S720** from the PPS referred to at the IRAP picture. Variable newW representing the width of the IRAP picture and the following pictures is set equal to `pps_pic_width_in_luma_samples`, and variable newH representing the height of the IRAP picture and the following pictures is set equal to `pps_pic_height_in_luma_samples`.

(993) Parameters `pps_pic_width_in_luma_samples` and `pps_pic_height_in_luma_samples` represent the resolution read from the PPS, and indicate the width and the height of each decoded picture referring to the PPS in units of luma samples.

(994) It is noted that this is only an example conforming with VVC. The present disclosure is not limited to VVC and may be implemented within or with any codec. Thus, the resolution may be also indicated in terms of chroma samples or in another manner (e.g. as a member of a predefined set of possible resolutions or the like).

(995) Then, it is checked in step **S730** whether new resolution $\text{newW} \times \text{newH}$ is the same as current resolution $\text{currW} \times \text{currH}$. When this is not the case (“no” in step **S730**), new post-processing resources are allocated in step **S740** based on newW, newH and the display resolution, and variables currW and currH are updated by setting them to newW and newH, respectively.

(996) The pictures within the next x seconds of the bitstream are then decoded and post-processed in step **S750**, where x represents the minimum allowed time interval between successive resolution changes, for example x=5 seconds in the example of FIG. **102** as well as FIG. **105** (encoder **100** which may produce the bitstream decoded and decodable by decoder **200** described herein).

(997) Then pictures are decoded and post-processed until the next IRAP picture in step **S760**.

(998) When it is determined that the new resolution $\text{newW} \times \text{newH}$ is the same as the current resolution $\text{currW} \times \text{currH}$ (“yes” in step **S730**), the step of allocating new post-processing resources can be skipped and pictures can be decoded and post-processed until the next IRAP picture in step **S760**. The process then loops back to checking the resolution signaled in the referred PPS at the IRAP picture (**S720**), unless the end of the bitstream has been reached in which case the process ends.

(999) For example, post-processing resources may include memory resources or computational resources. Post-processing may include one or more out of resampling the images, mapping the dynamic range or color space of the images to the capabilities of the display, or any algorithm to improve the picture quality, such as post-filtering, sharpening, addition of film grain noise, or the like.

(1000) With the configuration according to the present disclosure, there is a possibility to use reference picture resampling also for broadcast and broadband applications, which may improve coding efficiency in case of signals coming from different sources with different resolutions or in case of rate control for which it may be beneficial to modify the resolution of the signal to have better control of the bitrate of the bitstream, but without impacting the receiver too much by making sure that the receiver does not need to adapt its post-processing resources too often.

(1001) [Set of Predefined Resolutions (Aspect 2)]

(1002) The VVC specification has been designed to work with any value of scaling ratio between a picture to predict and its reference pictures, so that conforming decoders **200** should not have a problem to support any arbitrary resolution changes. However, displays and post-processing units may not be able to switch easily between any arbitrary resolutions.

(1003) According to an aspect of the present disclosure, it is only allowed to switch between a set of predefined resolutions. This aspect may be combined with the embodiments and examples of the aspect specifying constraint on resolution change location within the sequence of pictures. In particular, the present aspect may be an additional constraint to pose on the encoding and correspondingly on the decoding. However, the limitation of the resolutions to the set of predefined resolutions may also be employed separately and irrespectively of employing the previously described constraints.

(1004) According to an example of the present disclosure, encoder **100** for encoding a sequence of pictures includes a processing circuitry it may be the same processing circuitry as the examples mentioned above). The processing circuitry is configured to, in the encoding of the sequence of pictures, apply a constraint according to which a picture resolution change is allowed only to resolutions out of a preconfigured set of predetermined resolutions.

(1005) For example, the preconfigured set includes resolutions that are obtainable from a current resolution by scaling in horizontal or vertical direction by the ratio of 1, 2, or $3/2$. It may be advantageous to provide resolutions which can be resealed easily in order to limit the complexity of the receiver (decoder **200**) and/or at the transmitter (encoder **100**).

(1006) For example, Downsampling by the ratio of 2, for instance, can be performed by simply omitting (skipping) rows and/or columns of the current resolution. Upsampling by the ratio of 2 may be easily implemented e.g. by averaging the value of immediately adjacent (neighboring) samples. As there are mostly 2 or 4 neighbors, the division necessary for averaging may be simply implemented by a binary shift to the right.

(1007) According to an example of the present disclosure, the preconfigured set includes resolutions obtained from a current resolution by scaling in horizontal or vertical direction by a predetermined integer or fraction; and/or incrementing or decrementing the value of the width and/or height of a picture in number of samples. For example, the amount of the incremented or decremented values causes the obtained resolution to be an integer or fraction multiple of the current resolution.

(1008) In other words, the preconfigured set may include multiple resolutions determined by

scaling the current resolution by multiple fixed ratios, or multiple resolutions determined by applying addition or subtraction of fixed difference values to the current resolution.

(1009) In the following, some more details regarding this aspect are provided. The set of predefined resolutions here is a set which includes only predefined resolutions. Predefined here means, e.g. specified in a standard which may be a streaming standard or broadcasting standard or a proprietary streaming application or the like. It may be predefined by a user before the coding is performed. The set of predefined resolutions may be advantageously formed using common broadcast and broadband resolutions, such as 8K, 4K and HD. To select an optimal (or at least a beneficial) set of allowed predefined resolutions for reference picture resampling, a set of predefined ratios to a common predefined resolution can be used.

(1010) As an example, the VVC resampling interpolation filters used for motion compensation have been optimized assuming a reference picture that is twice the size of the current picture, one time and a half the size of the current picture, or the same size as the current picture. Since these ratios are likely to be the ones that offer the best compression efficiency when reference picture resampling is used, these ratios may be used to determine the predefined set of resolutions. Specifically, it may be only allowed to switch to a picture resolution that is either twice the size of the previous resolution, 1.5 times the size of the previous resolution, $\frac{2}{3}$ the size of the previous resolution, or half the size of the previous resolution in luma samples.

(1011) In the present aspect, size may mean only width in luma samples, or only height in luma samples, or both width and height in luma samples. As mentioned above, the present disclosure is not limited. For example it may be possible to count the size in chroma samples instead of luma samples, or in any other way.

(1012) Considering common resolutions used in broadcast and broadband applications, a broadcast or broadband VVC bitstream conforming to the present aspect may be restricted to changes between the following resolutions (two or more or all of them) only: 1) 7680×4320 (commonly called 8K or UHD-2) and 5120×2880, 2) 7680×4320 (commonly called 8K or UHD-2) and 3840×2160 (commonly called 4K or UHD-1), 3) 3840×2160 (commonly called 4K or UHD-1) and 2560×1440, 4) 3840×2160 (commonly called 4K or UHD-1) and 1920×1080 (commonly called full HD), 5) 1920×1080 (commonly called full HD) and 1280×720 (commonly called HD ready), and 6) 1920×1080 (commonly called full HD) and 960×540.

(1013) In the present aspect, receivers would only need to switch resolution by either increasing the resolution by one and a half times or decreasing the resolution by $\frac{2}{3}$ in cases 1), 3), and 5), or by increasing the resolution by two times or decreasing the resolution by half in cases 2), 4), and 6).

(1014) According to such configuration, a switch from resolution 7680×4320 down to resolution 1920×1080 would only be possible indirectly by combining the two allowed changes 2) and 4). First the resolution could be halved from 7680×4320 to 3840×2160 at an IRAP picture, after x seconds (such as 5 seconds for example) of video, a second switch could be performed by halving the resolution from 3840×2160 to 1920×1080 at another IRAP picture.

(1015) Alternatively, a direct switch between two resolutions which are allowed to be indirectly switched as described above may be allowed. In other words, a switch between two resolutions which are allowed to be indirectly switched may be performed at one IRAP picture.

(1016) Moreover, in spite of the combination of the above 1) to 6), a switch between any pair of the resolutions described in the above 1) to 6) may be allowed. Moreover, in spite of the combination of the above 1) and 6), one or more combinations of switchable resolutions may be newly determined from among the resolutions described in the above 1) to 6).

(1017) For example, a switch from resolution 5120×2880 down to resolution 3840×2160 and a switch from resolution 2560×1440 down to resolution 1920×1080 may be allowed. Such a switch corresponds to decreasing the resolution by $\frac{3}{4}$ times. In contrast, a switch from resolution 3840×2160 up to resolution 5120×2880 and a switch from resolution 1920×1080 up to resolution 2560×1440 may be allowed. Such a switch corresponds to increasing the resolution by $\frac{4}{3}$ times.

(1018) Another example to specify the predefined set of allowed resolutions may be to use a value to subtract or add from the width and/or height, whereby multiples of this value would also be allowed. Below are some examples of such values: 1) Reduction of width by multiples of 1920 would allow resolution width values: 7680, 5760, 3840, 1920; 2) Reduction of height by multiples of 1080 would allow resolution height values: 4320, 3240, 2160, 1080; 3) Reduction of width by multiples of 960 would allow resolution width values: 3840, 2880, 1920, 960; 4) Reduction of height by multiples of 540 would allow resolution height values: 2160, 1620, 1080, 540; 5) Reduction of width by multiples of 480 would allow resolution width values: 1920, 1440, 960, 480; and 6) Reduction of height by multiples of 270 would allow resolution height values: 1080, 810, 540, 270.

(1019) Only one of the above exemplary reductions may be allowed in some implementations. In other implementations, two or more or all of these reduction examples may be allowed. There may be further exemplary reduction examples construed in a similar manner. For example, in general, multiples of a predefined size S may be allowed and obtained either by multiplication or by one or multiple additions.

(1020) FIG. 106A and FIG. 106B is a flowchart illustrating an example process for a broadcast encoder (encoder 100) that can take as input a video signal that may be coming from several cameras using different resolutions and create a VVC bitstream out of it according to an exemplary implementation of the present aspect.

(1021) First, when starting encoding of a new piece of content, a variable $tResChange$ representing the time (time instant) of the last resolution change is set equal to $-x$ seconds (S805), where x is set to the value of the minimum allowed time interval between successive resolution changes, for example $x=5$ seconds in the example of FIG. 106A and FIG. 106B. Here as well as in the previous example, 5 s have been selected only for instance. As is clear to those skilled in the art, the actual value of x may be smaller (e.g. 2 or 3 seconds) or greater (e.g. 6, 7, or 10 or the like seconds) than 5.

(1022) Then the signal from camera $currC$ representing the current camera with a resolution of width $currW$ and height $currH$ is encoded in step S810. It is checked whether there are more pictures to be encoded from another camera $altC$ of a resolution $newW$ and $newH$ in step S815. For example, when the camera delivering the input video signal suddenly changes, more pictures arrive from a different camera than $currC$ denoted as $altC$ with a resolution of width $newW$ and height $newH$. In this case (“yes” in step S815), it is checked in step S818 whether the values of $currW$ and $newW$ are equal and whether the values of $currH$ and $newH$ are equal.

(1023) When this is the case (“yes” in step S818), the encoding can continue without resolution change and variable $currC$ denoting the current camera is simply set to $altC$ in step S820 (this step may be thus also omitted here as well as in the previous examples, as it leads to maintaining the same $currW$ and $currH$). Otherwise (“no” in step S818), variable t representing the current time media is determined in step S830. Typically, t would be calculated as the number of frames in display order before the first picture coming from camera $altC$ divided by the frame rate of the content.

(1024) Then, it is checked how big the time interval since the last change in resolution is in step S835. When the value of t minus $tResChange$ is less than x seconds (where x is equal to 5 in the example of FIG. 106A and FIG. 106B) (“no” in step S835), a resolution change is not allowed, so that the picture from camera $altC$ has to be resampled in step S840 to a resolution of width $currW$ and height $currH$, before continuing the encoding (S810) with $currC$ set to $altC$.

(1025) When the value of t minus $tResChange$ is greater than or equal to x seconds (where x is equal to 5 in the example of FIG. 106A and FIG. 106B) (“yes” in step S835), scaling ratio is calculated in step S850. More specifically, variable $ratioW$ representing the width scaling ratio between the width of a reference IRAP picture (new width) and the width of its leading pictures (previous width) is calculated as $newW$ divided by $currW$, and variable $ratioH$ representing the

height scaling ratio between the height of a reference IRAP picture (new height) and the height of its leading pictures (previous height) is calculated as newH divided by currH.

(1026) Then, it is checked, in step **S855**, whether ratioW and ratioH are equal to one of the allowed values, which are 2, 3/2, 2/3 or 1/2 in the example of FIG. **106A** and FIG. **106B**. These are the values which are allowed as they lead to the predefined resolution of the preconfigured set as was discussed above.

(1027) When this is the case (“yes” in step **S855**), an IRAP picture of resolution width newW and height newH is inserted into the bitstream (**S860**) and any remaining pictures from camera currC not yet encoded are encoded with the previous resolution of width currW and height currH as leading pictures of the IRAP picture using reference picture resampling. Then variables currW, currH and tResChange are updated to the new values of newW, newH and t, respectively in step **S865**, before continuing the encoding with currC set to altC in steps **S820** and **S810**.

(1028) When the values of ratioW or ratioH are not one of the allowed values (“no” in step **S855**), the closest allowed values are determined in variables selRatioW and selRatioH, respectively in step **S870**. The alternative width altW is calculated in step **S875** as selRatioW multiplied by currW, and the alternative height altH is calculated in step **S875** as selRatioH multiplied by currH.

(1029) The signal from camera altC is then resampled in step **S880** to the alternative resolution altW×altH, before step **S890** corresponding to inserting an IRAP picture of resolution width altW and height altH into the bitstream and encoding any remaining pictures from camera currC not yet encoded with the previous resolution of width currW and height currH as leading pictures of the IRAP picture using reference picture resampling.

(1030) Then variables currW, currH and tResChange are updated in step **S895** to values altW, altH and t, respectively, before continuing the encoding (**S810**) with currC set to altC (see also step **S820**). When there is no picture to be encoded from another camera altC, i.e., no more pictures from any camera arrives to encoder **100** (“no” in step **S815**), the process ends with the encoded bitstream as output.

(1031) [Rate Control in Case of Set of Predefined Resolutions]

(1032) According to an exemplary implementation, a rate matching uses the possibility of control rate by means of change of resolution, similarly to the example described above with reference to FIG. **104A** and FIG. **104B**. The rate matching can be further simplified when the above mentioned constraint on the set of predefined resolutions is applied.

(1033) FIG. **107A** and FIG. **107B** are a flowchart illustrating an example process for a broadcast encoder (encoder **100**) using rate control and taking as input a video signal with a resolution of width baseW and height baseH. It is assumed that resolution of width baseW and height baseH is one of the allowed resolutions for reference picture resampling. First the variables currW and currH representing the current resolution width and height, respectively, are set equal to baseW and baseH, respectively in step **S910**.

(1034) First, variable tResChange representing the time (time instant) of the last resolution change is set equal to -x seconds in step **S920**, where x is set to the value of the minimum allowed time interval between successive resolution changes. As an alternative, tResChange may also be initialised to zero. Variable BufferFullness representing the number of bits in the Coded Picture Buffer is set equal to zero in step **S920**.

(1035) Then, one Group Of Pictures (GOP) of the signal is encoded with a resolution of width currW and height currH in step **S930**, and the number of bits in the Coded Picture Buffer is tracked in the variable BufferFullness.

(1036) In step **S935**, it is tested whether a picture to be encoded remains. When a picture to be encoded remains (“yes” in step **S935**), it is checked in step **S938** whether the value of BufferFullness is within a desired range determined by the application.

(1037) The application may be for instance a particular streaming, broadcasting or video conferencing application, or the like. Such application may have requirements including a constant

bit rate or bitrate within a certain allowable range. Such requirements may be beneficial for instance in order to reserve appropriate amount of channel resources in case the encoded bitstream is to be transmitted. It may be also beneficial for hardware design of encoder **100** and/or decoder **200** for reproduction (e.g. play-back device or receiver or the like).

(1038) When this is the case (“yes” in step **S938**), the encoding (**S930**) can continue without resolution change and the next GOP is encoded with resolution of width currW and height currH. Otherwise (“no” in step **S938**), variable t representing the current time media is determined in step **S940** to find out whether a resolution change for increasing or decreasing the bitstream bitrate is appropriate. Typically, t would be calculated as the number of frames in display order coded so far divided by the frame rate of the content, as already described in the above mentioned examples.

(1039) Then, it is checked in step **S945** how big the time interval since the last change in resolution is. When the value of t minus tResChange is less than x seconds (“no” in step **S945**), a resolution change is not allowed, so that the encoding continues without resolution change and the next GOP is encoded with resolution of width currW and height currH (**S930**).

(1040) When the value of t minus tResChange is greater than or equal to x seconds (“yes” in step **S945**), it is checked in step **S948** more in details whether the current bitstream rate is appropriate by checking whether BufferFullness is below or above the desired range. When BufferFullness is below the desired range (“yes” in step **S948**), the bitstream rate is actually too low, so that a bigger resolution of width newW and height newH, with newW bigger than currW and newH bigger than currH, is selected in step **S950** from the set of allowed resolutions in order to increase the bitrate of the bitstream.

(1041) When BufferFullness is above the desired range (“no” in step **S948**), the resolution of the content should be decreased to decrease the bitrate of the bitstream. A smaller resolution of width newW and height newH, with newW smaller than currW and newH smaller than currH, is selected from the set of allowed resolutions in step **S960**.

(1042) Once a new resolution of width newW and height newH has been determined, the input video signal is resampled in step **S970** from its original resolution of width baseW and height baseH to the new resolution of width newW and height newH, before inserting an IRAP picture of resolution width newW and height newH in the bitstream (in step **S980**), and a new GOP with the new resolution using reference picture resampling is started.

(1043) Variables currW and currH are updated in step **S990** to the new resolution newW and newH, respectively, and variable tResChange is updated to the new value t at which the resolution change occurs in step **S990**, before continuing the encoding (**S930**) with resolution of width currW and height currH.

(1044) When no more pictures to be encoded remains (“no” in step **S935**), the process ends with the encoded bitstream as output.

(1045) With the configuration according to the present aspect relating to the constrained set of resolutions, there is a possibility to use reference picture resampling for common broadcast and broadband resolutions only in a limited set of allowed resolutions. Accordingly, a switch between resolutions can be simplified in displaying and post-processing and do not need to support arbitrary resolutions. Moreover, still an efficient rate control can be performed including employing the resolution change.

(1046) [Signaling of Resolution in Sequence Parameter Set (Aspect 3)]

(1047) The VVC specification allows signaling of a resolution in the SPS that is different from the resolution signaled in the PPS to enable reference picture resampling. In video coding standards, resolution is typically signaled in the Sequence Parameter Set (SPS) because it applies to the whole coded video sequence until a new IRAP picture overwrites the SPS information. However, when reference picture resampling is enabled, each picture in the coded video sequence can theoretically have a different resolution, so that resolution signaling is also necessary for every picture within the Picture Parameter Set (PPS).

(1048) The only constraint on the SPS and PPS syntax elements is that the resolution signaled in a referred PPS within the bitstream must be less than or equal to any resolution signaled in the SPS. In other words, the following two relationships must be satisfied:

$\text{sps_pic_width_max_in_luma_samples} \geq \text{pps_pic_width_in_luma_samples}$; and 1)

$\text{sps_pic_height_max_in_luma_samples} \geq \text{pps_pic_height_in_luma_samples}$. 2)

(1049) In the present aspect, an additional constraint is set that the SPS syntax elements signaling the picture resolution must be equal to the maximum of the PPS syntax elements signaling the picture resolutions for all referred PPSs, i.e. at least one picture within the bitstream has to use a resolution of width $\text{sps_pic_width_max_in_luma_samples}$ and $\text{sps_pic_height_max_in_luma_samples}$.

(1050) In other words, in the present aspect, the bitstream includes a sequence parameter set (SPS) which comprises a maximum picture resolution in vertical and/or horizontal direction for the sequence of pictures. The bitstream further includes at least one picture parameter set (PPS) which comprises a picture resolution in vertical and/or horizontal direction for a picture. The resolution signaled in the SPS is not larger than the maximum resolution possible to indicate in any of the PPSs referring to said SPS.

(1051) This aspect is applicable to any video codec which indicates parameters on sequence level (e.g. per SPS) and on picture level (e.g. PPS); it is not limited to the exemplified VVC. This constraint (as well as the previously described constraints) may be enforced, e.g., by a streaming standard configuring the video coding parameters.

(1052) Decoder **200** aware of the constraint may consider it for scheduling its processing resources, e.g. does not need to allocate more resources than necessary for decoding the bitstream. However, the present disclosure is not limited thereto and the constraint may also be embedded in the syntax format of a video codec, thereby providing for a more compact and efficient coding of the PPS and SPS.

(1053) It is noted that the present aspect is a constraint which may be but does not have to be combined with the above discussed constraints. For example, in another aspect of the present disclosure, the resolution can change depending on an IRAP picture. Moreover, the SPS can be determined in units of random access, i.e., in units of GOPs. Accordingly, when combined with the other aspects of the present disclosure, the resolution before or after the change depending on an IRAP picture may be limited so as to coincide with the resolution signaled in the SPS.

(1054) For example, when the resolution is decreased at the IRAP picture, the resolution of the leading pictures may be limited to width $\text{sps_pic_width_max_in_luma_samples}$ and height $\text{sps_pic_height_max_in_luma_samples}$. When the resolution is increased at the IRAP picture, the resolution of the IRAP and following pictures may be limited to width $\text{sps_pic_width_max_in_luma_samples}$ and height $\text{sps_pic_height_max_in_luma_samples}$.

(1055) With the configuration according to the present aspect, receivers (decoders **200**) do not need to allocate more decoding resources than necessary when decoding a bitstream using reference picture resampling.

(1056) As described in the above aspects, the present disclosure relates to constraining bitstreams using reference picture resampling to ease receivers' processing. In particular, the frequent occurrence of RPR may cause switching between sizes of subsequent-stage processes including displaying to occur frequently, which complicates control. The frequent occurrence of RPR causes switching between memory access methods for a current picture and a reference picture (switching between memory maps) to occur frequently, which complicates control.

(1057) In order to address this issue, in Aspect 1, the bitstream structure (and the corresponding encoder **100** and decoder **200** operation to generate and decode such bitstream structure) can include the resolution change only at random access pictures or their multiples. Accordingly, the receiver can reduce timings at which reallocation of resources is needed, to only timings at random-access points.

(1058) The frequency of resolution changes is kept below a certain value in this manner. On the other hand, a DASH receiver does not have to follow the constraints and may switch between DASH representations of different resolutions more often than the minimum allowed time interval between two resolution changes within a bitstream.

(1059) Another issue may be caused since a large number of switching patterns of RPR leads to the corresponding number of patterns of matching sizes of subsequent-stage processes including displaying, which complicates mounting. A large number of switching patterns of RPR may lead to the corresponding number of patterns of switching between memory access methods for a current picture and a reference picture (switching between memory maps), which complicates mounting.

(1060) In Aspect 2, in order to address this issue, reference picture resampling within a bitstream is only allowed between a set of predetermined resolutions. The allowed values for the scaling ratio between two resolutions within the set of predetermined resolutions may be constrained to a set of allowed values, such as $\times 2$, $\times 3/2$, $\times 2/3$ or $\times 1/2$. Resampling of a video signal at encoder **100** may be used to fulfil the constraints and/or enable rate control.

(1061) Another issue may be caused when a large size not actually used in a GOP is described in an SPS, decoder **200** secures memory resources etc. in accordance with the size of the SPS, and it is necessary to secure uselessly large resources. In Aspect 3, the resolution signaled in the Sequence Parameter Set is not allowed to be bigger than the maximum of the resolutions signaled in Picture Parameter Sets. Resolution is to be understood as width of the pictures only, or height of the pictures only, or both width and height in units of luma samples.

(1062) [Further Hardware and Software Implementations]

(1063) It is noted that the processing circuitry (the one at encoder **100** and/or the one at decoder **200**) may be any combination of software and/or hardware. For examples, the processing circuitry may be one or more processors which are configured by a software (code instructions) fetched from a memory. Correspondingly, encoder **100** and decoder **200** may comprise the memory. Exemplary implementations of encoder **100** and decoder **200** are described further below. The processing circuitries for encoder **100** and decoder **200** may further be configured to perform any steps described herein for the methods.

(1064) For example, encoder **100** may comprise circuitry and memory connected to the circuitry. The circuitry, in operation, uses reference picture resampling in bitstreams with receiver (decoder **200**) friendly constraints by allowing resolution change at a random-access intra-predicted picture only.

(1065) Moreover, decoder **200** may comprise circuitry and memory connected to the circuitry. The circuitry, in operation, decodes a bitstream using reference picture resampling in such a way that reallocation of resources is only needed at a random-access intra-predicted picture.

(1066) The present disclosure is not limited to implementations by one or more processors. There may be a dedicated hardware or programmable hardware included and/or a combination of such. In general, encoder **100** may comprise modules implemented in any hardware and/or software. Such modules may be, for example: 1) an input terminal into which at least one image is to be input; 2) a block splitter which splits a first image included in the at least one image into a plurality of blocks; 3) an intra predictor which predicts a block included in the first image, using a reference picture included in the first image; 4) an inter predictor which predicts a block included in the first image, using a reference block included in a second image different from the first image; 5) a loop filter which filters a block included in the first image; 6) a transformer which transforms a prediction error between an original signal and a prediction signal generated by the intra predictor or the inter predictor, to generate transform coefficients; 7) a quantizer which quantizes the transform coefficients to generate quantized coefficients; 8) an entropy encoder which variable encodes the quantized coefficients to generate an encoded bitstream; and 9) an output terminal from which the encoded bitstream includes the encoded quantized coefficients and control information is to be output.

(1067) Here, a bitstream using reference picture resampling is created with receiver (decoder **200**) friendly constraints by allowing resolution change at a random-access intra-predicted picture only. (1068) Correspondingly, decoder **200** may be provided with the following modules: 1) an input terminal into which an encoded bitstream is to be input; 2) a decoder (such as an entropy decoder) which decodes an encoded bitstream to output quantized coefficients; 3) an inverse quantizer which inverse quantizes the quantized coefficients to output transform coefficients; 4) an inverse transformer which inverse transforms the transform coefficients to output a prediction error; 5) an intra predictor which predicts a block included in the image, using a reference picture included in a first image; 6) an inter predictor which predicts a block included in the first image, using a reference block included in a second image different from the first image; 7) a loop filter which filters a block included in the first image; and 8) an output terminal from which a picture includes the first image is to be output.

(1069) Here, a bitstream using reference picture resampling is decoded in such a way that reallocation of resources is only needed at a random-access intra-predicted picture.

(1070) Similarly, an encoding method may be provided, comprising creation of a bitstream using reference picture resampling with receiver (decoder **200**) friendly constraints by allowing resolution change at a random-access intra-predicted picture only. Moreover, a decoding method may be provided, comprising the decoding of a bitstream using reference picture resampling in such a way that reallocation of resources is only needed at a random-access intra-predicted picture.

(1071) [Representative Example of Configuration and Processing]

(1072) The following shows a representative example of the configuration and the processing of each of encoder **100** and decoder **200**.

(1073) FIG. **108** is a flow chart indicating an operation performed by encoder **100**. For example, encoder **100** includes circuitry and memory coupled to the circuitry. The circuitry and memory included in encoder **100** may correspond to processor a1 and memory a2 illustrated in FIG. **8**. The circuitry of encoder **100** performs the following steps in operation.

(1074) For example, the circuitry of encoder **100** controls, according to a constraint, whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order (S**1081**). This constraint allows the changing only when the picture is a random access picture among one or more random access pictures.

(1075) Moreover, when a resolution of a reference picture is different from a resolution of an inter-predicted picture, the circuitry of encoder **100** resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture. Here, the reference picture is a reference picture to be used in encoding of the inter-predicted picture. Then, the circuitry of encoder **100** encodes an image in the inter-predicted picture using the resampled reference image (S**1082**).

(1076) With this, in encoder **100**, it may be possible to prevent the frequent resolution change. Accordingly, in encoder **100**, it may be possible to reduce the consumption of the computational resource and memory resource, and also reduce the load of the encoder. Moreover, in encoder **100**, it may be possible to prevent the computational resource and the memory resource supporting the resolution change from being constantly secured. Moreover, in encoder **100**, it may be possible to reduce the difficulty of the test.

(1077) Moreover, for example, the constraint may allow the changing of the resolution of the picture only when the picture is a random access picture among the one or more random access pictures at every k-th random access picture. Here, k is an integer greater than 1.

(1078) With this, in encoder **100**, it may be possible to further prevent the frequent resolution change. Accordingly, in encoder **100**, it may be possible to further reduce the consumption of the computational resource and memory resource, and also further reduce the load of the encoder.

(1079) Moreover, for example, the circuitry of encoder **100** may sequentially encode pictures into a bitstream. Moreover, the circuitry of encoder **100** may monitor an amount of bits in the bitstream.

Then, the circuitry of encoder **100** may control the changing of the resolution of the picture according to the monitored amount of bits.

(1080) With this, in encoder **100**, it may be possible to change the resolution according to the amount of bits in the bitstream and adjust the amount of bits in the bitstream according to the resolution change.

(1081) Moreover, for example, the circuitry of encoder **100** may encode at least one first trailing picture belonging to a first group of pictures. Subsequently, the circuitry of encoder **100** may encode a random access picture belonging to a second group of pictures following the first group of pictures in the display order.

(1082) Subsequently, the circuitry of encoder **100** may encode at least one leading picture belonging to the first group of pictures. Subsequently, the circuitry of encoder **100** may encodes at least one second trailing picture belonging to the second group of pictures.

(1083) In the constraint, a resolution of the at least one leading picture need not be allowed to be different from a resolution of the at least one first trailing picture. Moreover, a resolution of the random access picture may be allowed to be different from the resolution of the at least one leading picture. Moreover, a resolution of the at least one second trailing picture need not be allowed to be different from the resolution of the random access picture.

(1084) With this, in encoder **100**, it may be possible to change the resolution only at a random access picture in the display order. Accordingly, in encoder **100**, it may be possible to smoothly perform the resolution change along the display order.

(1085) Moreover, for example, the circuitry of encoder **100** may encode at least one first trailing picture belonging to a first group of pictures. Subsequently, the circuitry of encoder **100** may encode a random access picture belonging to a second group of pictures following the first group of pictures in the display order.

(1086) Subsequently, the circuitry of encoder **100** may encode at least one leading picture belonging to the first group of pictures. Subsequently, the circuitry of encoder **100** may encodes at least one second trailing picture belonging to the second group of pictures.

(1087) In the constraint, a resolution of the random access picture may be allowed to be different from a resolution of the at least one first trailing picture. Moreover, a resolution of the at least one leading picture need not be allowed to be different from the resolution of the random access picture. Moreover, a resolution of the at least one second trailing picture need not be allowed to be different from the resolution of the at least one leading picture.

(1088) With this, in encoder **100**, it may be possible to change the resolution only at a random access picture in the encoding order. Accordingly, in encoder **100**, it may be possible to smoothly perform the resolution change along the encoding order.

(1089) Moreover, for example, for each of segments determined using Dynamic Adaptive Streaming over HTTP (DASH) protocol, the constraint need not allow the changing of the resolution of the picture in the segment.

(1090) With this, in encoder **100**, it may be possible to constrain the resolution change in units of segments. With this, in encoder **100**, it may be possible to prevent the frequent resolution change.

(1091) Moreover, for example, for each of representations determined using DASH protocol, the constraint need not allow the changing of the resolution of the picture to be performed at intervals shorter than a threshold.

(1092) With this, in encoder **100**, for the same representation, it may be possible to prevent the resolution change from being performed in short intervals. Accordingly, with this, in encoder **100**, it may be possible to further prevent the frequent resolution change.

(1093) Moreover, for example, the constraint may limit the resolution of the picture to a resolution candidate among a plurality of resolution candidates.

(1094) With this, in encoder **100**, it may be possible to limit the resolution to any of resolution candidates. Accordingly, in encoder **100**, it may be possible to reduce the complexity of the

processing. Moreover, in encoder **100**, it may be possible to reduce the difficulty of the test. (1095) Moreover, for example, the plurality of resolution candidates may include resolution candidates determined by scaling the resolution of the previous picture by a plurality of fixed ratios.

(1096) With this, in encoder **100**, it may be possible to smoothly change the resolution from the resolution before the change using the fixed ratio. Moreover, in encoder **100**, it may be possible to smoothly resample the reference image using the fixed ratio. Accordingly, in encoder **100**, it may be possible to reduce the complexity of the processing.

(1097) Moreover, for example, the plurality of fixed ratios may include at least one of 2 times, 3/2 times, 4/3 times, 3/4 times, 2/3 times, or 1/2 times.

(1098) With this, in encoder **100**, it may be possible to change the resolution using a ratio easy to resample the reference image. Accordingly, in encoder **100**, it may be possible to reduce the complexity of the processing.

(1099) Moreover, for example, the plurality of resolution candidates may include resolution candidates determined as 7680×4320 pixels, 5120×2880 pixels, 3840×2160 pixels, 2560×1440 pixels, 1920×1080 pixels, 1280×720 pixels, and 960×540 pixels.

(1100) With this, in encoder **100**, it may be possible to use a general resolution, and maintain the versatility of the encoder.

(1101) Moreover, for example, the plurality of resolution candidates may include resolution candidates determined by applying addition or subtraction of fixed difference values to the resolution of the previous picture.

(1102) With this, in encoder **100**, it may be possible to appropriately limit the resolution after the change according to the resolution before the change and the fixed difference value, and reduce the complexity of the processing.

(1103) Moreover, for example, the constraint may limit a maximum resolution specified as an upper limit in a sequence parameter set to be equal to a resolution specified as a resolution of at least one picture in a picture parameter set. This picture parameter set is a picture parameter set applying to the at least one picture among pictures to which the sequence parameter set is applied.

(1104) Then, the circuitry of encoder **100** may encode, into the sequence parameter set, the maximum resolution determined according to the constraint, and encode, into the picture parameter set, the resolution specified as the resolution of the at least one picture.

(1105) With this, in encoder **100**, it may be possible to prevent a too high maximum resolution from being specified in the sequence parameter set. Accordingly, in encoder **100**, it may be possible to prevent too much computational resource and memory resource from being secured.

(1106) Moreover, for example, inter predictor **126** of encoder **100** may perform the operation described above as the circuitry of encoder **100**. Inter predictor **126** also may perform the operation described above in cooperation with the other constituent elements.

(1107) FIG. **109** is a flow chart indicating an operation performed by decoder **200**. For example, decoder **200** includes circuitry and memory coupled to the circuitry. The circuitry and memory included in decoder **200** may correspond to processor **b1** and memory **b2** illustrated in FIG. **68**. The circuitry of decoder **200** performs the following in operation.

(1108) For example, the circuitry of decoder **200** controls, according to a constraint, whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and decoding order (**S1091**). This constraint allows the changing only when the picture is a random access picture among one or more random access pictures.

(1109) Moreover, when a resolution of a reference picture is different from a resolution of an inter-predicted picture, the circuitry of decoder **200** resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture. Here, the reference picture is a reference picture to be used in decoding of the inter-predicted picture. Then, the circuitry of decoder **200** decodes an image in the inter-

predicted picture using the resampled reference image (S1092).

(1110) With this, in decoder **200**, it may be possible to prevent the frequent resolution change. Accordingly, in decoder **200**, it may be possible to reduce the consumption of the computational resource and memory resource, and also reduce the load of the decoder. Moreover, in decoder **200**, it may be possible to prevent the computational resource and the memory resource supporting the resolution change from being constantly secured. Moreover, in decoder **200**, it may be possible to reduce the difficulty of the test.

(1111) Moreover, for example, the circuitry of decoder **200** may allocate, according to the constraint, at least one of a computational resource or a memory resource to post-processing for displaying a decoded image.

(1112) With this, in decoder **200**, it may be possible to prepare, according to the constraint, computational resource and memory resource necessary and sufficient for the post-processing for displaying the decoded image.

(1113) Moreover, for example, the constraint may allow the changing of the resolution of the picture only when the picture is a random access picture among the one or more random access pictures at every k-th random access picture. Here, k is an integer greater than 1.

(1114) With this, in decoder **200**, it may be possible to further prevent the frequent resolution change. Accordingly, in decoder **200**, it may be possible to further reduce the consumption of the computational resource and memory resource, and also further reduce the load of the decoder.

(1115) Moreover, for example, the circuitry of decoder **200** may sequentially decode pictures from a bitstream. Moreover, the circuitry of decoder **200** may monitor an amount of bits in the bitstream. Then, the circuitry of decoder **200** may control the changing of the resolution of the picture according to the monitored amount of bits.

(1116) With this, in decoder **200**, it may be possible to change the resolution according to the amount of bits in the bitstream and adjust the amount of bits in the bitstream according to the resolution change.

(1117) Moreover, for example, the circuitry of decoder **200** may decode at least one first trailing picture belonging to a first group of pictures. Subsequently, the circuitry of decoder **200** may decode a random access picture belonging to a second group of pictures following the first group of pictures in the display order.

(1118) Subsequently, the circuitry of decoder **200** may decode at least one leading picture belonging to the first group of pictures. Subsequently, the circuitry of decoder **200** may decode at least one second trailing picture belonging to a second group of pictures.

(1119) In the constraint, a resolution of the at least one leading picture need not be allowed to be different from a resolution of the at least one first trailing picture. Moreover, a resolution of the random access picture may be allowed to be different from the resolution of the at least one leading picture. Moreover, a resolution of the at least one second trailing picture need not be allowed to be different from the resolution of the random access picture.

(1120) With this, in decoder **200**, it may be possible to change the resolution only at a random access picture in the display order. Accordingly, in decoder **200**, it may be possible to smoothly perform the resolution change along the display order.

(1121) Moreover, for example, the circuitry of decoder **200** may decode at least one first trailing picture belonging to a first group of pictures. Subsequently, the circuitry of decoder **200** may decode a random access picture belonging to a second group of pictures following the first group of pictures in the display order.

(1122) Subsequently, the circuitry of decoder **200** may decode at least one leading picture belonging to the first group of pictures. Subsequently, the circuitry of decoder **200** may decode at least one second trailing picture belonging to a second group of pictures.

(1123) In the constraint, a resolution of the random access picture may be allowed to be different from a resolution of the at least one first trailing picture. Moreover, a resolution of the at least one

leading picture need not be allowed to be different from the resolution of the random access picture. Moreover, a resolution of the at least one second trailing picture need not be allowed to be different from the resolution of the at least one leading picture.

(1124) With this, in decoder **200**, it may be possible to change the resolution only at a random access picture in the decoding order. Accordingly, in decoder **200**, it may be possible to smoothly perform the resolution change along the display order.

(1125) Moreover, for example, for each of segments determined using Dynamic Adaptive Streaming over HTTP (DASH) protocol, the constraint need not allow the changing of the resolution of the picture in the segment.

(1126) With this, in decoder **200**, it may be possible to constrain the resolution change in units of segments. Accordingly, with this, in decoder **200**, it may be possible to prevent the frequent resolution change.

(1127) Moreover, for example, for each of representations determined using DASH protocol, the constraint need not allow the changing of the resolution of the picture to be performed at intervals shorter than a threshold.

(1128) With this, in decoder **200**, for the same representation, it may be possible to prevent the resolution change from being performed in short intervals. Accordingly, with this, in decoder **200**, it may be possible to further prevent the frequent resolution change.

(1129) Moreover, for example, the constraint may limit the resolution of the picture to a resolution candidate among a plurality of resolution candidates.

(1130) With this, in decoder **200**, it may be possible to limit the resolution to any of resolution candidates. Accordingly, in decoder **200**, it may be possible to reduce the complexity of the processing. Moreover, in decoder **200**, it may be possible to reduce the difficulty of the test.

(1131) Moreover, for example, the plurality of resolution candidates may include resolution candidates determined by scaling the resolution of the previous picture by a plurality of fixed ratios.

(1132) With this, in decoder **200**, it may be possible to smoothly change the resolution from the resolution before the change using the fixed ratio. Moreover, in decoder **200**, it may be possible to smoothly resample the reference image using the fixed ratio. Accordingly, in decoder **200**, it may be possible to reduce the complexity of the processing.

(1133) Moreover, for example, the plurality of fixed ratios may include at least one of 2 times, $3/2$ times, $4/3$ times, $3/4$ times, $2/3$ times, or $1/2$ times.

(1134) With this, in decoder **200**, it may be possible to change the resolution using a ratio easy to resample the reference image. Accordingly, in decoder **200**, it may be possible to reduce the complexity of the processing.

(1135) Moreover, for example, the plurality of resolution candidates may include resolution candidates determined as 7680×4320 pixels, 5120×2880 pixels, 3840×2160 pixels, 2560×1440 pixels, 1920×1080 pixels, 1280×720 pixels, and 960×540 pixels.

(1136) With this, in decoder **200**, it may be possible to use a general resolution, and maintain the versatility of the decoder.

(1137) Moreover, for example, the plurality of resolution candidates may include resolution candidates determined by applying addition or subtraction of fixed difference values to the resolution of the previous picture.

(1138) With this, in decoder **200**, it may be possible to appropriately limit the resolution after the change according to the resolution before the change and the fixed difference value, and reduce the complexity of the processing.

(1139) Moreover, for example, the constraint may limit a maximum resolution specified as an upper limit in a sequence parameter set to be equal to a resolution specified as a resolution of at least one picture in a picture parameter set. This picture parameter set is a picture parameter set applying to the at least one picture among pictures to which the sequence parameter set is applied.

(1140) Then, the circuitry of decoder **200** may decode, from the sequence parameter set, the maximum resolution determined according to the constraint, and decode, from the picture parameter set, the resolution specified as the resolution of the at least one picture.

(1141) With this, in decoder **200**, it may be possible to prevent a too high maximum resolution from being specified in the sequence parameter set. Accordingly, in decoder **200**, it may be possible to prevent too much computational resource and memory resource from being secured.

(1142) Moreover, for example, inter predictor **218** of decoder **200** may perform the operation described above as the circuitry of decoder **200**. Moreover, inter predictor **218** also may perform the operation described above in cooperation with the other constituent elements.

(1143) Moreover, for example, a non-transitory computer readable medium according to one aspect of the present disclosure may store a bitstream. The bitstream may include a plurality of pictures, and a plurality of parameters to perform a decoding process for decoding the plurality of pictures from the bitstream.

(1144) In the decoding process, in operation according to the plurality of parameters, according to a constraint, it is controlled whether a resolution of a picture is changed from a resolution of a previous picture preceding the picture in one of display order and decoding order, and an image in the inter-predicted picture may be decoded. The constraint may allow the changing only when the picture is a random access picture among one or more random access pictures.

(1145) When a resolution of a reference picture is different from a resolution of the inter-predicted picture, a reference image in the reference picture may be resampled according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture. Here, the reference picture is a reference picture to be used in decoding of the inter-predicted picture. Then, an image in the inter-predicted picture may be decoded using the resampled reference image.

(1146) With this, it may be possible to prevent the frequent resolution change. Accordingly, it may be possible to reduce the consumption of the computational resource and memory resource, and also reduce the load. Moreover, it may be possible to prevent the computational resource and the memory resource supporting the resolution change from being constantly secured. Moreover, it may be possible to reduce the difficulty of the test.

(1147) Moreover, for example, the plurality of parameters may include: a first parameter indicating whether changing of a resolution of one or more pictures among the plurality of pictures is allowed; and a second parameter indicating the resolution of the picture. When (i) the changing of the resolution of the one or more pictures is allowed according to the first parameter, (ii) the picture is included in the one or more pictures, and (iii) the changing of the resolution of the picture is allowed according to the constraint, the resolution of the picture may be changed according to the second parameter.

(1148) With this, it may be possible to change the resolution according to the parameters and the constraint related to the resolution change while preventing the frequent resolution change.

(1149) It is to be noted that the reference image to be resampled may be a reference image of a reference block in a reference picture. Moreover, the image to be encoded or decoded using the reference image may be an image of a block in the inter-predicted picture.

(1150) Moreover, in the resampling of the reference image, the reference image in the reference picture may be scaled by the ratio of the resolution of the inter-predicted picture to the resolution of the reference picture. For example, when the ratio of the resolution of the inter-predicted picture to the resolution of the reference picture is x times, the reference image in the reference picture may be scaled to x times.

(1151) Moreover, the resolution candidate to limit the resolution of the picture may correspond to the predefined resolution.

(1152) Moreover, the first parameter may correspond to `sps_res_change_in_clvs_allowed_flag`. The second parameter may correspond to one or more of `pps_pic_width_in_luma_samples`,

pps_pic_height_in_luma_samples, pps_scaling_win_left_offset, pps_scaling_win_right_offset, pps_scaling_win_top_offset, and pps_scaling_win_bottom_offset.

Other Examples

(1153) Encoder **100** and decoder **200** in each of the above-described examples may be used as an image encoder and an image decoder, respectively, or may be used as a video encoder and a video decoder, respectively.

(1154) Alternatively, each of encoder **100** and decoder **200** may be used as an inter prediction device. In other words, encoder **100** and decoder **200** may correspond to only inter predictor **126** and inter predictor **218**, respectively. The other constituent elements may be included in other devices.

(1155) Moreover, encoder **100** may include an inputter and an outputter. For example, one or more pictures are input to an inputter of encoder **100**, and an encoded bitstream is output from an outputter of encoder **100**. Decoder **200** may also include an inputter and an outputter. For example, an encoded bitstream is input to an inputter of decoder **200**, and one or more pictures are output from an outputter of decoder **200**. The encoded bitstream may include quantized coefficients to which variable length encoding has been applied and control information.

(1156) Moreover, the term encode may be replaced with another term such as store, include, write, describe, signal, send out, notice, or hold. For example, encoding information may be including information in a bitstream. Moreover, encoding information into a bitstream may mean that information is encoded to generate a bitstream including the encoded information.

(1157) Moreover, the term decode may be replaced with another term such as retrieve, parse, read, load, derive, obtain, receive, extract, or restore. For example, decoding information may be obtaining information from a bitstream. Moreover, decoding information from a bitstream may mean that a bitstream is decoded to obtain information included in the bitstream.

(1158) In addition, at least part of each of the examples described above may be used as an encoding method or a decoding method, may be used as an inter prediction method, or may be used as another method.

(1159) In addition, each constituent element may be configured with dedicated hardware, or may be implemented by executing a software program suitable for the constituent element. Each component may be implemented by causing a program executor such as a CPU or a processor to read out and execute a software program stored on a medium such as a hard disk or a semiconductor memory.

(1160) More specifically, each of encoder **100** and decoder **200** may include processing circuitry and storage which is electrically connected to the processing circuitry and is accessible from the processing circuitry. For example, the processing circuitry corresponds to processor **a1** or **b1**, and the storage corresponds to memory **a2** or **b2**.

(1161) The processing circuitry includes at least one of a dedicated hardware and a program executor, and performs processing using the storage. Moreover, when the processing circuitry includes the program executor, the storage stores a software program to be executed by the program executor. Here, an example of the software program described above is a bitstream. The bitstream includes picture data to be decoded and syntaxes to perform a decoding process for decoding picture data. The bitstream causes a decoder to execute the process according to the syntaxes, and thus it is possible to decode the picture data.

(1162) Here, the software which implements either encoder **100**, decoder **200**, or the like described above is a program indicated below.

(1163) For example, this program may cause a computer to execute an encoding method including: controlling whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in encoding of an inter-

predicted picture is different from a resolution of the inter-predicted picture, resampling a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and encoding an image in the inter-predicted picture using the reference image resampled. It is to be noted that “preceding” described here may refer to preceding in the display order or in the encoding order.

(1164) Moreover, for example, this program may cause a computer to execute a decoding method including: controlling whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and decoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in decoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resampling a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and decoding an image in the inter-predicted picture using the reference image resampled. It is to be noted that “preceding” described here may refer to preceding in the display order or in the decoding order.

(1165) Moreover, each constituent element may be a circuit as described above. The circuits may compose circuitry as a whole, or may be separate circuits. Alternatively, each constituent element may be implemented as a general processor, or may be implemented as a dedicated processor.

(1166) Moreover, the process that is executed by a particular constituent element may be executed by another constituent element. Moreover, the processing execution order may be modified, or a plurality of processes may be executed in parallel. Moreover, an encoder and decoder may include encoder **100** and decoder **200**.

(1167) In addition, the ordinal numbers such as “first” and “second” used for explanation may be changed appropriately. Moreover, the ordinal number may be newly assigned to a component, etc., or may be deleted from a component, etc. Moreover, the ordinal numbers may be assigned to components to differentiate between the components, and may not correspond to the meaningful order.

(1168) Moreover, the expression of at least a threshold and the expression of greater than a threshold may be replaced with each other. Moreover, the expression of at most a threshold and the expression of less than a threshold may be replaced with each other.

(1169) Although aspects of encoder **100** and decoder **200** have been described based on a plurality of examples, aspects of encoder **100** and decoder **200** are not limited to these examples. The scope of the aspects of encoder **100** and decoder **200** may encompass embodiments obtainable by adding, to any of these embodiments, various kinds of modifications that a person skilled in the art would conceive and embodiments configurable by combining constituent elements in different embodiments, without deviating from the scope of the present disclosure.

(1170) The present aspect may be performed by combining one or more aspects disclosed herein with at least part of other aspects according to the present disclosure. In addition, the present aspect may be performed by combining, with the other aspects, part of the processes indicated in any of the flow charts according to the aspects, part of the configuration of any of the devices, part of syntaxes, etc.

(1171) [Implementations and Applications]

(1172) As described in each of the above embodiments, each functional or operational block may typically be realized as an MPU (micro processing unit) and memory, for example. Moreover, processes performed by each of the functional blocks may be realized as a program execution unit, such as a processor which reads and executes software (a program) recorded on a medium such as ROM. The software may be distributed. The software may be recorded on a variety of media such as semiconductor memory. Note that each functional block can also be realized as hardware (dedicated circuit).

(1173) The processing described in each of the embodiments may be realized via integrated

processing using a single apparatus (system), and, alternatively, may be realized via decentralized processing using a plurality of apparatuses. Moreover, the processor that executes the above-described program may be a single processor or a plurality of processors. In other words, integrated processing may be performed, and, alternatively, decentralized processing may be performed.

(1174) Embodiments of the present disclosure are not limited to the above exemplary embodiments; various modifications may be made to the exemplary embodiments, the results of which are also included within the scope of the embodiments of the present disclosure.

(1175) Next, application examples of the moving picture encoding method (image encoding method) and the moving picture decoding method (image decoding method) described in each of the above embodiments will be described, as well as various systems that implement the application examples. Such a system may be characterized as including an image encoder that employs the image encoding method, an image decoder that employs the image decoding method, or an image encoder-decoder that includes both the image encoder and the image decoder. Other configurations of such a system may be modified on a case-by-case basis.

Usage Examples

(1176) FIG. 110 illustrates an overall configuration of content providing system ex100 suitable for implementing a content distribution service. The area in which the communication service is provided is divided into cells of desired sizes, and base stations ex106, ex107, ex108, ex109, and ex110, which are fixed wireless stations in the illustrated example, are located in respective cells.

(1177) In content providing system ex100, devices including computer ex111, gaming device ex112, camera ex113, home appliance ex114, and smartphone ex115 are connected to internet ex101 via internet service provider ex102 or communications network ex104 and base stations ex106 through ex110. Content providing system ex100 may combine and connect any of the above devices. In various implementations, the devices may be directly or indirectly connected together via a telephone network or near field communication, rather than via base stations ex106 through ex110. Further, streaming server ex103 may be connected to devices including computer ex111, gaming device ex112, camera ex113, home appliance ex114, and smartphone ex115 via, for example, internet ex101. Streaming server ex103 may also be connected to, for example, a terminal in a hotspot in airplane ex117 via satellite ex116.

(1178) Note that instead of base stations ex106 through ex110, wireless access points or hotspots may be used. Streaming server ex103 may be connected to communications network ex104 directly instead of via internet ex101 or internet service provider ex102, and may be connected to airplane ex117 directly instead of via satellite ex116.

(1179) Camera ex113 is a device capable of capturing still images and video, such as a digital camera. Smartphone ex115 is a smartphone device, cellular phone, or personal handyphone system (PHS) phone that can operate under the mobile communications system standards of the 2G, 3G, 3.9G, and 4G systems, as well as the next-generation 5G system.

(1180) Home appliance ex114 is, for example, a refrigerator or a device included in a home fuel cell cogeneration system.

(1181) In content providing system ex100, a terminal including an image and/or video capturing function is capable of, for example, live streaming by connecting to streaming server ex103 via, for example, base station ex106. When live streaming, a terminal (e.g., computer ex111, gaming device ex112, camera ex113, home appliance ex114, smartphone ex115, or a terminal in airplane ex117) may perform the encoding processing described in the above embodiments on still-image or video content captured by a user via the terminal, may multiplex video data obtained via the encoding and audio data obtained by encoding audio corresponding to the video, and may transmit the obtained data to streaming server ex103. In other words, the terminal functions as the image encoder according to one aspect of the present disclosure.

(1182) Streaming server ex103 streams transmitted content data to clients that request the stream. Client examples include computer ex111, gaming device ex112, camera ex113, home appliance

ex114, smartphone ex115, and terminals inside airplane ex117, which are capable of decoding the above-described encoded data. Devices that receive the streamed data decode and reproduce the received data. In other words, the devices may each function as the image decoder, according to one aspect of the present disclosure.

(1183) [Decentralized Processing]

(1184) Streaming server ex103 may be realized as a plurality of servers or computers between which tasks such as the processing, recording, and streaming of data are divided. For example, streaming server ex103 may be realized as a content delivery network (CDN) that streams content via a network connecting multiple edge servers located throughout the world. In a CDN, an edge server physically near a client is dynamically assigned to the client. Content is cached and streamed to the edge server to reduce load times. In the event of, for example, some type of error or change in connectivity due, for example, to a spike in traffic, it is possible to stream data stably at high speeds, since it is possible to avoid affected parts of the network by, for example, dividing the processing between a plurality of edge servers, or switching the streaming duties to a different edge server and continuing streaming.

(1185) Decentralization is not limited to just the division of processing for streaming; the encoding of the captured data may be divided between and performed by the terminals, on the server side, or both. In one example, in typical encoding, the processing is performed in two loops. The first loop is for detecting how complicated the image is on a frame-by-frame or scene-by-scene basis, or detecting the encoding load. The second loop is for processing that maintains image quality and improves encoding efficiency. For example, it is possible to reduce the processing load of the terminals and improve the quality and encoding efficiency of the content by having the terminals perform the first loop of the encoding and having the server side that received the content perform the second loop of the encoding. In such a case, upon receipt of a decoding request, it is possible for the encoded data resulting from the first loop performed by one terminal to be received and reproduced on another terminal in approximately real time. This makes it possible to realize smooth, real-time streaming.

(1186) In another example, camera ex113 or the like extracts a feature amount from an image, compresses data related to the feature amount as metadata, and transmits the compressed metadata to a server. For example, the server determines the significance of an object based on the feature amount and changes the quantization accuracy accordingly to perform compression suitable for the meaning (or content significance) of the image. Feature amount data is particularly effective in improving the precision and efficiency of motion vector prediction during the second compression pass performed by the server. Moreover, encoding that has a relatively low processing load, such as variable length coding (VLC), may be handled by the terminal, and encoding that has a relatively high processing load, such as context-adaptive binary arithmetic coding (CABAC), may be handled by the server.

(1187) In yet another example, there are instances in which a plurality of videos of approximately the same scene are captured by a plurality of terminals in, for example, a stadium, shopping mall, or factory. In such a case, for example, the encoding may be decentralized by dividing processing tasks between the plurality of terminals that captured the videos and, if necessary, other terminals that did not capture the videos, and the server, on a per-unit basis. The units may be, for example, groups of pictures (GOP), pictures, or tiles resulting from dividing a picture. This makes it possible to reduce load times and achieve streaming that is closer to real time.

(1188) Since the videos are of approximately the same scene, management and/or instructions may be carried out by the server so that the videos captured by the terminals can be cross-referenced. Moreover, the server may receive encoded data from the terminals, change the reference relationship between items of data, or correct or replace pictures themselves, and then perform the encoding. This makes it possible to generate a stream with increased quality and efficiency for the individual items of data.

(1189) Furthermore, the server may stream video data after performing transcoding to convert the encoding format of the video data. For example, the server may convert the encoding format from MPEG to VP (e.g., VP9), and may convert H.264 to H.265.

(1190) In this way, encoding can be performed by a terminal or one or more servers. Accordingly, although the device that performs the encoding is referred to as a “server” or “terminal” in the following description, some or all of the processes performed by the server may be performed by the terminal, and likewise some or all of the processes performed by the terminal may be performed by the server. This also applies to decoding processes.

(1191) [3D, Multi-Angle]

(1192) There has been an increase in usage of images or videos combined from images or videos of different scenes concurrently captured, or of the same scene captured from different angles, by a plurality of terminals such as camera ex113 and/or smartphone ex115. Videos captured by the terminals are combined based on, for example, the separately obtained relative positional relationship between the terminals, or regions in a video having matching feature points.

(1193) In addition to the encoding of two-dimensional moving pictures, the server may encode a still image based on scene analysis of a moving picture, either automatically or at a point in time specified by the user, and transmit the encoded still image to a reception terminal. Furthermore, when the server can obtain the relative positional relationship between the video capturing terminals, in addition to two-dimensional moving pictures, the server can generate three-dimensional geometry of a scene based on video of the same scene captured from different angles. The server may separately encode three-dimensional data generated from, for example, a point cloud and, based on a result of recognizing or tracking a person or object using three-dimensional data, may select or reconstruct and generate a video to be transmitted to a reception terminal, from videos captured by a plurality of terminals.

(1194) This allows the user to enjoy a scene by freely selecting videos corresponding to the video capturing terminals, and allows the user to enjoy the content obtained by extracting a video at a selected viewpoint from three-dimensional data reconstructed from a plurality of images or videos. Furthermore, as with video, sound may be recorded from relatively different angles, and the server may multiplex audio from a specific angle or space with the corresponding video, and transmit the multiplexed video and audio.

(1195) In recent years, content that is a composite of the real world and a virtual world, such as virtual reality (VR) and augmented reality (AR) content, has also become popular. In the case of VR images, the server may create images from the viewpoints of both the left and right eyes, and perform encoding that tolerates reference between the two viewpoint images, such as multi-view coding (MVC), and, alternatively, may encode the images as separate streams without referencing. When the images are decoded as separate streams, the streams may be synchronized when reproduced, so as to recreate a virtual three-dimensional space in accordance with the viewpoint of the user.

(1196) In the case of AR images, the server superimposes virtual object information existing in a virtual space onto camera information representing a real-world space, based on a three-dimensional position or movement from the perspective of the user. The decoder may obtain or store virtual object information and three-dimensional data, generate two-dimensional images based on movement from the perspective of the user, and then generate superimposed data by seamlessly connecting the images. Alternatively, the decoder may transmit, to the server, motion from the perspective of the user in addition to a request for virtual object information. The server may generate superimposed data based on three-dimensional data stored in the server, in accordance with the received motion, and encode and stream the generated superimposed data to the decoder. Note that superimposed data includes, in addition to RGB values, a value indicating transparency, and the server sets a value for sections other than the object generated from three-dimensional data to, for example, 0, and may perform the encoding while those sections are

transparent. Alternatively, the server may set the background to a determined RGB value, such as a chroma key, and generate data in which areas other than the object are set as the background. (1197) Decoding of similarly streamed data may be performed by the client (i.e., the terminals), on the server side, or divided therebetween. In one example, one terminal may transmit a reception request to a server, the requested content may be received and decoded by another terminal, and a decoded signal may be transmitted to a device having a display. It is possible to reproduce high image quality data by decentralizing processing and appropriately selecting content regardless of the processing ability of the communications terminal itself. In yet another example, while a TV, for example, is receiving image data that is large in size, a region of a picture, such as a tile obtained by dividing the picture, may be decoded and displayed on a personal terminal or terminals of a viewer or viewers of the TV. This makes it possible for the viewers to share a big-picture view as well as for each viewer to check his or her assigned area, or inspect a region in further detail up close.

(1198) In situations in which a plurality of wireless connections are possible over near, mid, and far distances, indoors or outdoors, it may be possible to seamlessly receive content using a streaming system standard such as MPEG Dynamic Adaptive Streaming over HTTP (MPEG-DASH). The user may switch between data in real time while freely selecting a decoder or display apparatus including the user's terminal, displays arranged indoors or outdoors, etc. Moreover, using, for example, information on the position of the user, decoding can be performed while switching which terminal handles decoding and which terminal handles the displaying of content. This makes it possible to map and display information, while the user is on the move in route to a destination, on the wall of a nearby building in which a device capable of displaying content is embedded, or on part of the ground. Moreover, it is also possible to switch the bit rate of the received data based on the accessibility to the encoded data on a network, such as when encoded data is cached on a server quickly accessible from the reception terminal, or when encoded data is copied to an edge server in a content delivery service.

(1199) [Web Page Optimization]

(1200) FIG. 111 illustrates an example of a display screen of a web page on computer ex111, for example. FIG. 112 illustrates an example of a display screen of a web page on smartphone ex115, for example. As illustrated in FIG. 111 and FIG. 112, a web page may include a plurality of image links that are links to image content, and the appearance of the web page differs depending on the device used to view the web page. When a plurality of image links are viewable on the screen, until the user explicitly selects an image link, or until the image link is in the approximate center of the screen or the entire image link fits in the screen, the display apparatus (decoder) may display, as the image links, still images included in the content or I pictures; may display video such as an animated gif using a plurality of still images or I pictures; or may receive only the base layer, and decode and display the video.

(1201) When an image link is selected by the user, the display apparatus performs decoding while giving the highest priority to the base layer. Note that if there is information in the Hyper Text Markup Language (HTML) code of the web page indicating that the content is scalable, the display apparatus may decode up to the enhancement layer. Further, in order to guarantee real-time reproduction, before a selection is made or when the bandwidth is severely limited, the display apparatus can reduce delay between the point in time at which the leading picture is decoded and the point in time at which the decoded picture is displayed (that is, the delay between the start of the decoding of the content to the displaying of the content) by decoding and displaying only forward reference pictures (I picture, P picture, forward reference B picture). Still further, the display apparatus may purposely ignore the reference relationship between pictures, and coarsely decode all B and P pictures as forward reference pictures, and then perform normal decoding as the number of pictures received over time increases.

(1202) [Autonomous Driving]

(1203) When transmitting and receiving still image or video data such as two- or three-dimensional map information for autonomous driving or assisted driving of an automobile, the reception terminal may receive, in addition to image data belonging to one or more layers, information on, for example, the weather or road construction as metadata, and associate the metadata with the image data upon decoding. Note that metadata may be assigned per layer and, alternatively, may simply be multiplexed with the image data.

(1204) In such a case, since the automobile, drone, airplane, etc., containing the reception terminal is mobile, the reception terminal may seamlessly receive and perform decoding while switching between base stations among base stations ex106 through ex110 by transmitting information indicating the position of the reception terminal. Moreover, in accordance with the selection made by the user, the situation of the user, and/or the bandwidth of the connection, the reception terminal may dynamically select to what extent the metadata is received, or to what extent the map information, for example, is up dated.

(1205) In content providing system ex100, the client may receive, decode, and reproduce, in real time, encoded information transmitted by the user.

(1206) [Streaming of Individual Content]

(1207) In content providing system ex100, in addition to high image quality, long content distributed by a video distribution entity, unicast or multicast streaming of low image quality, and short content from an individual are also possible. Such content from individuals is likely to further increase in popularity. The server may first perform editing processing on the content before the encoding processing, in order to refine the individual content. This may be achieved using the following configuration, for example.

(1208) In real time while capturing video or image content, or after the content has been captured and accumulated, the server performs recognition processing based on the raw data or encoded data, such as capture error processing, scene search processing, meaning analysis, and/or object detection processing. Then, based on the result of the recognition processing, the server—either when prompted or automatically—edits the content, examples of which include: correction such as focus and/or motion blur correction; removing low-priority scenes such as scenes that are low in brightness compared to other pictures, or out of focus; object edge adjustment; and color tone adjustment. The server encodes the edited data based on the result of the editing. It is known that excessively long videos tend to receive fewer views. Accordingly, in order to keep the content within a specific length that scales with the length of the original video, the server may, in addition to the low-priority scenes described above, automatically clip out scenes with low movement, based on an image processing result. Alternatively, the server may generate and encode a video digest based on a result of an analysis of the meaning of a scene.

(1209) There may be instances in which individual content may include content that infringes a copyright, moral right, portrait rights, etc. Such instance may lead to an unfavorable situation for the creator, such as when content is shared beyond the scope intended by the creator. Accordingly, before encoding, the server may, for example, edit images so as to blur faces of people in the periphery of the screen or blur the inside of a house, for example. Further, the server may be configured to recognize the faces of people other than a registered person in images to be encoded, and when such faces appear in an image, may apply a mosaic filter, for example, to the face of the person. Alternatively, as pre- or post-processing for encoding, the user may specify, for copyright reasons, a region of an image including a person or a region of the background to be processed. The server may process the specified region by, for example, replacing the region with a different image, or blurring the region. If the region includes a person, the person may be tracked in the moving picture, and the person's head region may be replaced with another image as the person moves.

(1210) Since there is a demand for real-time viewing of content produced by individuals, which tends to be small in data size, the decoder first receives the base layer as the highest priority, and

performs decoding and reproduction, although this may differ depending on bandwidth. When the content is reproduced two or more times, such as when the decoder receives the enhancement layer during decoding and reproduction of the base layer, and loops the reproduction, the decoder may reproduce a high image quality video including the enhancement layer. If the stream is encoded using such scalable encoding, the video may be low quality when in an unselected state or at the start of the video, but it can offer an experience in which the image quality of the stream progressively increases in an intelligent manner. This is not limited to just scalable encoding; the same experience can be offered by configuring a single stream from a low quality stream reproduced for the first time and a second stream encoded using the first stream as a reference.

Other Implementation and Application Examples

(1211) The encoding and decoding may be performed by LSI (large scale integration circuitry) ex500 (see FIG. 110), which is typically included in each terminal. LSI ex500 may be configured of a single chip or a plurality of chips. Software for encoding and decoding moving pictures may be integrated into some type of a medium (such as a CD-ROM, a flexible disk, or a hard disk) that is readable by, for example, computer ex111, and the encoding and decoding may be performed using the software. Furthermore, when smartphone ex115 is equipped with a camera, video data obtained by the camera may be transmitted. In this case, the video data is coded by LSI ex500 included in smartphone ex115.

(1212) Note that LSI ex500 may be configured to download and activate an application. In such a case, the terminal first determines whether it is compatible with the scheme used to encode the content, or whether it is capable of executing a specific service. When the terminal is not compatible with the encoding scheme of the content, or when the terminal is not capable of executing a specific service, the terminal first downloads a codec or application software and then obtains and reproduces the content.

(1213) Aside from the example of content providing system ex100 that uses internet ex101, at least the moving picture encoder (image encoder) or the moving picture decoder (image decoder) described in the above embodiments may be implemented in a digital broadcasting system. The same encoding processing and decoding processing may be applied to transmit and receive broadcast radio waves superimposed with multiplexed audio and video data using, for example, a satellite, even though this is geared toward multicast, whereas unicast is easier with content providing system ex100.

(1214) [Hardware Configuration]

(1215) FIG. 113 illustrates further details of smartphone ex115 shown in FIG. 110. FIG. 114 illustrates a configuration example of smartphone ex115. Smartphone ex115 includes antenna ex450 for transmitting and receiving radio waves to and from base station ex110, camera ex465 capable of capturing video and still images, and display ex458 that displays decoded data, such as video captured by camera ex465 and video received by antenna ex450. Smartphone ex115 further includes user interface ex466 such as a touch panel, audio output unit ex457 such as a speaker for outputting speech or other audio, audio input unit ex456 such as a microphone for audio input, memory ex467 capable of storing decoded data such as captured video or still images, recorded audio, received video or still images, and mail, as well as decoded data, and slot ex464 which is an interface for Subscriber Identity Module (SIM) ex468 for authorizing access to a network and various data. Note that external memory may be used instead of memory ex467.

(1216) Main controller ex460, which comprehensively controls display ex458 and user interface ex466, power supply circuit ex461, user interface input controller ex462, video signal processor ex455, camera interface ex463, display controller ex459, modulator/demodulator ex452, multiplexer/demultiplexer ex453, audio signal processor ex454, slot ex464, and memory ex467 are connected via bus ex470.

(1217) When the user turns on the power button of power supply circuit ex461, smartphone ex115 is powered on into an operable state, and each component is supplied with power from a battery

pack.

(1218) Smartphone ex115 performs processing for, for example, calling and data transmission, based on control performed by main controller ex460, which includes a CPU, ROM, and RAM. When making calls, an audio signal recorded by audio input unit ex456 is converted into a digital audio signal by audio signal processor ex454, to which spread spectrum processing is applied by modulator/demodulator ex452 and digital-analog conversion and frequency conversion processing are applied by transmitter/receiver ex451, and the resulting signal is transmitted via antenna ex450. The received data is amplified, frequency converted, and analog-digital converted, inverse spread spectrum processed by modulator/demodulator ex452, converted into an analog audio signal by audio signal processor ex454, and then output from audio output unit ex457. In data transmission mode, text, still-image, or video data is transmitted by main controller ex460 via user interface input controller ex462 based on operation of user interface ex466 of the main body, for example. Similar transmission and reception processing is performed. In data transmission mode, when sending a video, still image, or video and audio, video signal processor ex455 compression encodes, by the moving picture encoding method described in the above embodiments, a video signal stored in memory ex467 or a video signal input from camera ex465, and transmits the encoded video data to multiplexer/demultiplexer ex453. Audio signal processor ex454 encodes an audio signal recorded by audio input unit ex456 while camera ex465 is capturing a video or still image, and transmits the encoded audio data to multiplexer/demultiplexer ex453. Multiplexer/demultiplexer ex453 multiplexes the encoded video data and encoded audio data using a determined scheme, modulates and converts the data using modulator/demodulator (modulator/demodulator circuit) ex452 and transmitter/receiver ex451, and transmits the result via antenna ex450.

(1219) When a video appended in an email or a chat, or a video linked from a web page, is received, for example, in order to decode the multiplexed data received via antenna ex450, multiplexer/demultiplexer ex453 demultiplexes the multiplexed data to divide the multiplexed data into a bitstream of video data and a bitstream of audio data, supplies the encoded video data to video signal processor ex455 via synchronous bus ex470, and supplies the encoded audio data to audio signal processor ex454 via synchronous bus ex470. Video signal processor ex455 decodes the video signal using a moving picture decoding method corresponding to the moving picture encoding method described in the above embodiments, and video or a still image included in the linked moving picture file is displayed on display ex458 via display controller ex459. Audio signal processor ex454 decodes the audio signal and outputs audio from audio output unit ex457. Since real-time streaming is becoming increasingly popular, there may be instances in which reproduction of the audio may be socially inappropriate, depending on the user's environment. Accordingly, as an initial value, a configuration in which only video data is reproduced, i.e., the audio signal is not reproduced, may be preferable; and audio may be synchronized and reproduced only when an input is received from the user clicking video data, for instance.

(1220) Although smartphone ex115 was used in the above example, three other implementations are conceivable: a transceiver terminal including both an encoder and a decoder; a transmitter terminal including only an encoder; and a receiver terminal including only a decoder. In the description of the digital broadcasting system, an example is given in which multiplexed data obtained as a result of video data being multiplexed with audio data is received or transmitted. The multiplexed data, however, may be video data multiplexed with data other than audio data, such as text data related to the video. Further, the video data itself rather than multiplexed data may be received or transmitted.

(1221) Although main controller ex460 including a CPU is described as controlling the encoding or decoding processes, various terminals often include Graphics Processing Units (GPUs). Accordingly, a configuration is acceptable in which a large area is processed at once by making use of the performance ability of the GPU via memory shared by the CPU and GPU, or memory

including an address that is managed so as to allow common usage by the CPU and GPU. This makes it possible to shorten encoding time, maintain the real-time nature of streaming, and reduce delay. In particular, processing relating to motion estimation, deblocking filtering, sample adaptive offset (SAO), and transformation/quantization can be effectively carried out by the GPU, instead of the CPU, in units of pictures, for example, all at once.

(1222) Although only some exemplary embodiments of the present disclosure have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the present disclosure.

INDUSTRIAL APPLICABILITY

(1223) The present disclosure is applicable to, for example, television receivers, digital video recorders, car navigation systems, mobile phones, digital cameras, digital video cameras, teleconferencing systems, electronic mirrors, etc.

Claims

1. An encoder comprising: circuitry; and memory coupled to the circuitry, wherein in operation, the circuitry: controls whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; when a resolution of a reference picture to be used in encoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and encodes an image in the inter-predicted picture using the reference image resampled; encodes at least one first trailing picture belonging to a first group of pictures; subsequently encodes a random access picture belonging to a second group of pictures following the first group of pictures in the display order, the random access picture being included in the one or more random access pictures; subsequently encodes at least one leading picture belonging to the first group of pictures; and subsequently encodes at least one second trailing picture belonging to the second group of pictures, and the constraint: allows a resolution of the random access picture to be different from a resolution of the at least one first trailing picture; does not allow a resolution of the at least one leading picture to be different from the resolution of the random access picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the at least one leading picture.

2. A decoder comprising: circuitry; and memory coupled to the circuitry, wherein in operation, the circuitry: controls whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and decoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; when a resolution of a reference picture to be used in decoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resamples a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and decodes an image in the inter-predicted picture using the reference image resampled; decodes at least one first trailing picture belonging to a first group of pictures; subsequently decodes a random access picture belonging to a second group of pictures following the first group of pictures in the display order, the random access picture being included in the one or more random access pictures; subsequently decodes at least one leading picture belonging to the first group of pictures; and subsequently decodes at least one second trailing picture belonging to the second group of pictures, and the constraint: allows a resolution of the random access picture to be different from a resolution of the at least one first

trailing picture; does not allow a resolution of the at least one leading picture to be different from the resolution of the random access picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the at least one leading picture.

3. An encoding method comprising: controlling whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and encoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; when a resolution of a reference picture to be used in encoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resampling a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and encoding an image in the inter-predicted picture using the reference image resampled; encoding at least one first trailing picture belonging to a first group of pictures; subsequently encoding a random access picture belonging to a second group of pictures following the first group of pictures in the display order, the random access picture being included in the one or more random access pictures; subsequently encoding at least one leading picture belonging to the first group of pictures; subsequently encoding at least one second trailing picture belonging to the second group of pictures, wherein the constraint: allows a resolution of the random access picture to be different from a resolution of the at least one first trailing picture; does not allow a resolution of the at least one leading picture to be different from the resolution of the random access picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the at least one leading picture.

4. A decoding method comprising: controlling whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and decoding order, according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; when a resolution of a reference picture to be used in decoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, resampling a reference image in the reference picture according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and decoding an image in the inter-predicted picture using the reference image resampled; decoding at least one first trailing picture belonging to a first group of pictures; subsequently decoding a random access picture belonging to a second group of pictures following the first group of pictures in the display order, the random access picture being included in the one or more random access pictures; subsequently decoding at least one leading picture belonging to the first group of pictures; and subsequently decoding at least one second trailing picture belonging to the second group of pictures, wherein the constraint: allows a resolution of the random access picture to be different from a resolution of the at least one first trailing picture; does not allow a resolution of the at least one leading picture to be different from the resolution of the random access picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the at least one leading picture.

5. A non-transitory computer readable medium storing a bitstream, the bitstream including: a plurality of pictures; and a plurality of parameters to perform a decoding process for decoding the plurality of pictures from the bitstream, wherein in the decoding process, in operation according to the plurality of parameters, whether to change a resolution of a picture from a resolution of a previous picture preceding the picture in one of display order and decoding order is controlled according to a constraint that allows the changing only when the picture is a random access picture among one or more random access pictures; and when a resolution of a reference picture to be used in decoding of an inter-predicted picture is different from a resolution of the inter-predicted picture, a reference image in the reference picture is resampled according to a difference between the resolution of the reference picture and the resolution of the inter-predicted picture, and an image in the inter-predicted picture is decoded using the reference image resampled; at least one first trailing

picture belonging to a first group of pictures is decoded; a random access picture belonging to a second group of pictures following the first group of pictures in the display order is subsequently decoded, the random access picture being included in the one or more random access pictures; at least one leading picture belonging to the first group of pictures is subsequently decoded; and at least one second trailing picture belonging to the second group of pictures is subsequently decoded, wherein the constraint: allows a resolution of the random access picture to be different from a resolution of the at least one first trailing picture; does not allow a resolution of the at least one leading picture to be different from the resolution of the random access picture; and does not allow a resolution of the at least one second trailing picture to be different from the resolution of the at least one leading picture.
